

This is a collection of slides on various topics in Kubernetes. It is intended to be used as a starting point for your own talks, rather than a self-contained presentation. If you find issues with it or would like to add to this deck, please contact thockin@google.com

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Title

Subtitle

Contact info

Kubernetes

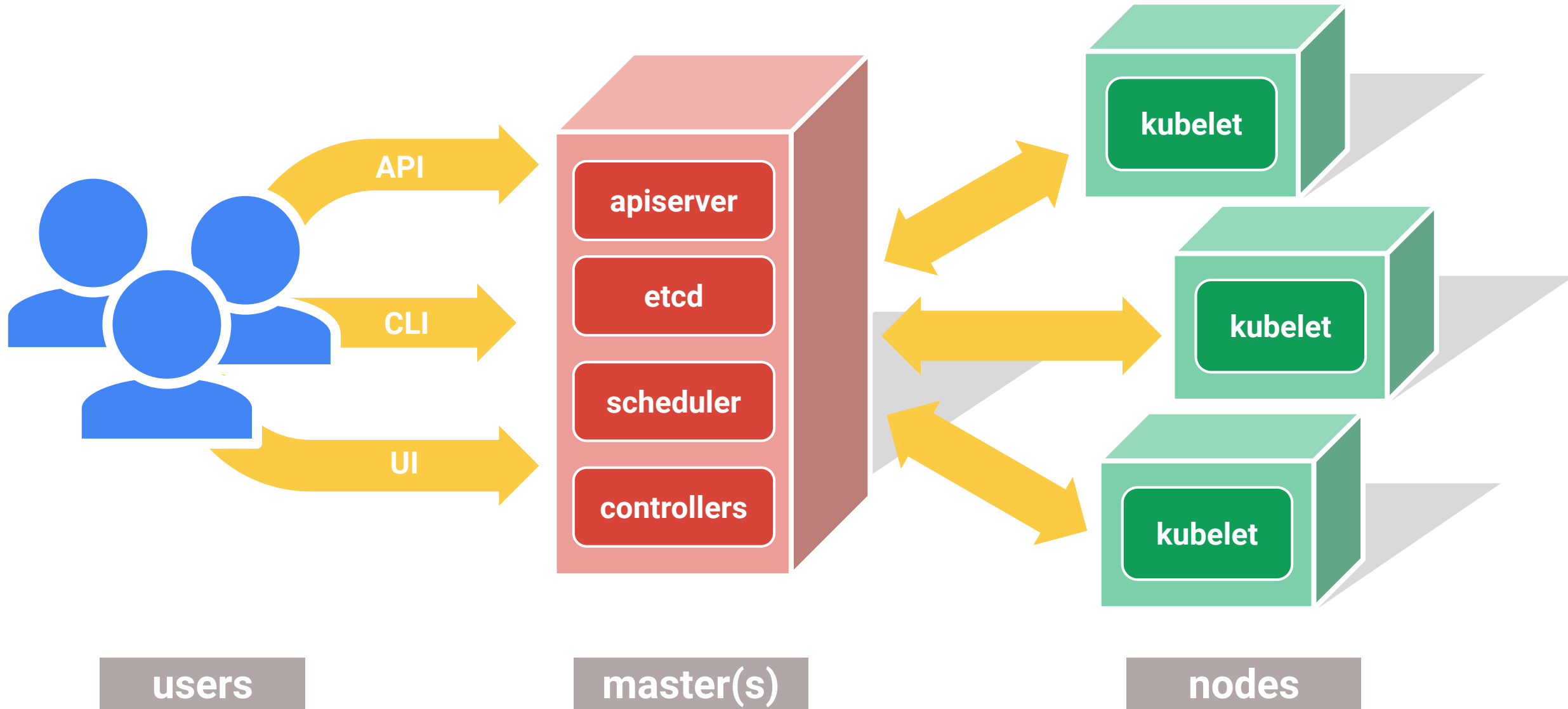
Greek for “*Helmsman*”; also the root of the words “*governor*” and “*cybernetic*”

- Manages container clusters
- Inspired and informed by Google’s experiences and internal systems
- Supports multiple cloud and bare-metal environments
- Supports multiple container runtimes
- **100% Open source**, written in Go

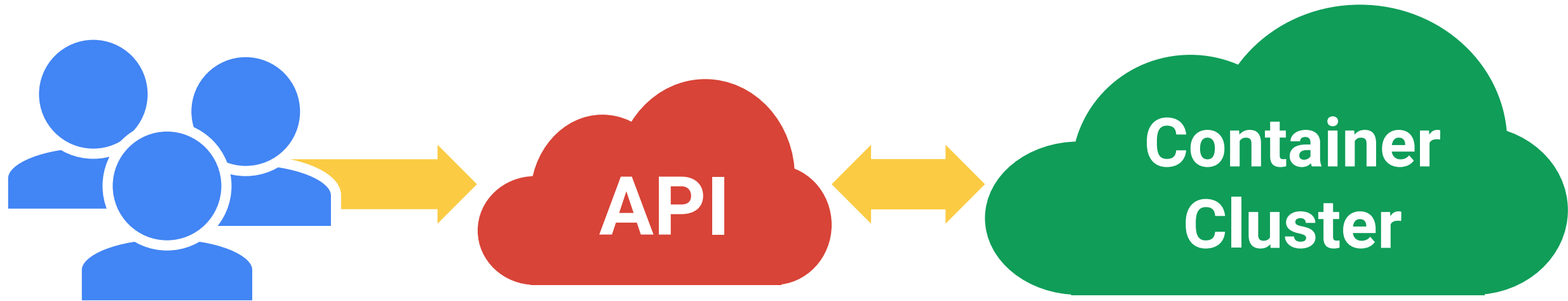
Manage applications, not machines



The 10000 foot view



All you really care about



Container clusters: A story in two parts

Container clusters: A story in two parts

1. Setting up the cluster

- Choose a cloud: GCE, AWS, Azure, Rackspace, on-premises, ...
- Choose a node OS: CoreOS, Atomic, RHEL, Debian, CentOS, Ubuntu, ...
- Provision machines: Boot VMs, install and run kube components, ...
- Configure networking: IP ranges for Pods, Services, SDN, ...
- Start cluster services: DNS, logging, monitoring, ...
- Manage nodes: kernel upgrades, OS updates, hardware failures...

Not the easy or fun part, but unavoidable

Container clusters: A story in two parts

2. Using the cluster

- Run Pods & Containers
- Replication controllers
- Services
- Volumes

This is the fun part!

A distinct set of problems from cluster setup and management

Don't make developers deal with cluster administration!

Accelerate development by focusing on the applications, not the cluster

Workload Portability

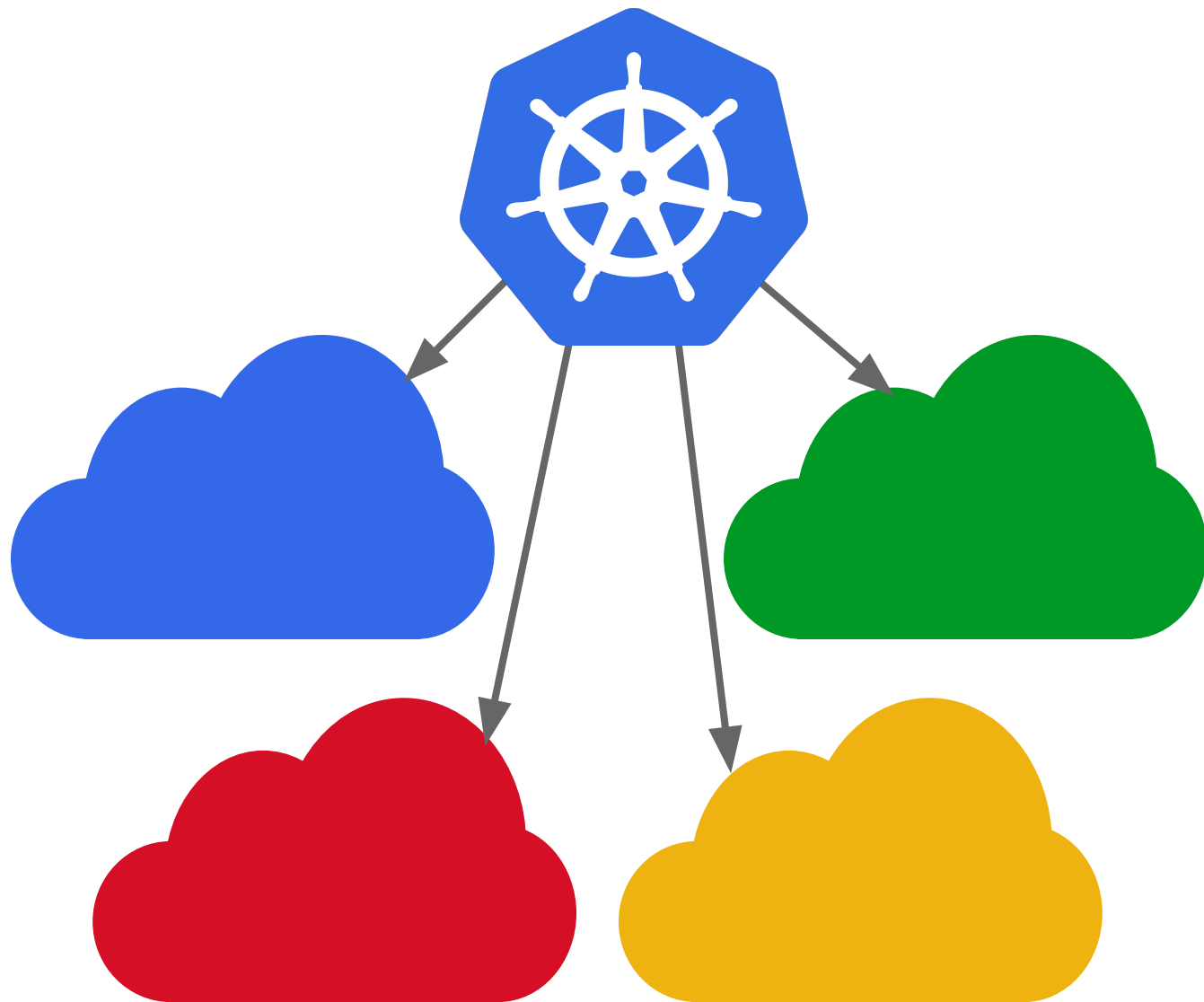
Workload portability

Goal: Avoid vendor lock-in

Runs in many environments, including “the cloud”, “bare metal”, and “your laptop”

The API and the implementation are 100% open

The whole system is modular and replaceable



Workload portability

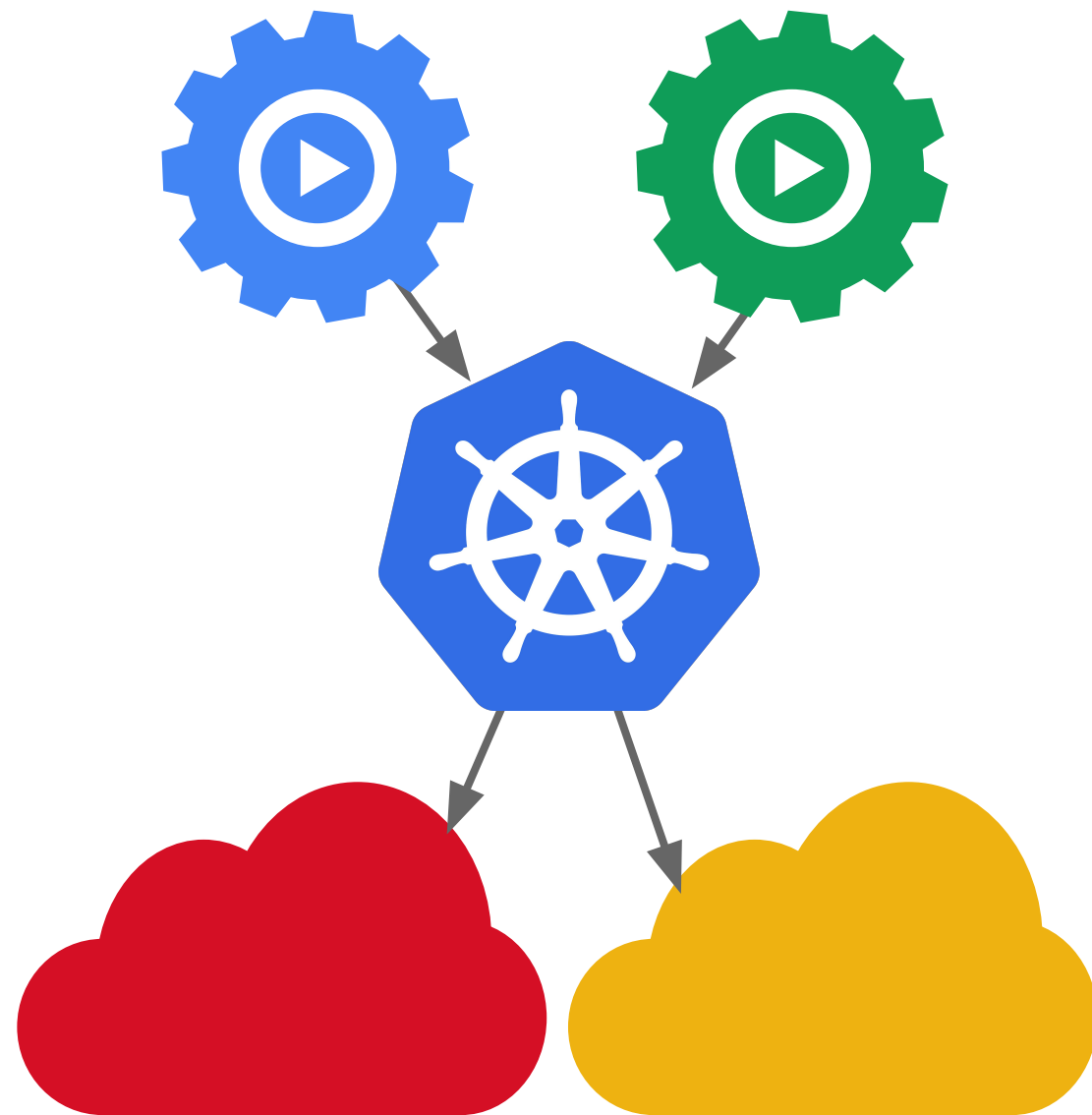
Goal: Write once, run anywhere*

Don't force apps to know about concepts that are cloud-provider-specific

Examples of this:

- Network model
- Ingress
- Service load-balancers
- PersistentVolumes

** approximately*



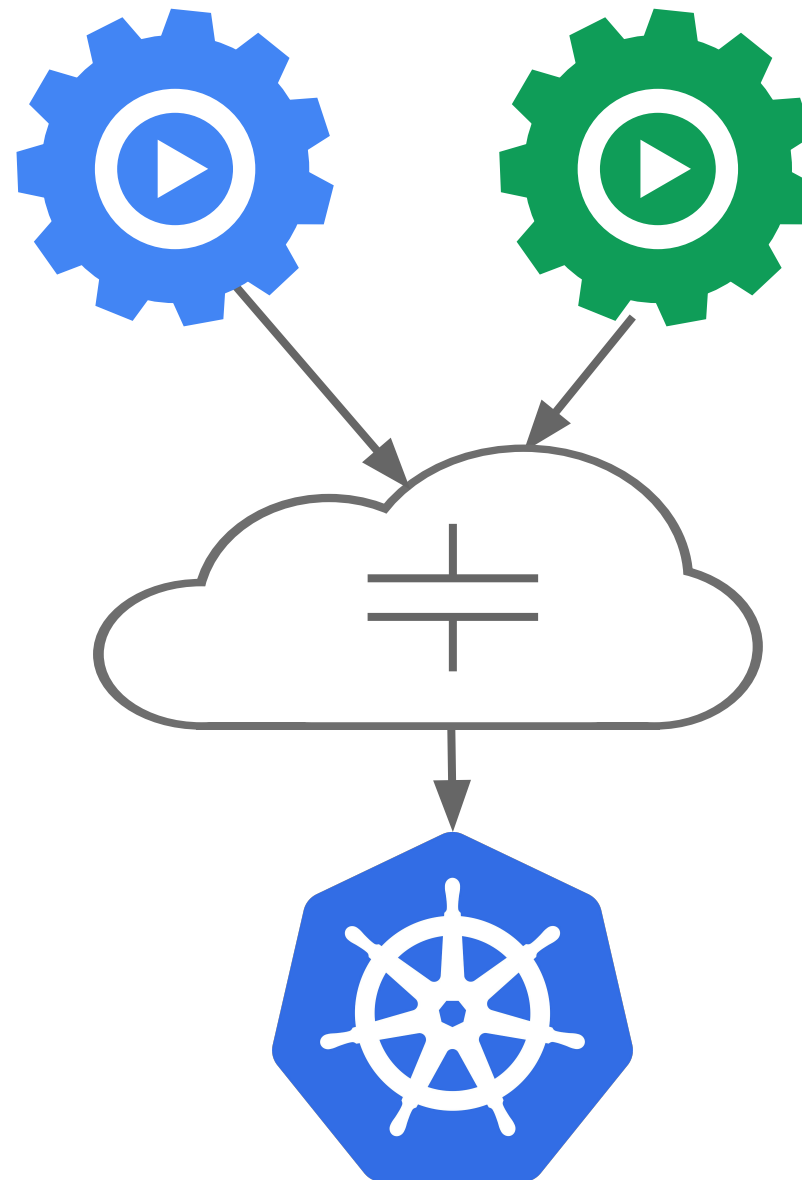
Workload portability

Goal: Avoid coupling

Don't force apps to know about concepts that are Kubernetes-specific

Examples of this:

- Namespaces
- Services / DNS
- Downward API
- Secrets / ConfigMaps



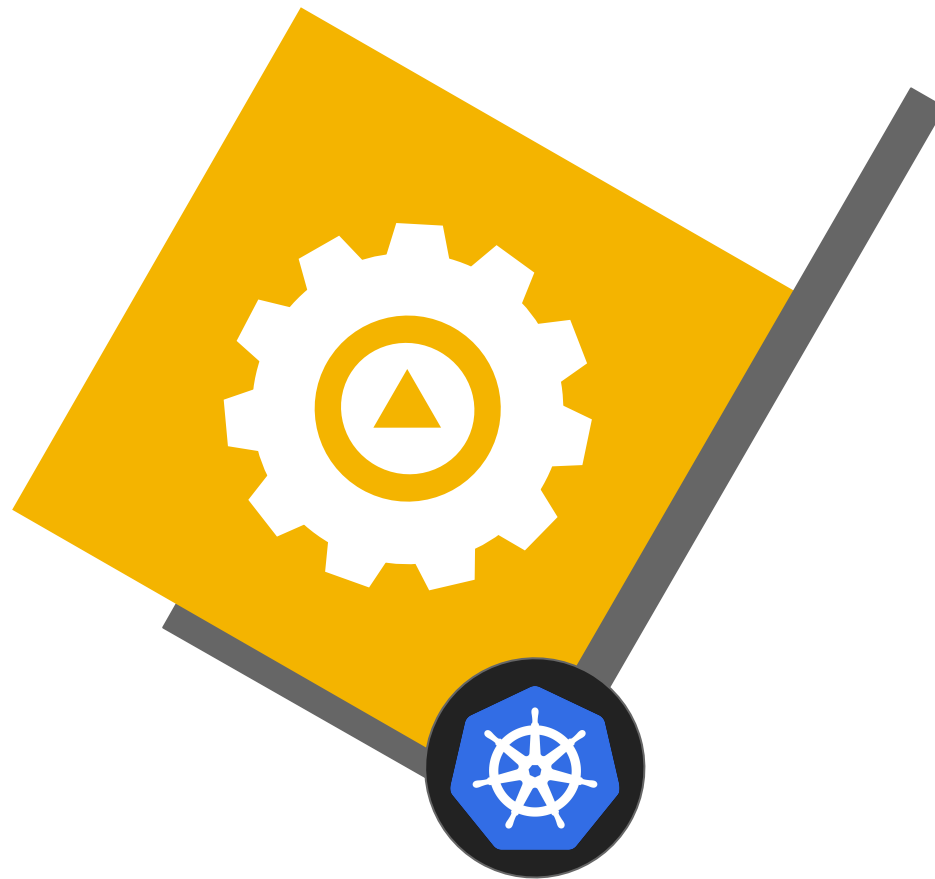
Workload portability

Result: Portability

Build your apps on-prem, lift-and-shift into cloud when you are ready

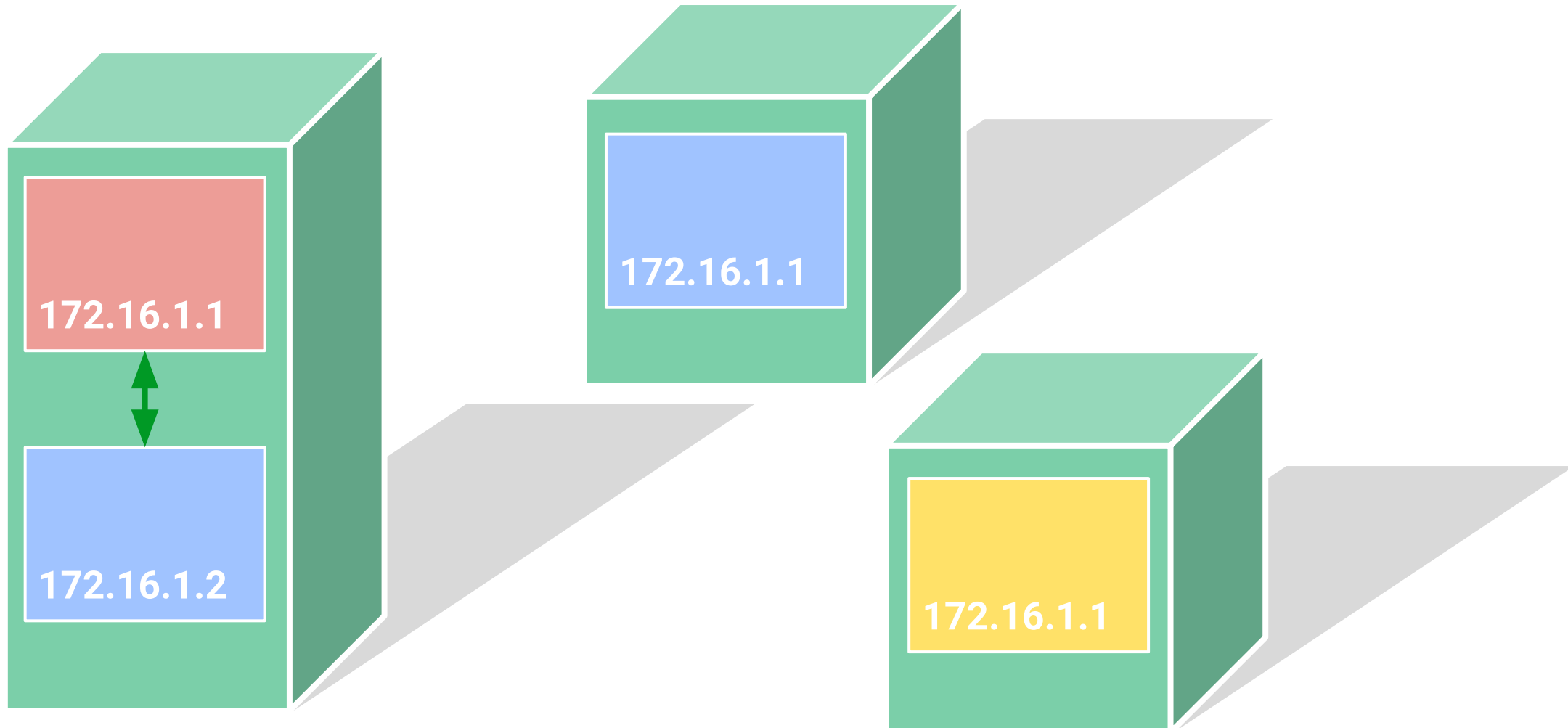
Don't get stuck with a platform that doesn't work for you

Put your app on wheels and move it whenever and wherever you need

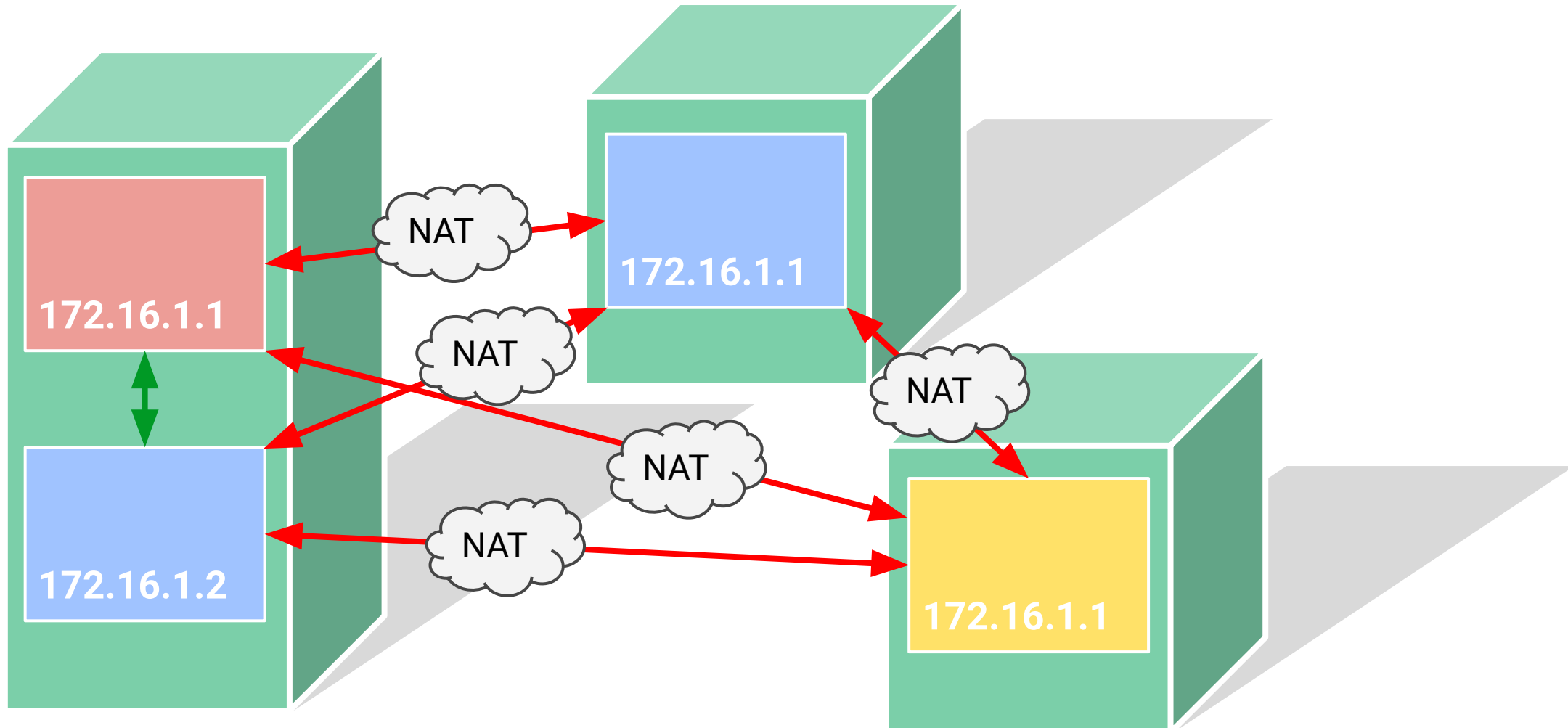


Networking

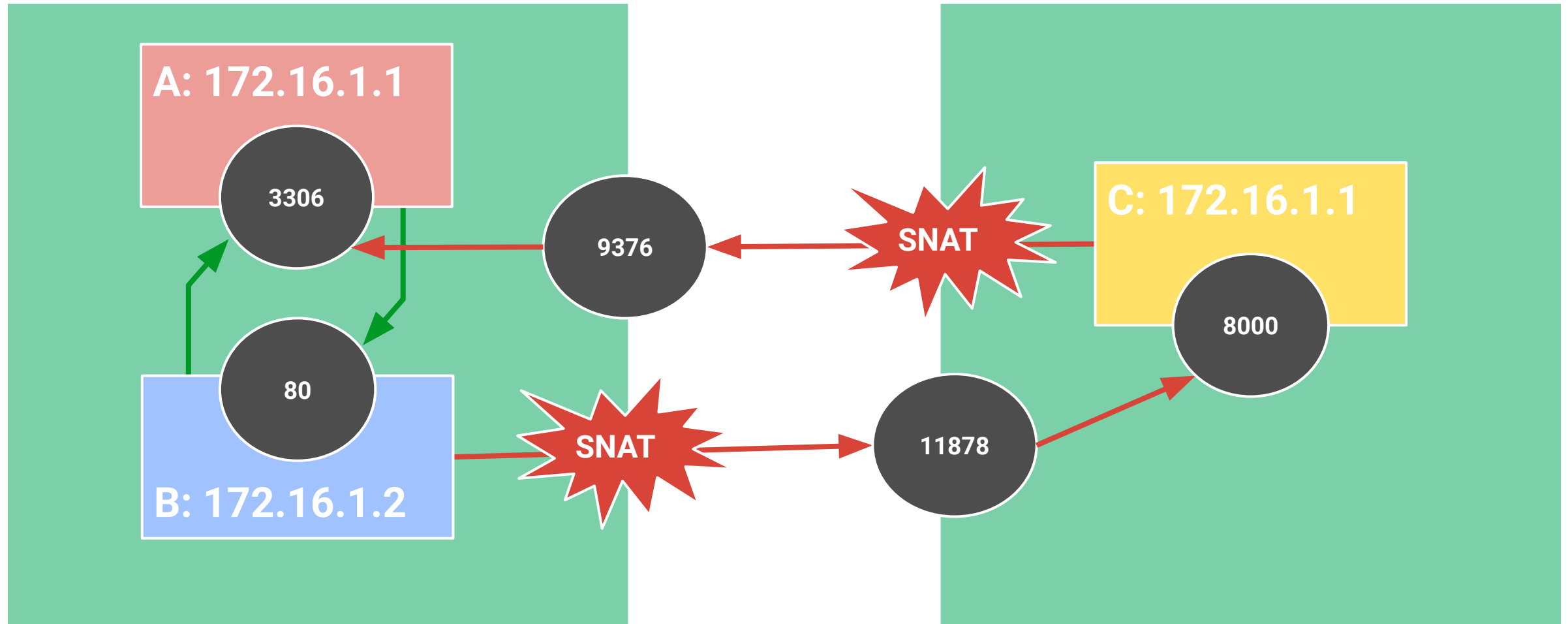
Docker networking (default)



Docker networking (default)



Port mapping



Kubernetes networking

IPs are **cluster-scoped**

Pods can reach each other directly

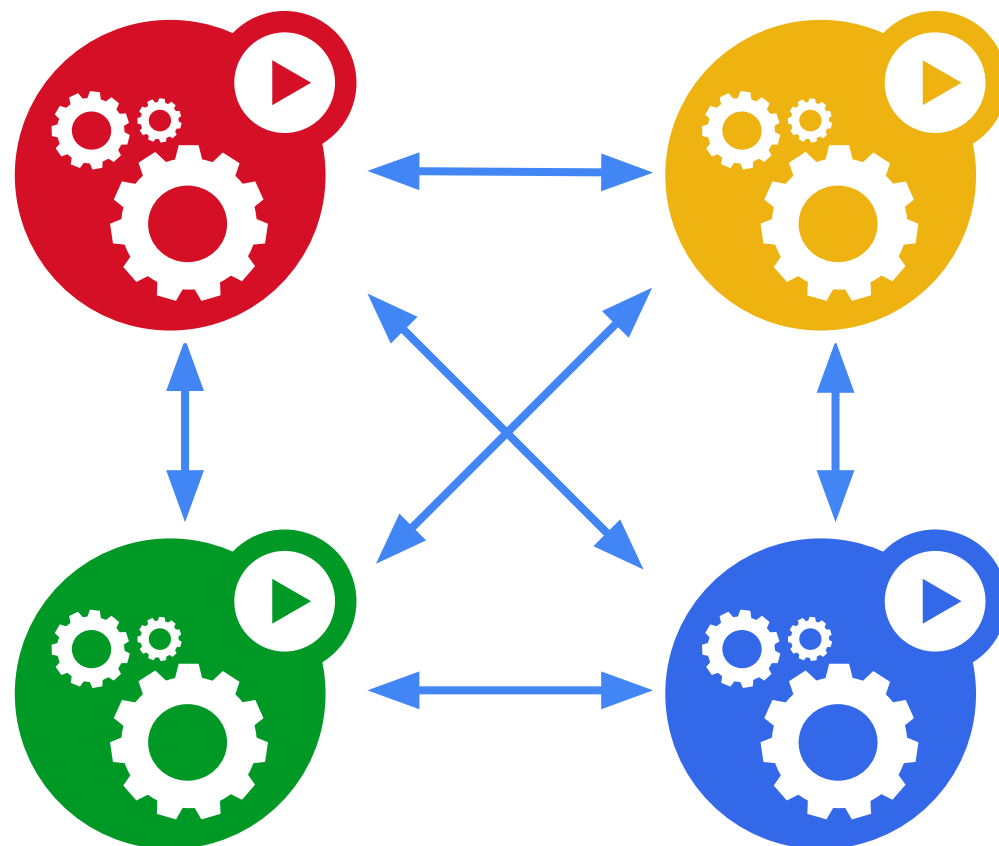
- even across nodes

No brokering of port numbers

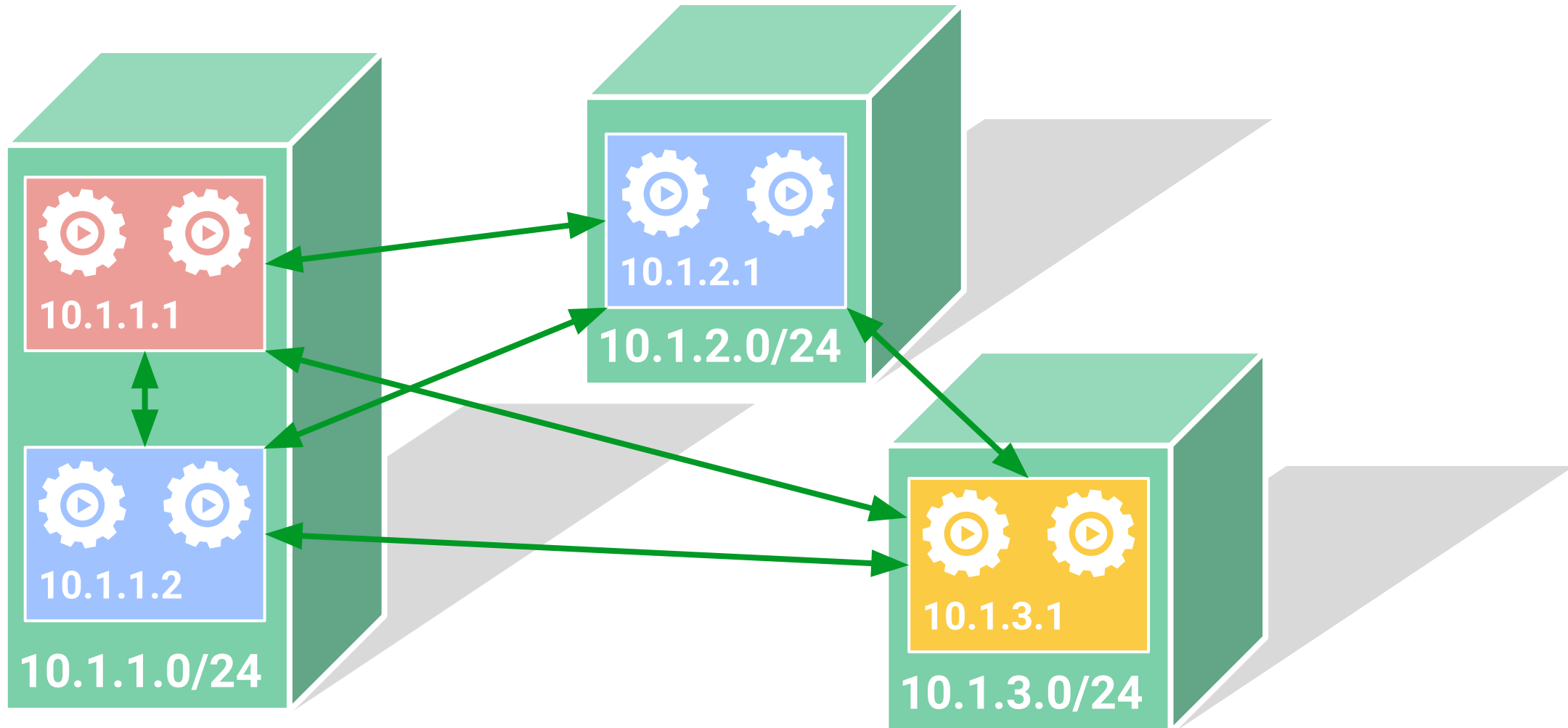
- too complex, why bother?

This is a fundamental requirement

- can be L3 routed
- can be underlayed (cloud)
- can be overlayed (SDN)



Kubernetes networking



Pods

Pods

Small group of containers & volumes

Tightly coupled

The **atom** of scheduling & placement

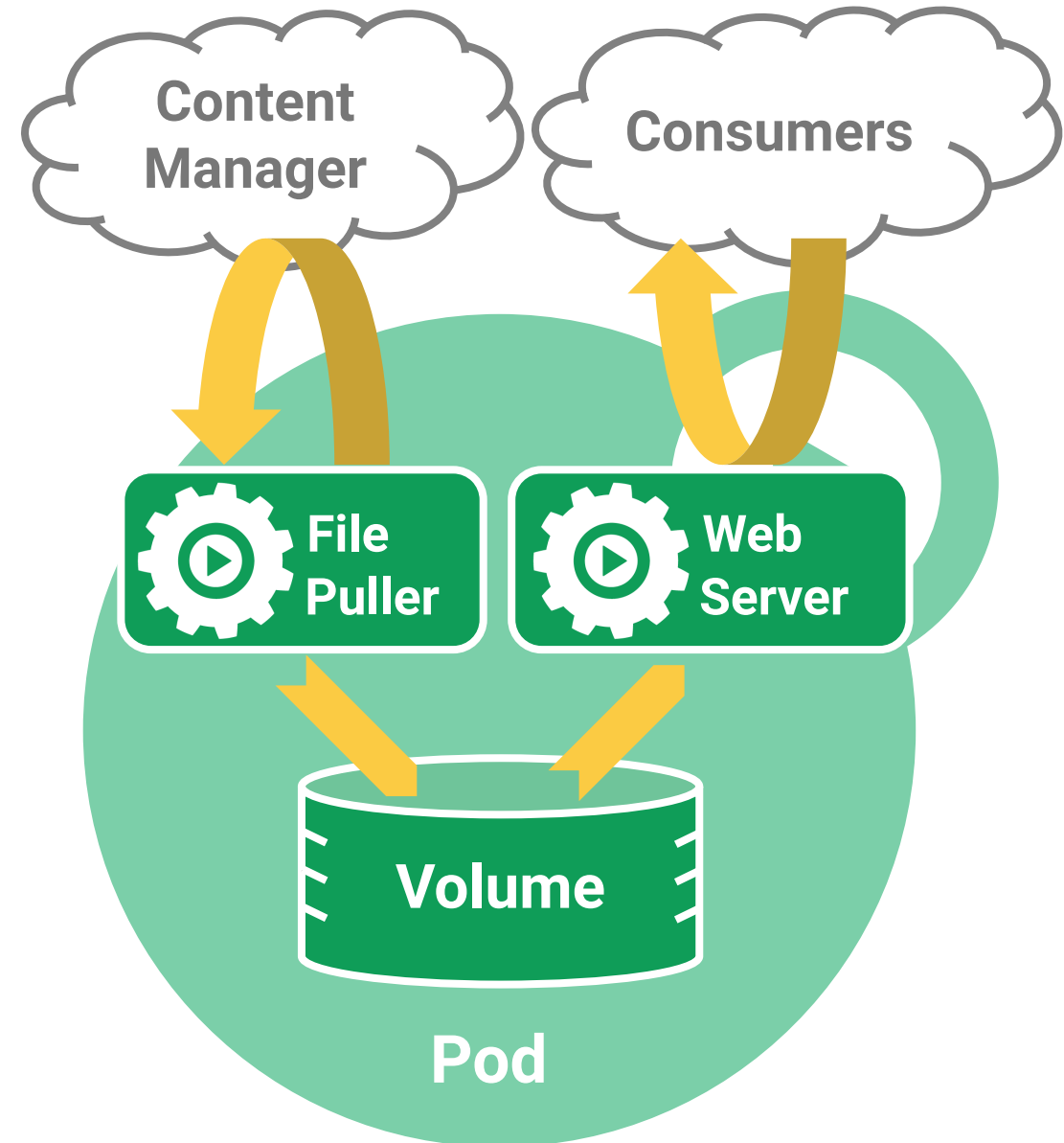
Shared namespace

- share IP address & localhost
- share IPC, etc.

Managed lifecycle

- bound to a node, restart in place
- can die, cannot be reborn with same ID

Example: data puller & web server



Volumes

Pod-scoped storage

Support many types of volume plugins

- Empty dir (and tmpfs)
- Host path
- Git repository
- GCE Persistent Disk
- AWS Elastic Block Store
- Azure File Storage
- iSCSI
- NFS
- Photon
- Portworx
- Quobyte
- vSphere
- GlusterFS
- Ceph File and RBD
- Cinder
- FibreChannel
- ScaleIO
- StorageOS
- Secret, ConfigMap, DownwardAPI
- Flex (exec a binary)
- ...



Labels & Selectors



Labels

Arbitrary metadata

Attached to **any API object**

Generally represent **identity**

Queryable by **selectors**

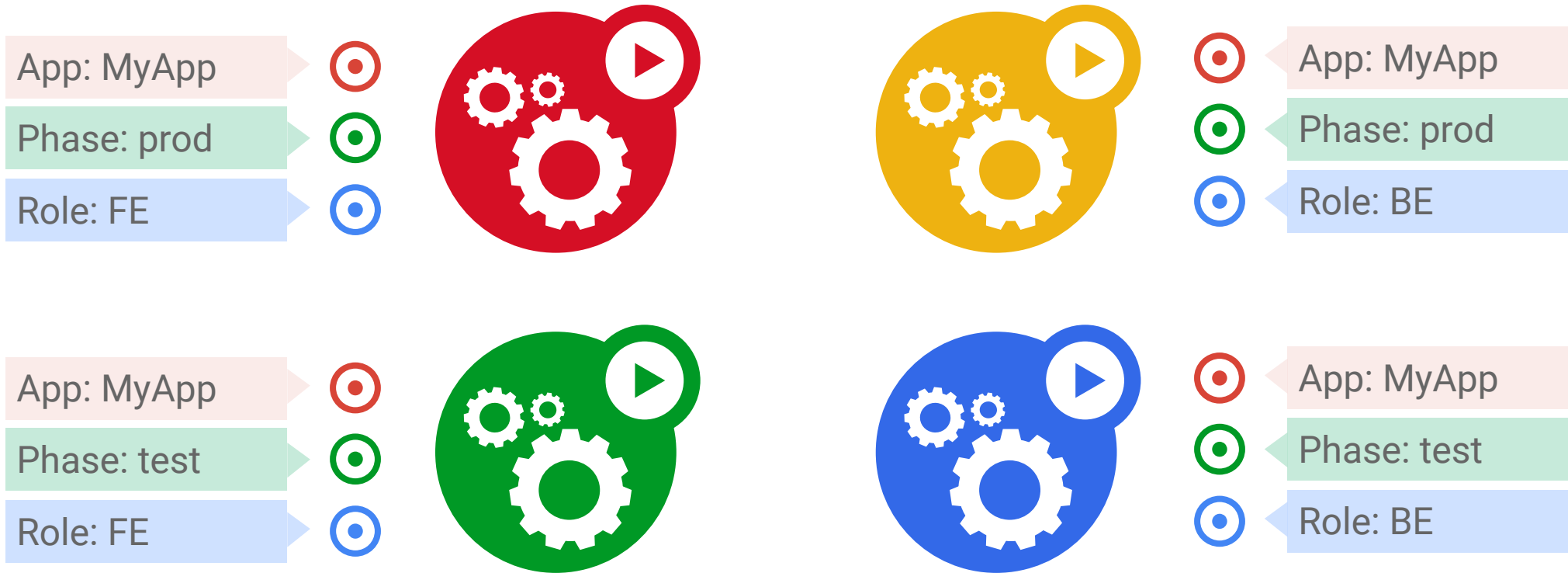
- think SQL *'select ... where ...'*

The **only** grouping mechanism

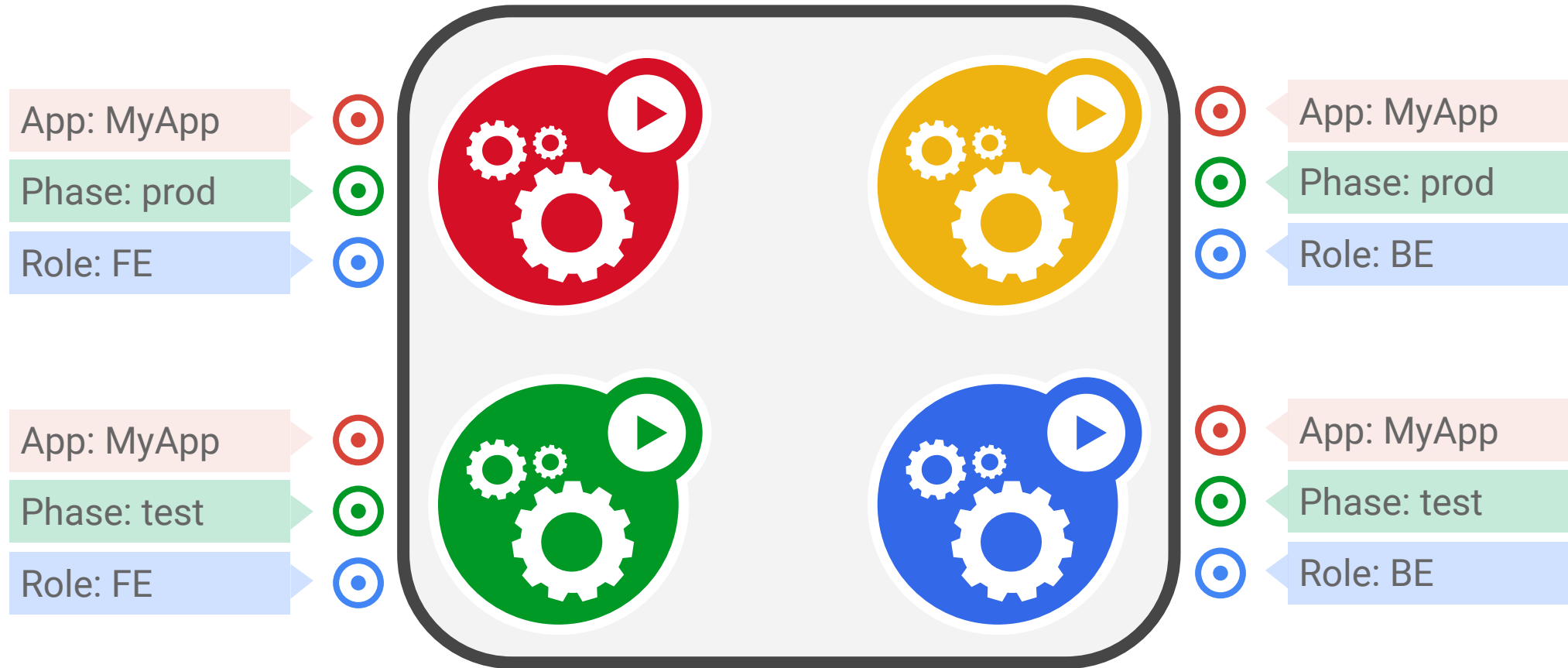
- pods in a Deployment
- pods in a Service
- capabilities of a node (constraints)



Selectors

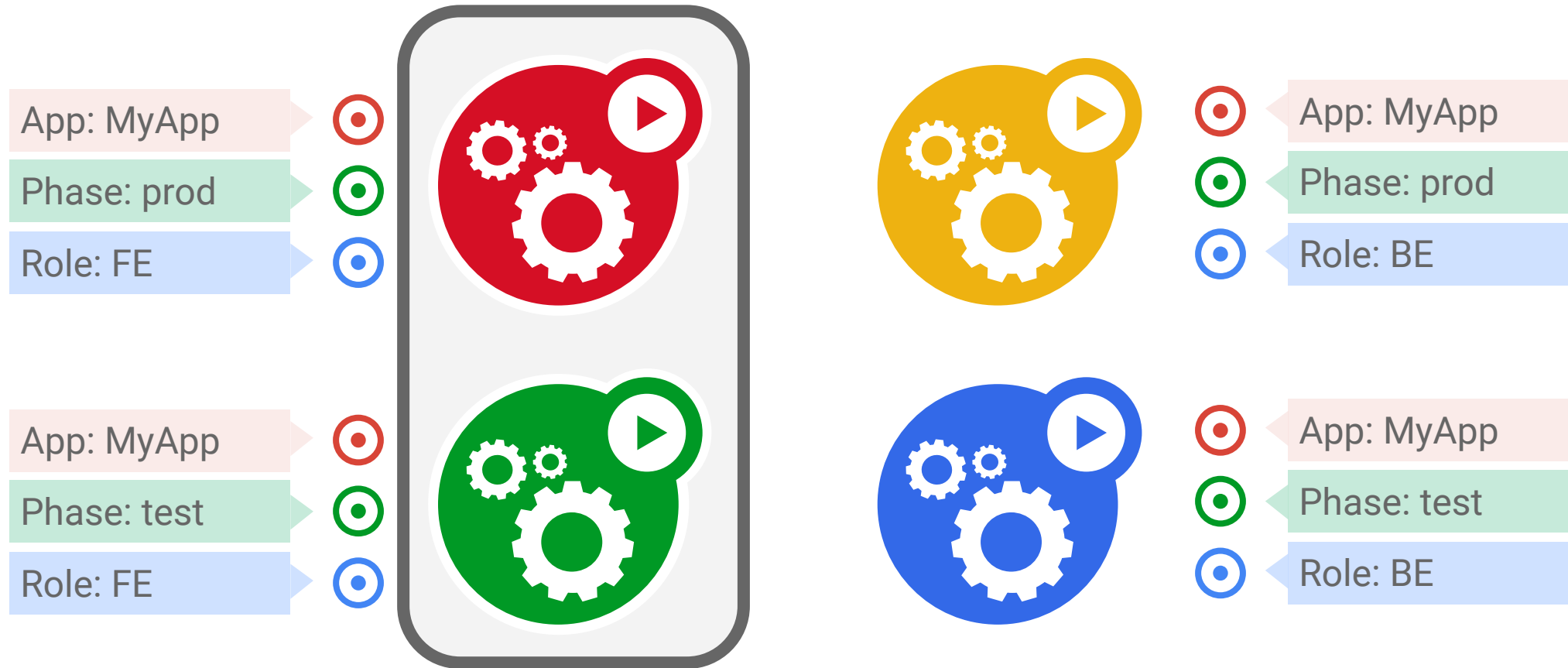


Selectors



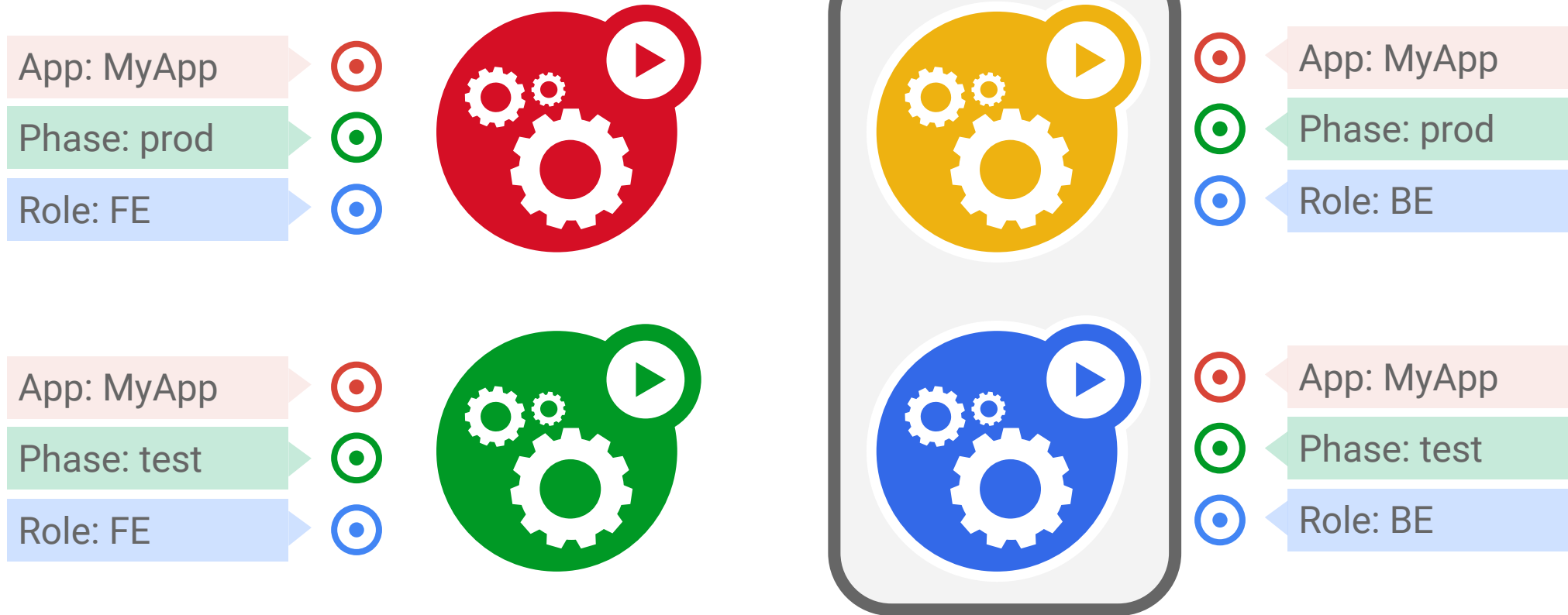
App = MyApp

Selectors



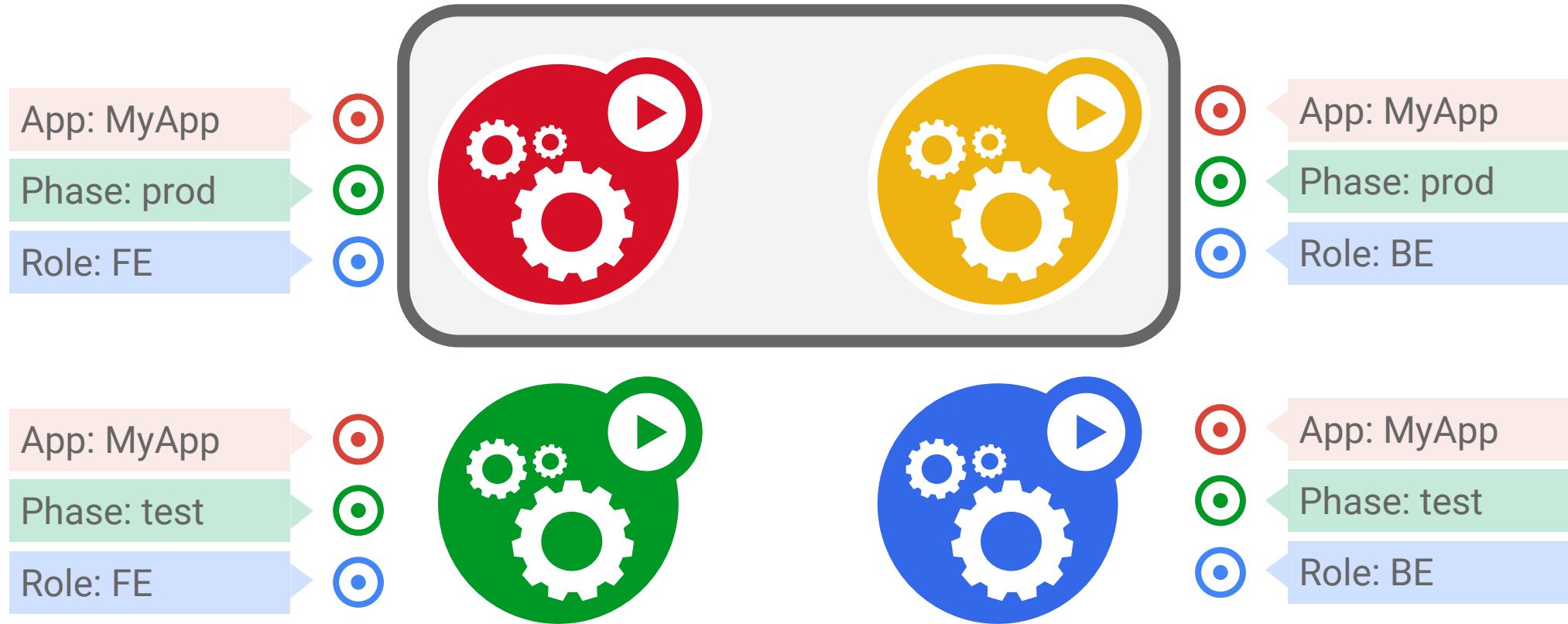
App = MyApp, Role = FE

Selectors



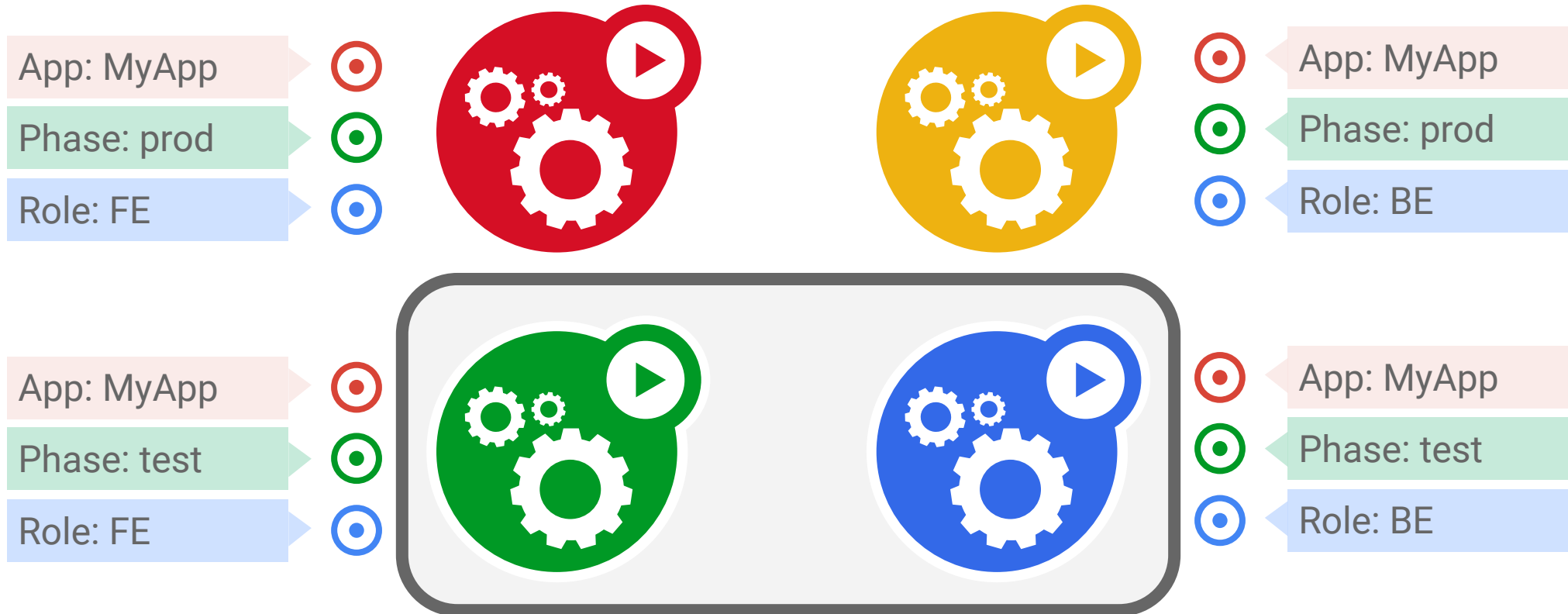
App = MyApp, Role = BE

Selectors



App = MyApp, Phase = prod

Selectors



App = MyApp, Phase = test

Running Pods

ReplicaSets

A simple controller loop

Runs out-of-process wrt API server

One job: ensure N copies of a pod

- grouped by a selector
- too few? start some
- too many? kill some

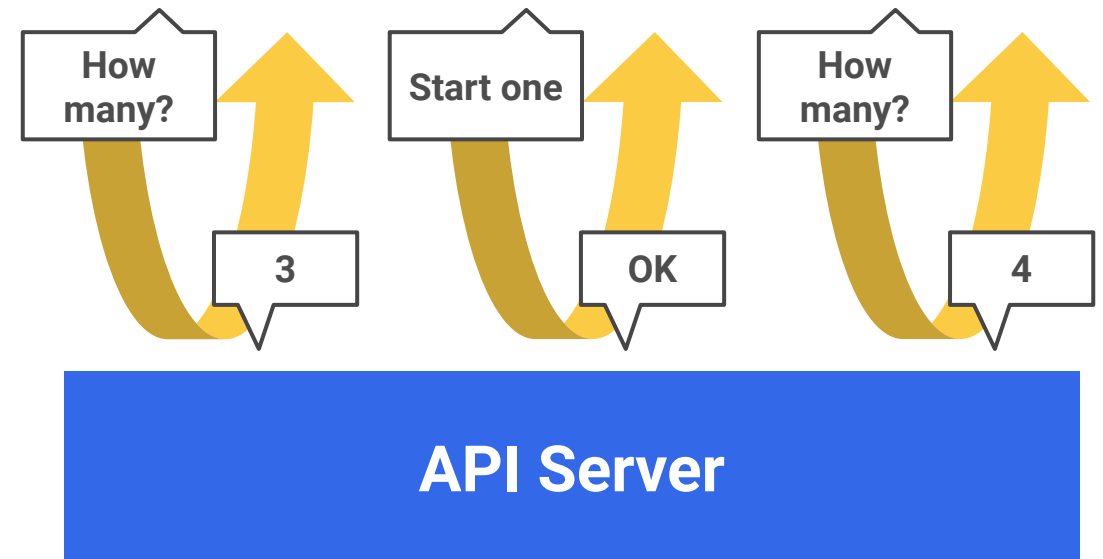
Layered on top of the public Pod API

Replicated pods are **fungible**

- Have no order or identity

ReplicaSet

- name = "my-rc"
- selector = {"App": "MyApp"}
- template = { ... }
- replicas = 4



Rolling Updates



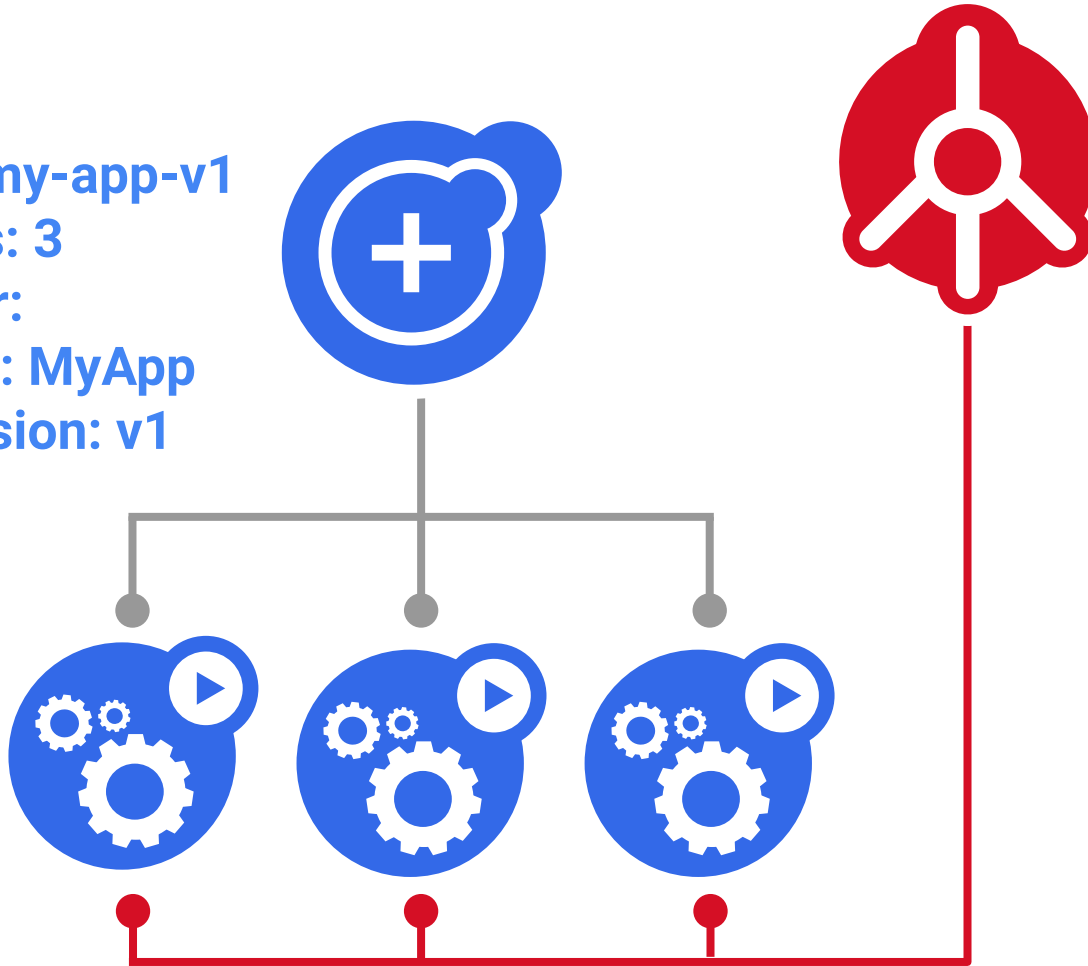
Rolling Update

Service

- app: MyApp

ReplicaSet

- name: my-app-v1
- replicas: 3
- selector:
 - app: MyApp
 - version: v1

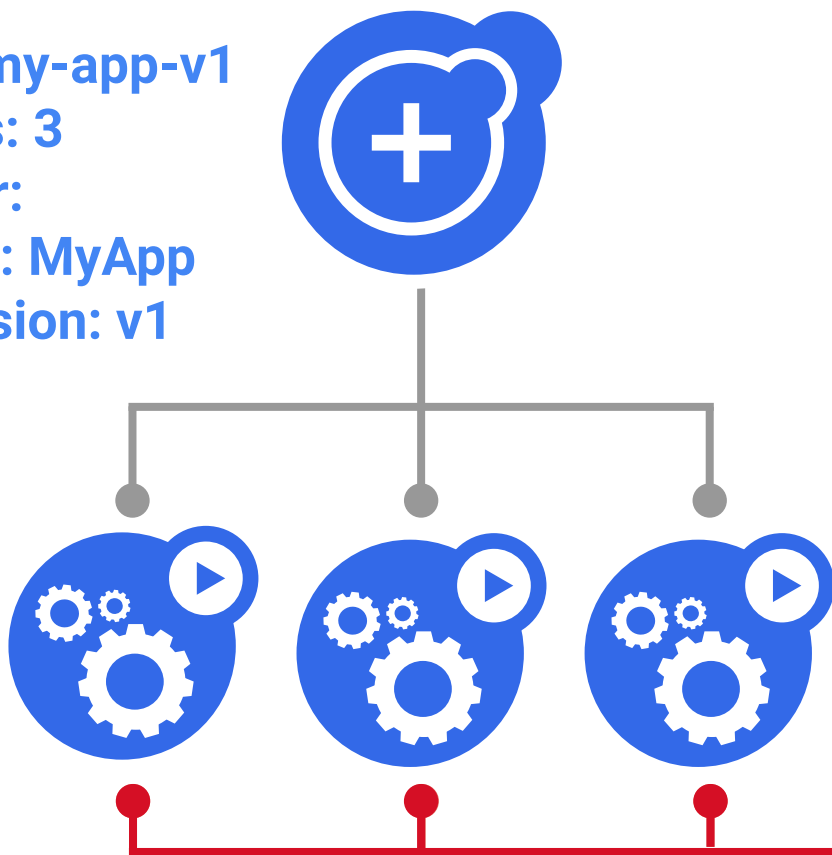


Rolling Update

Service
- app: MyApp

ReplicaSet

- name: my-app-v1
- replicas: 3
- selector:
 - app: MyApp
 - version: v1



ReplicaSet

- name: my-app-v2
- replicas: 0
- selector:
 - app: MyApp
 - version: v2

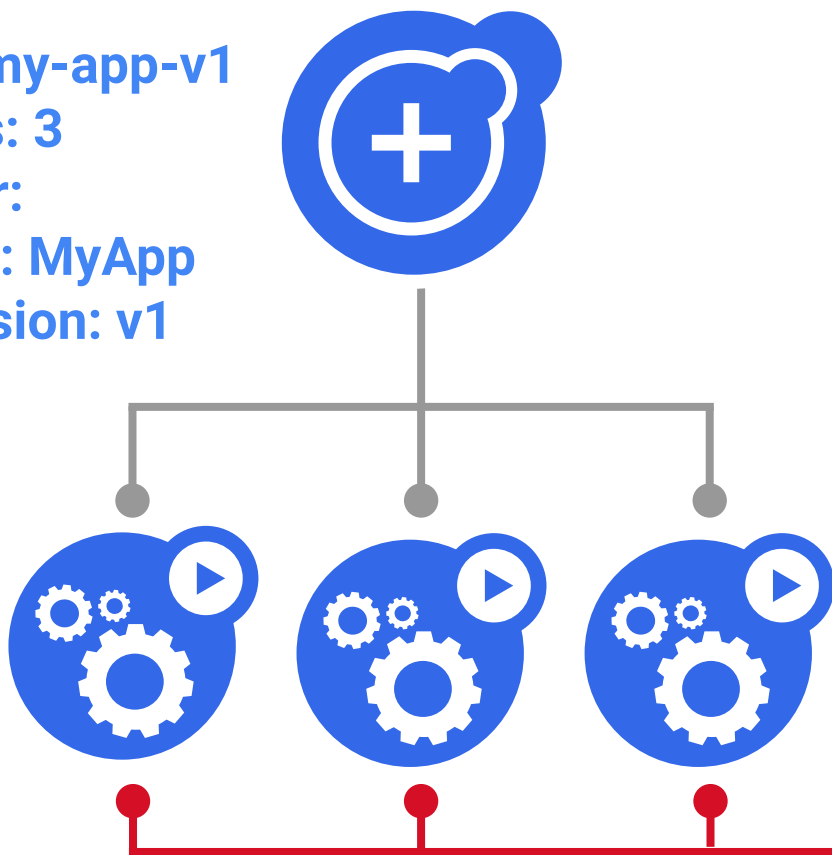


Rolling Update

Service
- app: MyApp

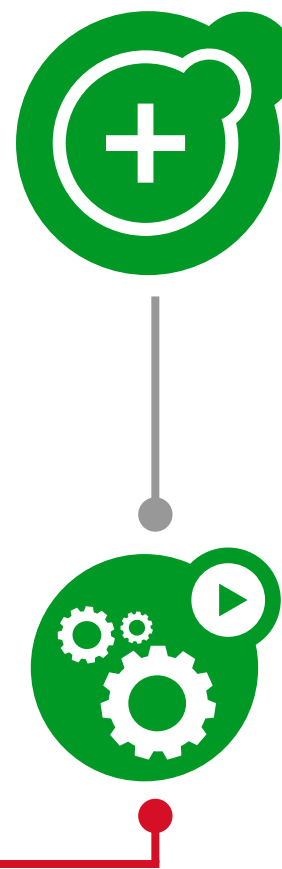
ReplicaSet

- name: my-app-v1
- replicas: 3
- selector:
 - app: MyApp
 - version: v1



ReplicaSet

- name: my-app-v2
- replicas: 1
- selector:
 - app: MyApp
 - version: v2

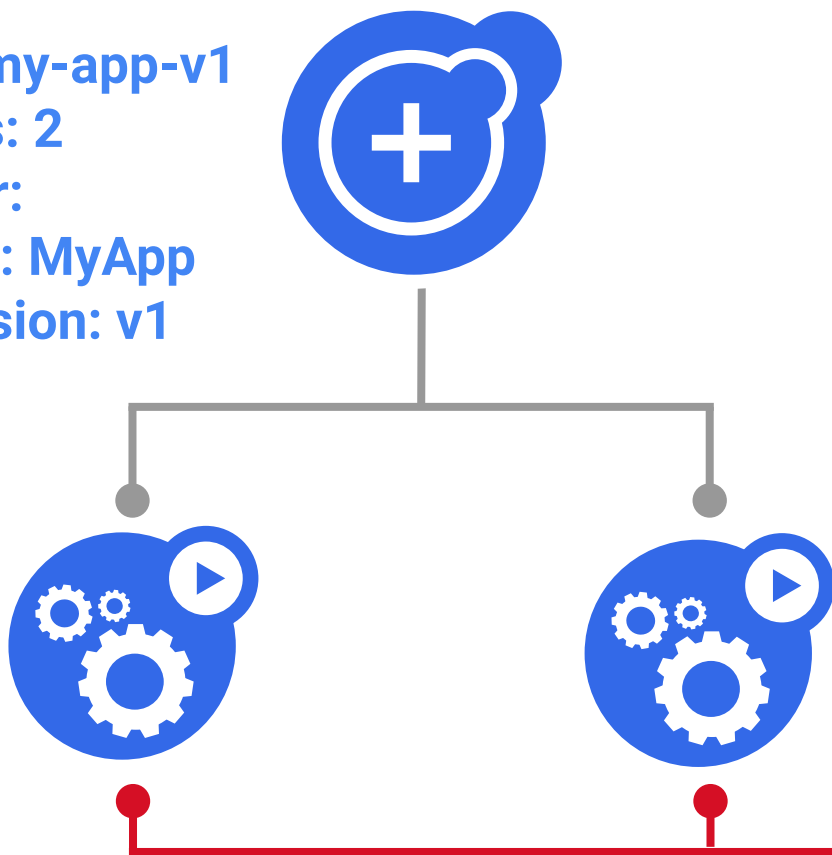


Rolling Update

Service
- app: MyApp

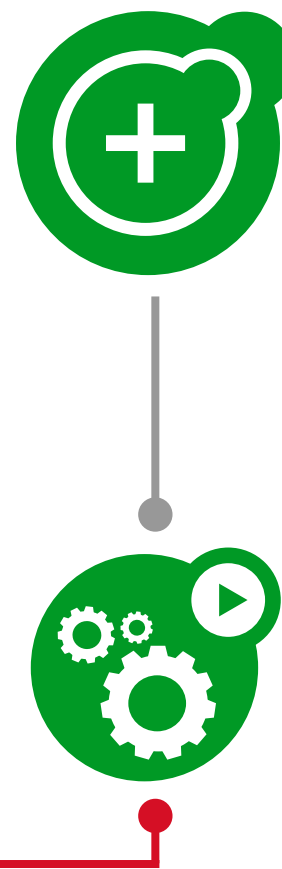
ReplicaSet

- name: my-app-v1
- replicas: 2
- selector:
 - app: MyApp
 - version: v1



ReplicaSet

- name: my-app-v2
- replicas: 1
- selector:
 - app: MyApp
 - version: v2

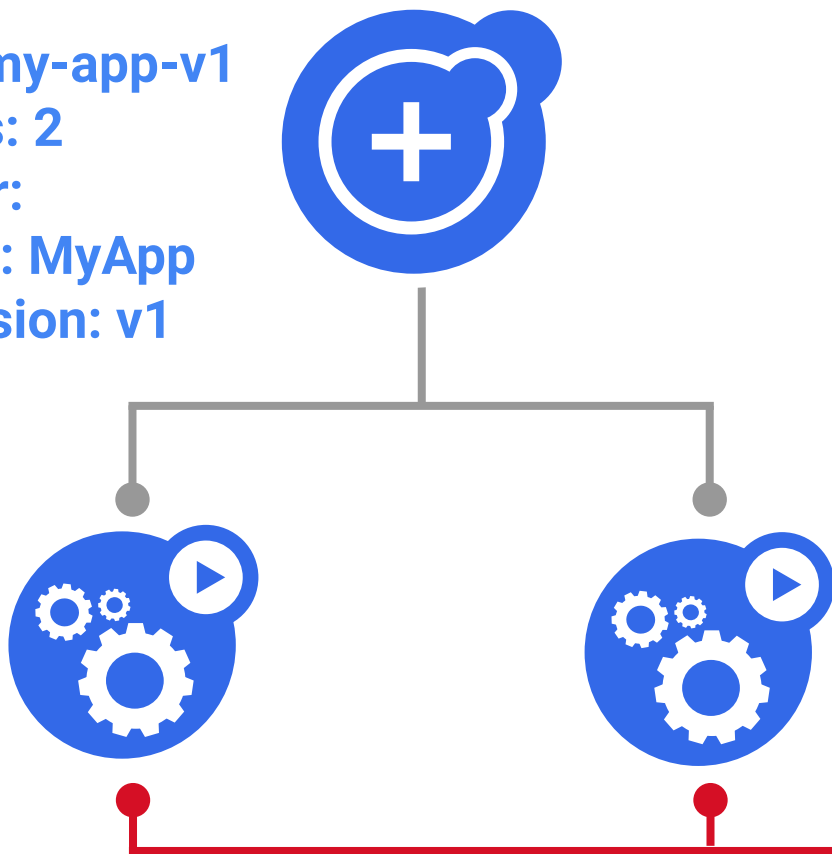


Rolling Update

Service
- app: MyApp

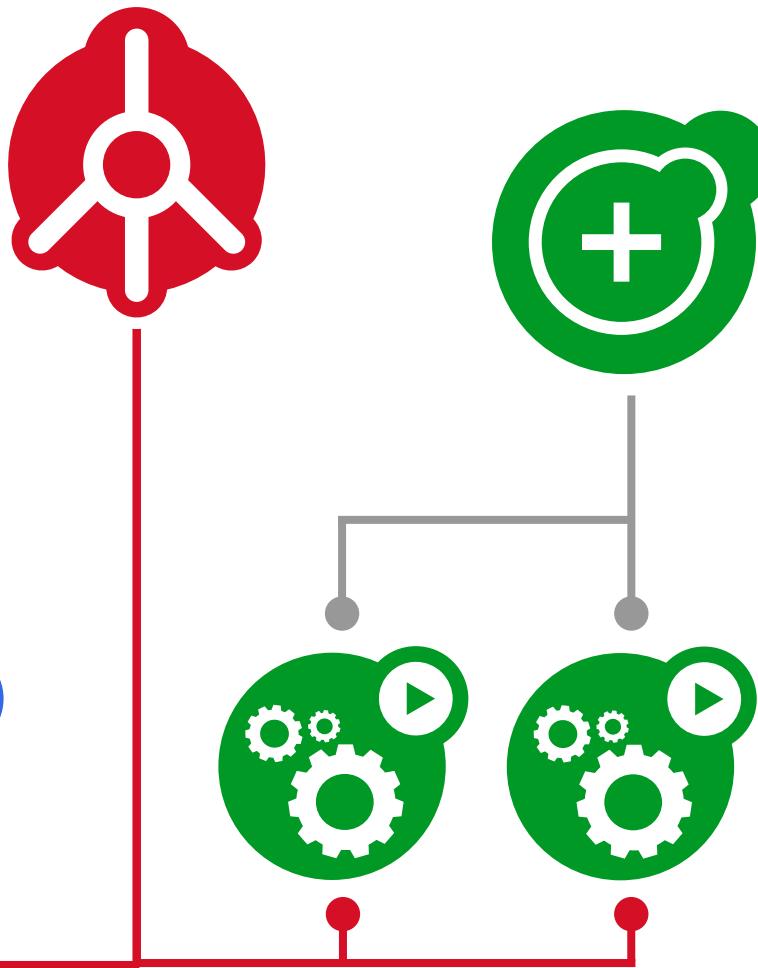
ReplicaSet

- name: my-app-v1
- replicas: 2
- selector:
 - app: MyApp
 - version: v1



ReplicaSet

- name: my-app-v2
- replicas: 2
- selector:
 - app: MyApp
 - version: v2



Rolling Update

Service
- app: MyApp

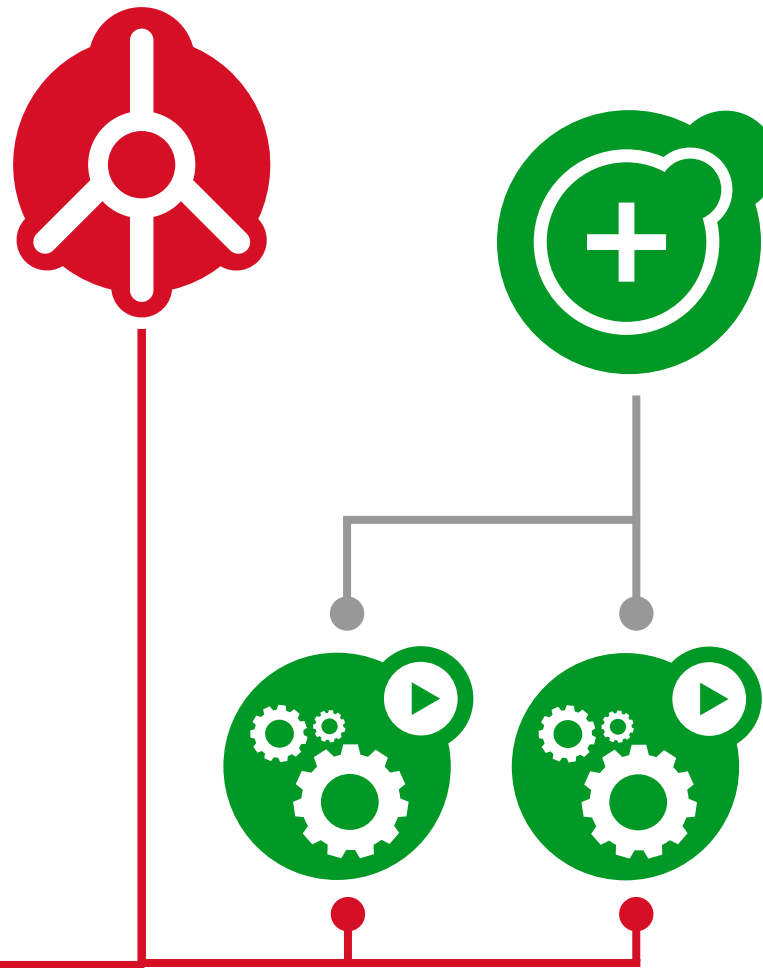
ReplicaSet

- name: my-app-v1
- replicas: 1
- selector:
 - app: MyApp
 - version: v1



ReplicaSet

- name: my-app-v2
- replicas: 2
- selector:
 - app: MyApp
 - version: v2



Rolling Update

Service
- app: MyApp

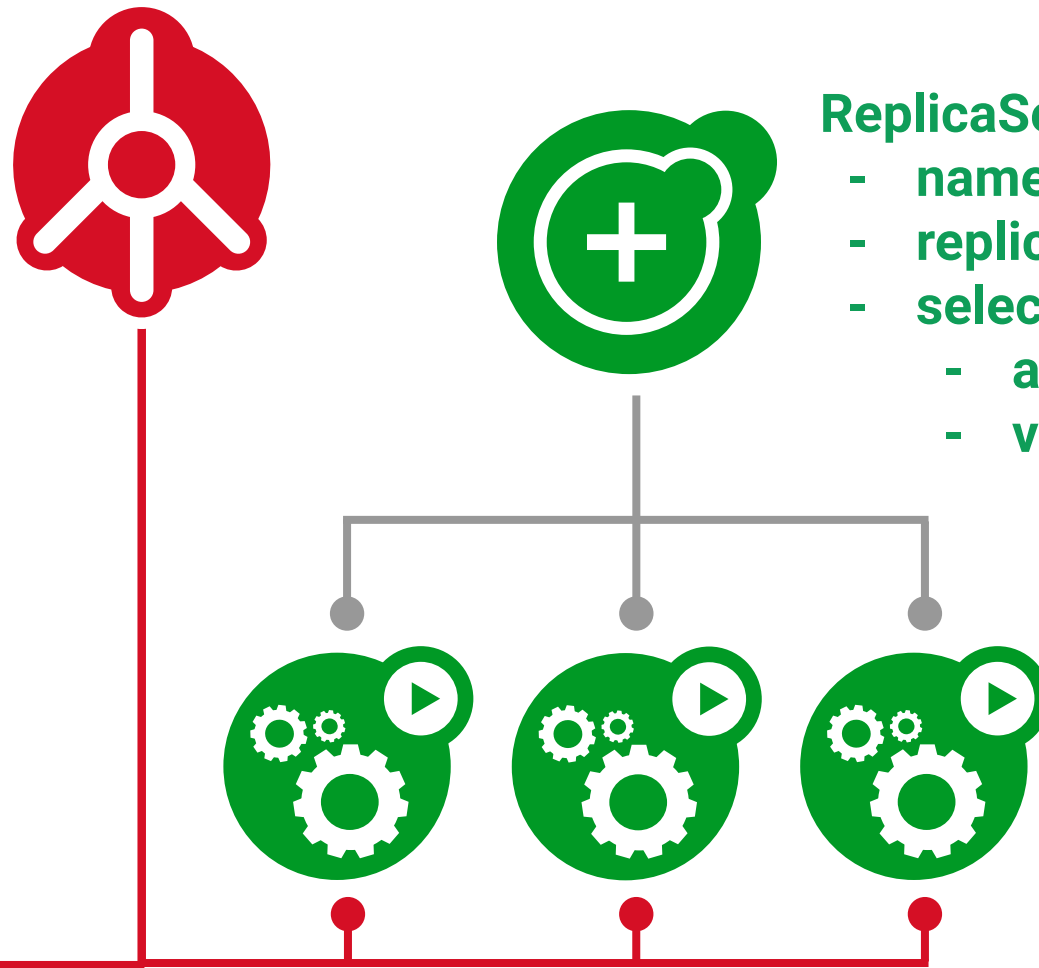
ReplicaSet

- name: my-app-v1
- replicas: 1
- selector:
 - app: MyApp
 - version: v1



ReplicaSet

- name: my-app-v2
- replicas: 3
- selector:
 - app: MyApp
 - version: v2



Rolling Update

ReplicaSet

- name: my-app-v1
- replicas: 0
- selector:
 - app: MyApp
 - version: v1



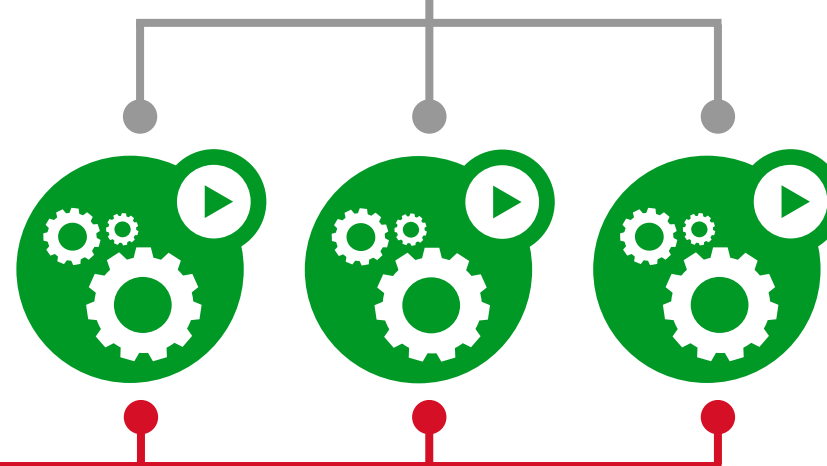
Service

- app: MyApp



ReplicaSet

- name: my-app-v2
- replicas: 3
- selector:
 - app: MyApp
 - version: v2



Rolling Update

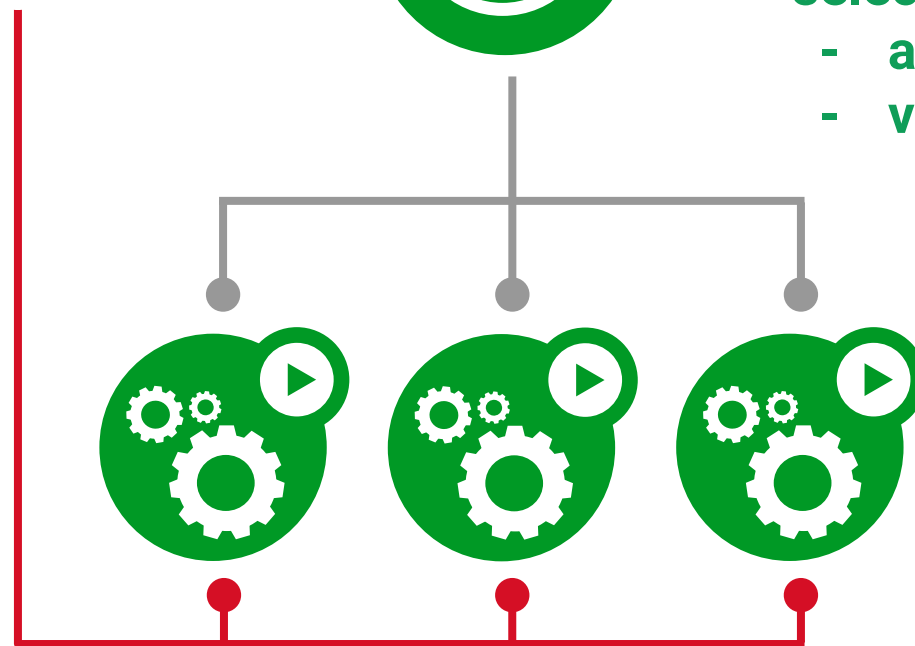
Service

- app: MyApp



ReplicaSet

- name: my-app-v2
- replicas: 3
- selector:
 - app: MyApp
 - version: v2



Deployments



Deployments

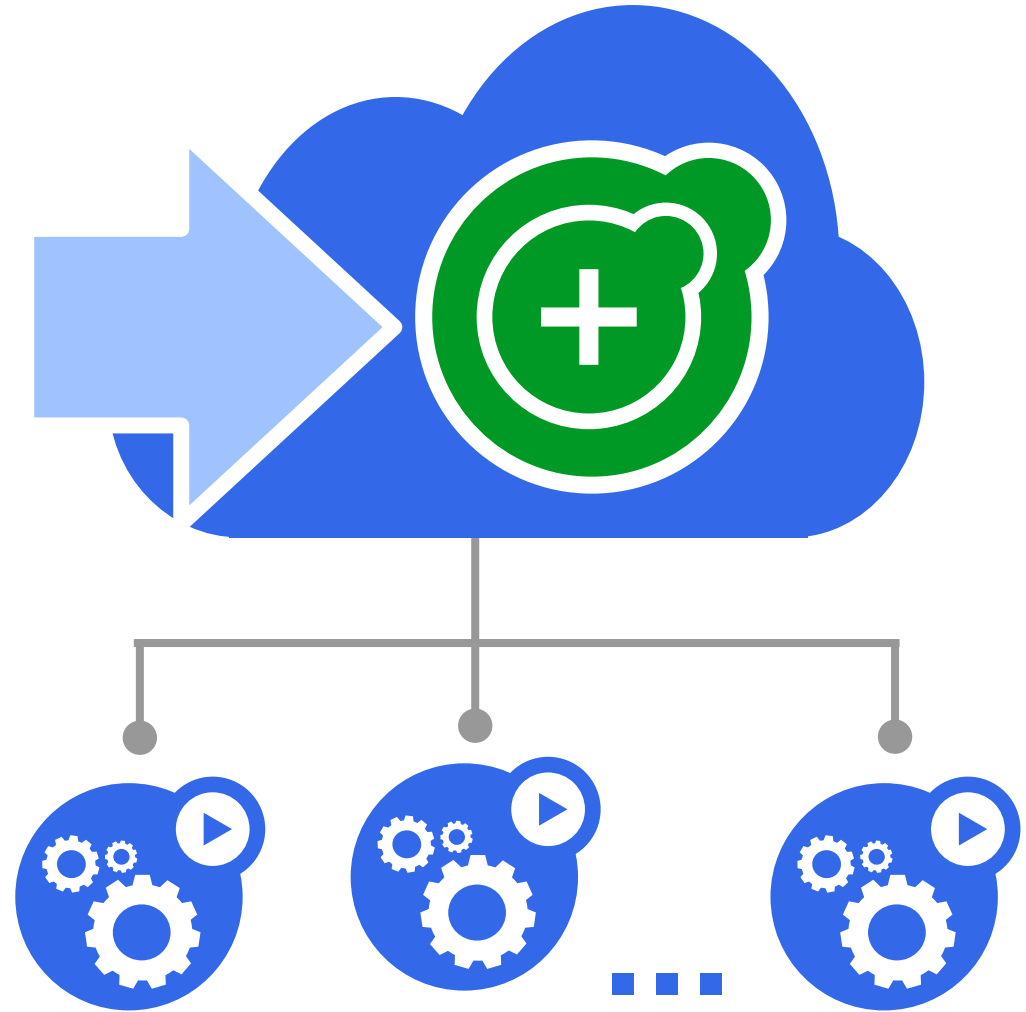
Updates-as-a-service

- Rolling update is imperative, client-side

Deployment manages rollouts for you

- stable object name in API
- updates are configurable, server-side
- `kubectl edit`, `kubectl apply`, etc.

Can have multiple updates in flight



Services

Services

A group of **pods that work together**

- grouped by a selector

Defines access policy

- “load balanced” or “headless”, external IP, etc.

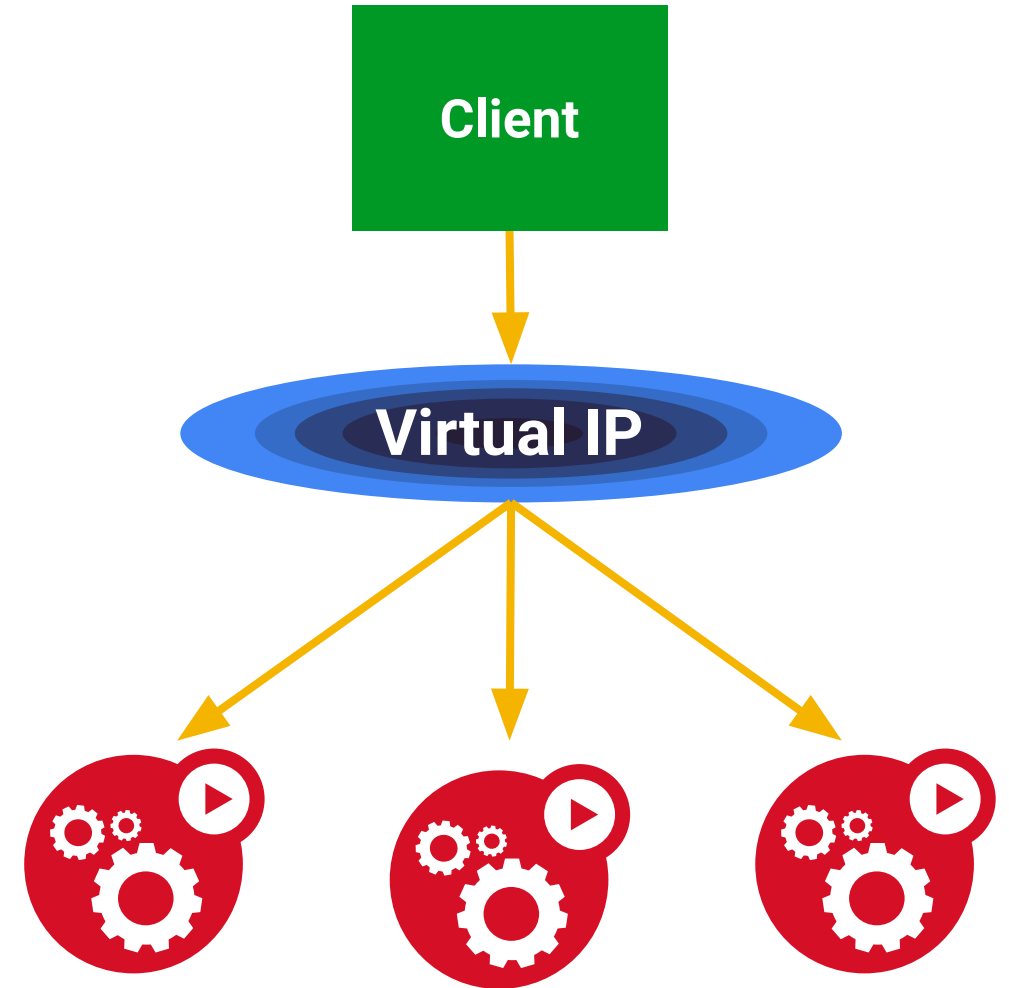
Can have a stable **virtual IP** and port(s)

- also a DNS name

VIP is managed by *kube-proxy*

- watches all services
- updates iptables when backends change
- default implementation - can be replaced!

Hides complexity



External services

Services VIPs are only available **inside** the cluster

Need to receive traffic from “the outside world”

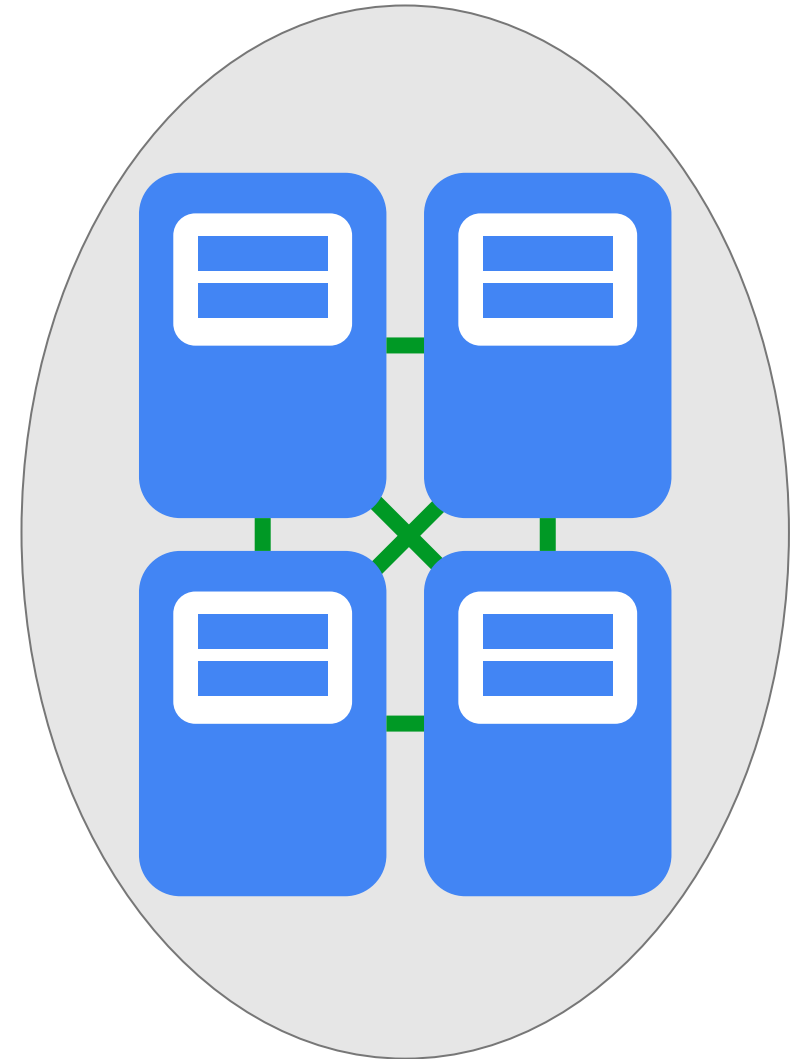
Service “type”

- NodePort: expose on a port on every node
- LoadBalancer: provision a cloud load-balancer

DiY load-balancer solutions

- socat (for NodePort remapping)
- haproxy
- nginx

Ingress (L7 LB)



Ingress (L7 LB)

Many apps are HTTP/HTTPS

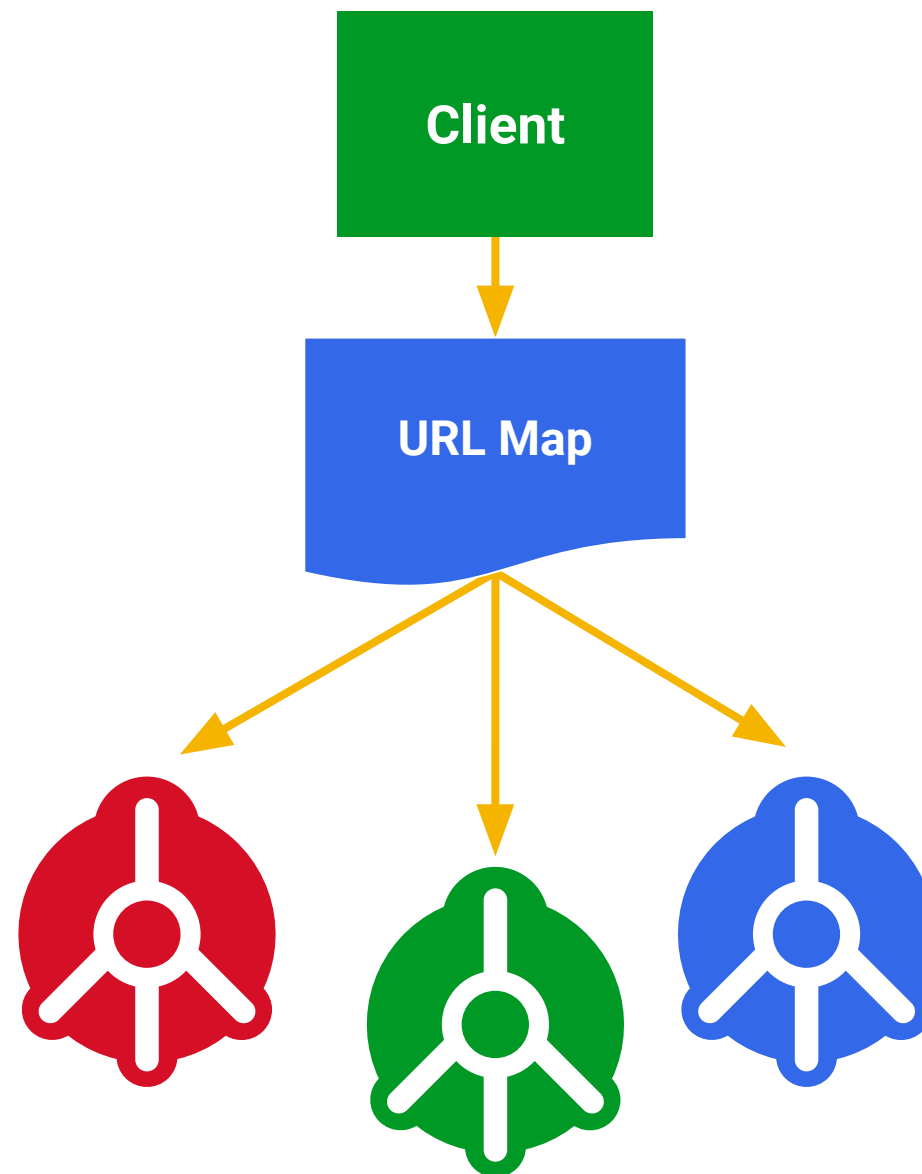
Services are L4 (IP + port)

Ingress maps incoming traffic to backend services

- by HTTP host headers
- by HTTP URL paths

HAProxy, NGINX, AWS and GCE implementations in progress

Now with SSL!



Running Daemons

DaemonSets

Problem: how to run a Pod on every node?

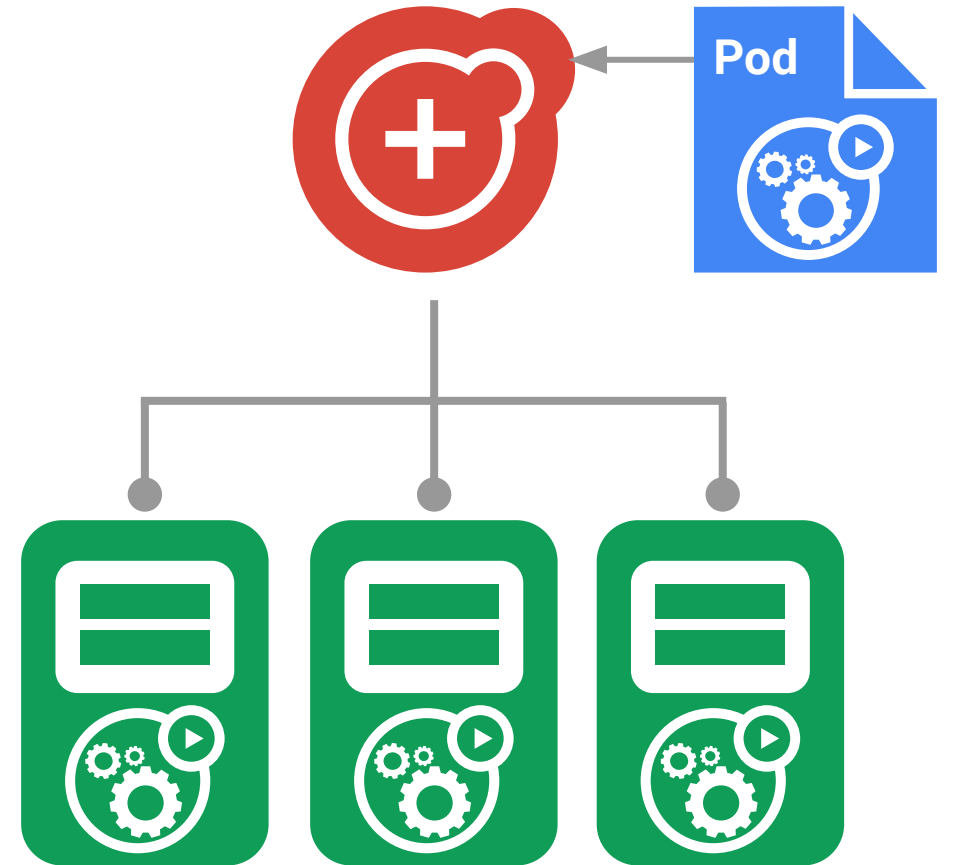
- or a subset of nodes

Similar to ReplicaSet

- principle: do one thing, don't overload

“Which nodes?” is a selector

Uses consistent tools and patterns



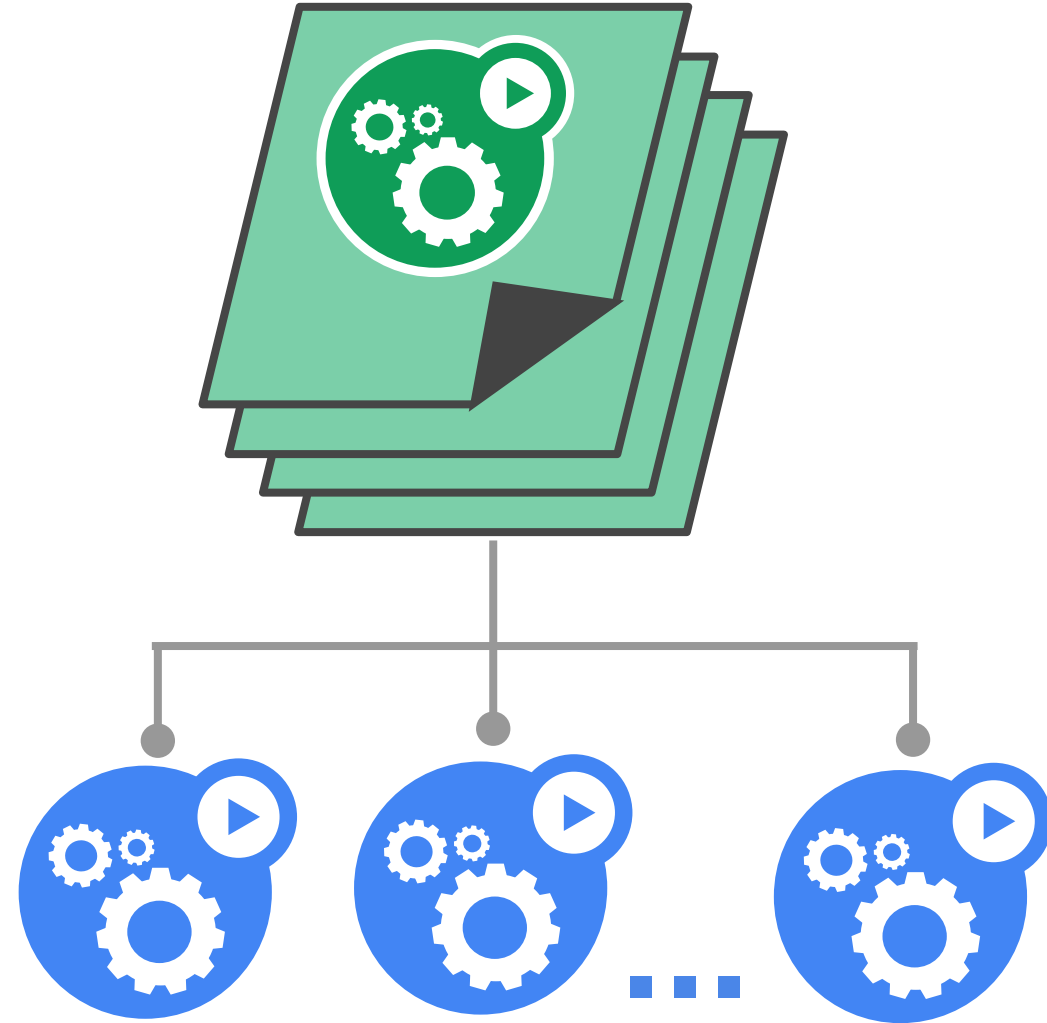
Running Jobs

Run-to-completion, as opposed to run-forever

- Express parallelism & required completions
- Workflow: restart on failure
- Build/test: don't restart on failure

Aggregates success/failure counts

Built for batch and big-data work



Running With State

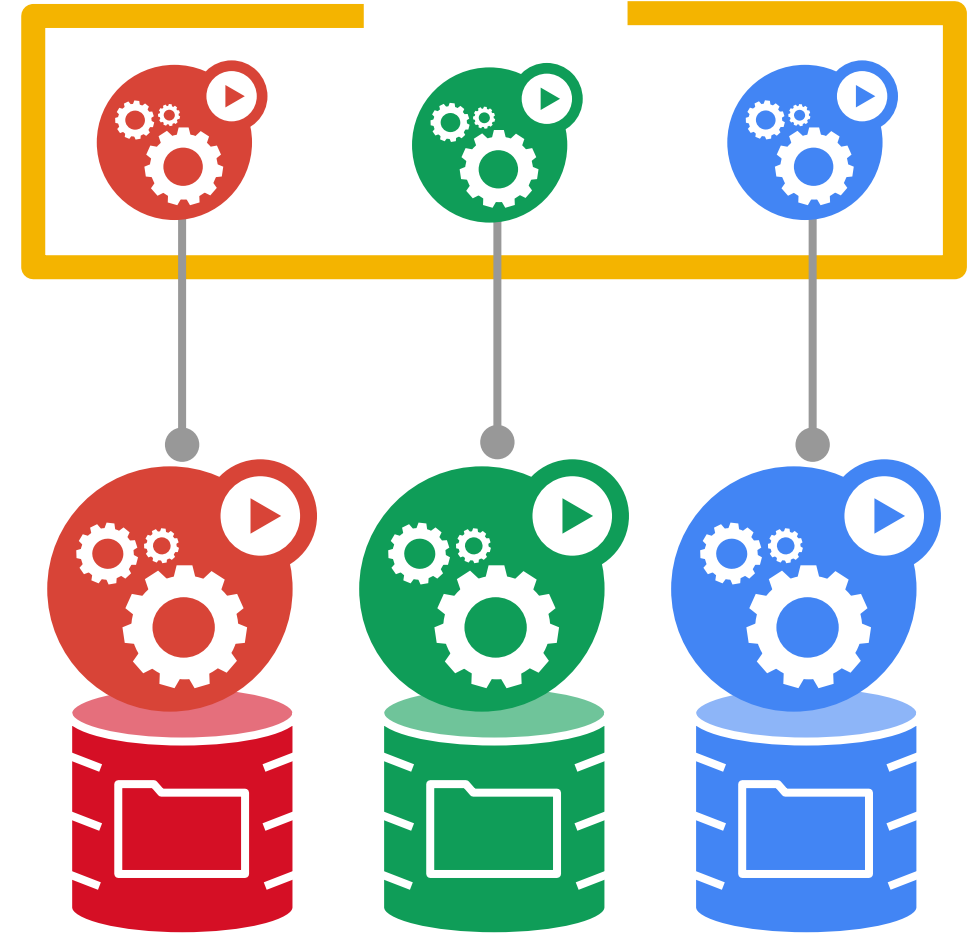
StatefulSets

Goal: enable clustered software on Kubernetes

- mysql, redis, zookeeper, ...

Clustered apps need “identity” and have sequencing constraints

- stable hostname, available in DNS
- an ordinal index
- stable storage: linked to the ordinal & hostname
- discovery of peers for quorum
- startup/teardown ordering



ConfigMaps



ConfigMaps

Goal: manage app configuration

- ...without making overly-brittle container images

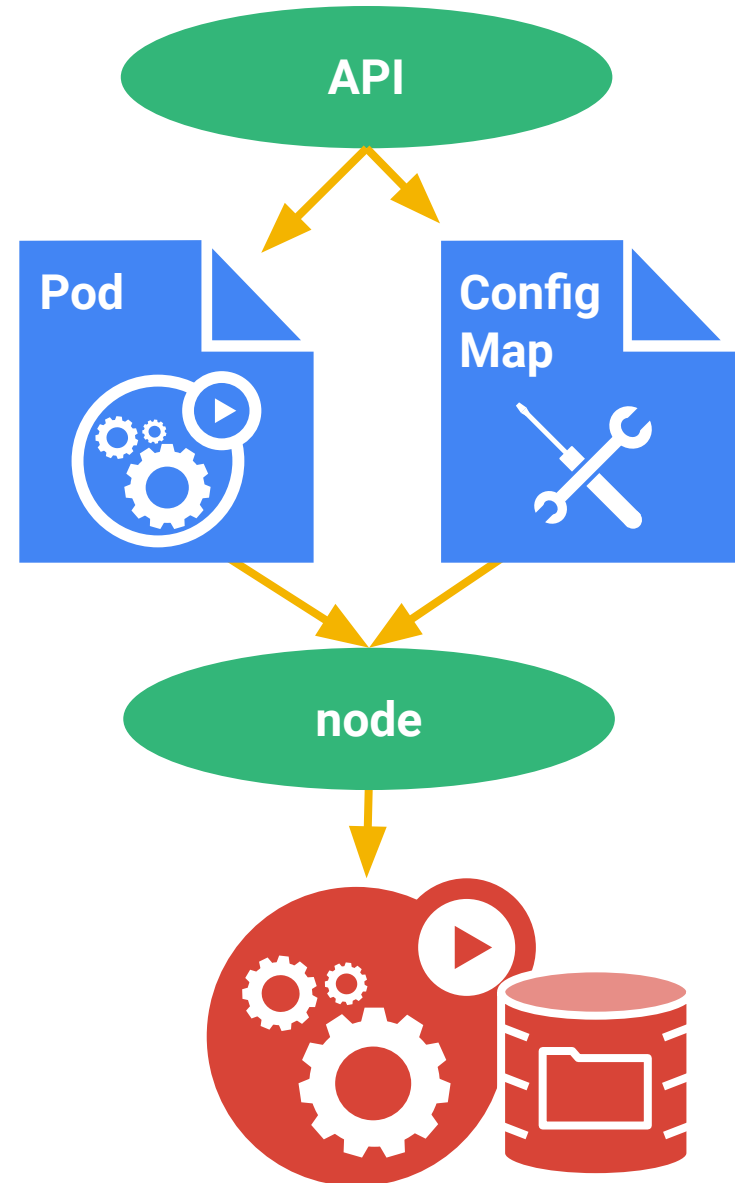
[12-factor](#) says config comes from the environment

- Kubernetes is the environment

Manage config via the Kubernetes API

Inject config as a virtual volume into your Pods

- late-binding, live-updated (atomic)
- also available as env vars



Secrets



Secrets

Goal: grant a pod access to a secured *something*

- don't put secrets in the container image!

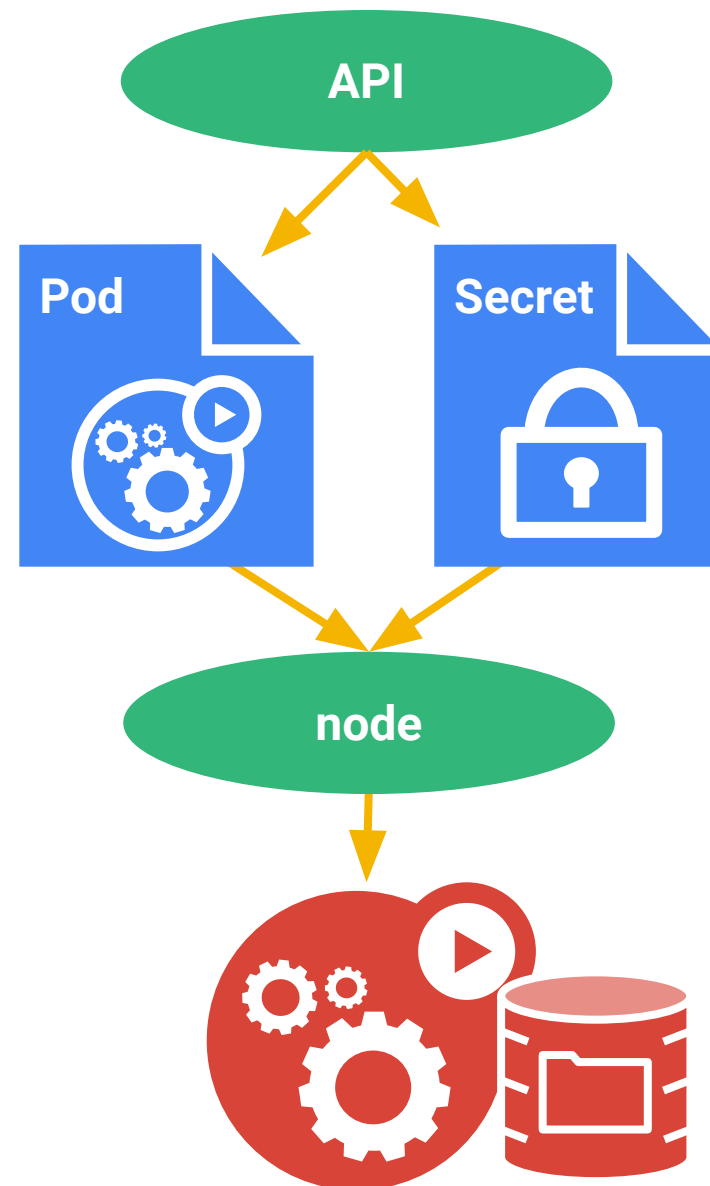
12-factor says config comes from the environment

- Kubernetes is the environment

Manage secrets via the Kubernetes API

Inject secrets as virtual volumes into your Pods

- late-binding, tmpfs - never touches disk
- also available as env vars



Graceful Termination

Graceful Termination

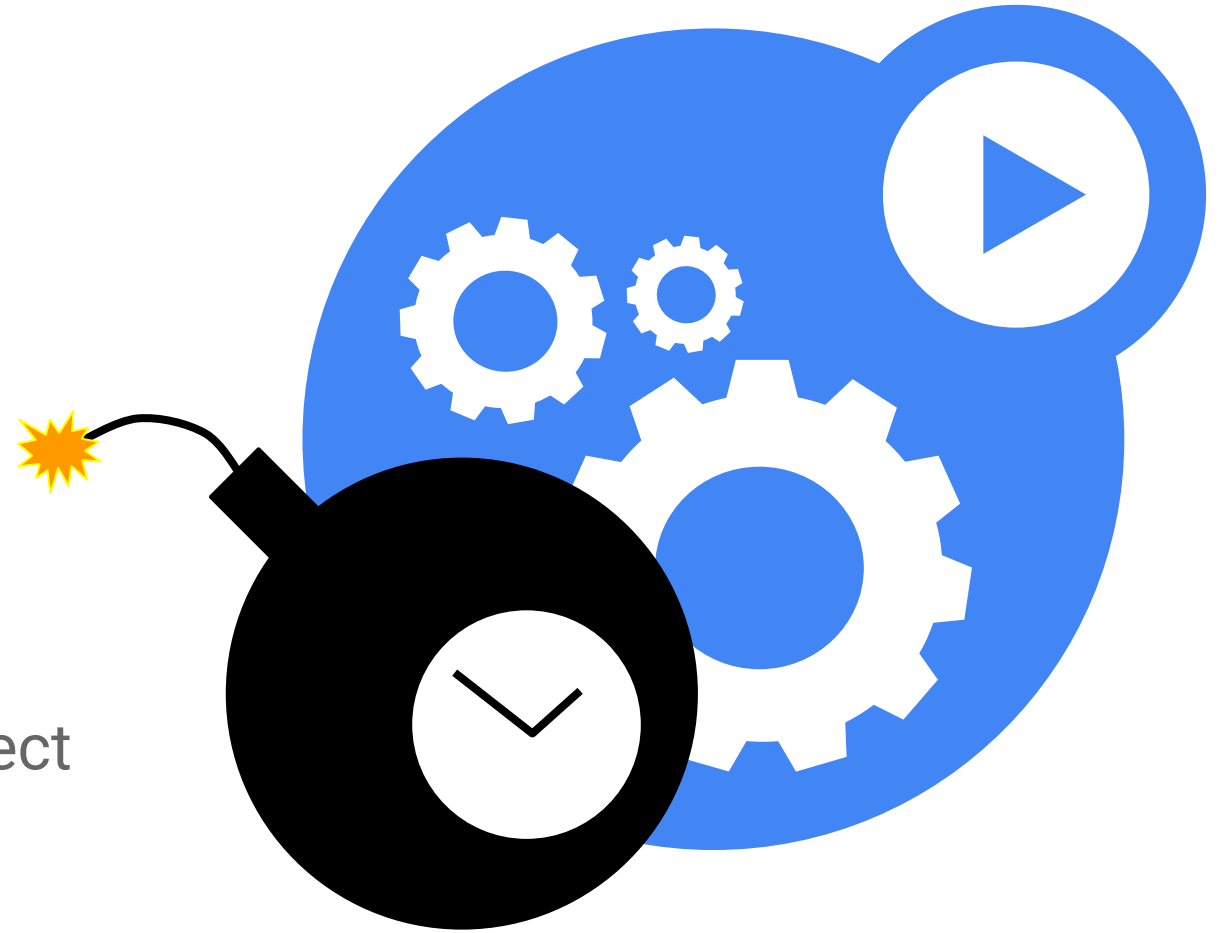
Goal: Give pods time to clean up

- finish in-flight operations
- log state
- flush to disk
- 30 seconds by default

Catch **SIGTERM**, cleanup, exit ASAP

Pod status “Terminating”

Declarative: ‘DELETE’ appears as an object field in the API



Auto-Scaling



HorizontalPodAutoScalers

Goal: Automatically scale pods as needed

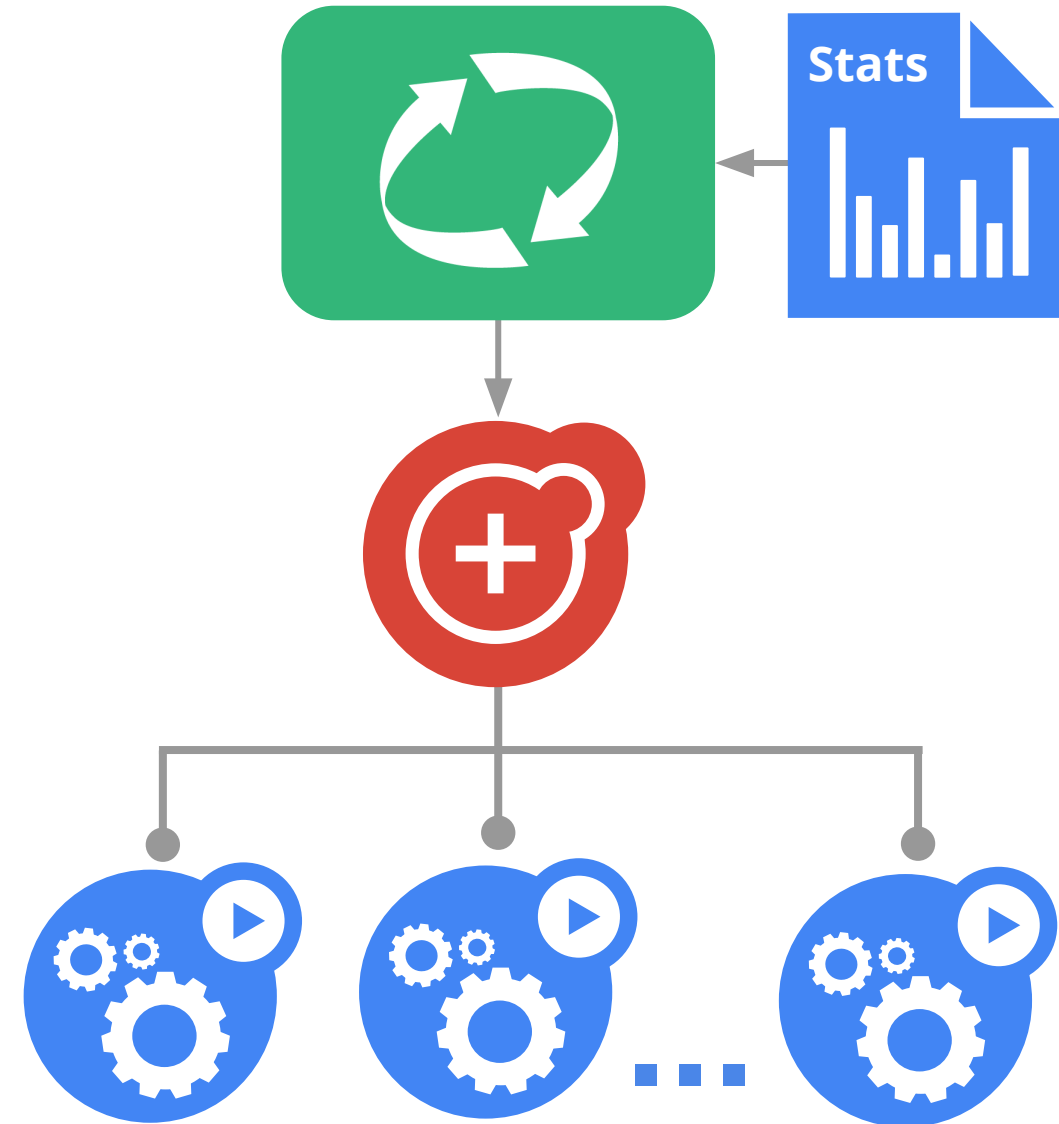
- based on CPU utilization (for now)
- custom metrics in Alpha

Efficiency now, capacity when you need it

Operates within user-defined min/max bounds

Set it and forget it

Status: Stable in kubernetes 1.9, see Heapster



PersistentVolumes

PersistentVolumes

A higher-level storage abstraction

- insulation from any one cloud environment

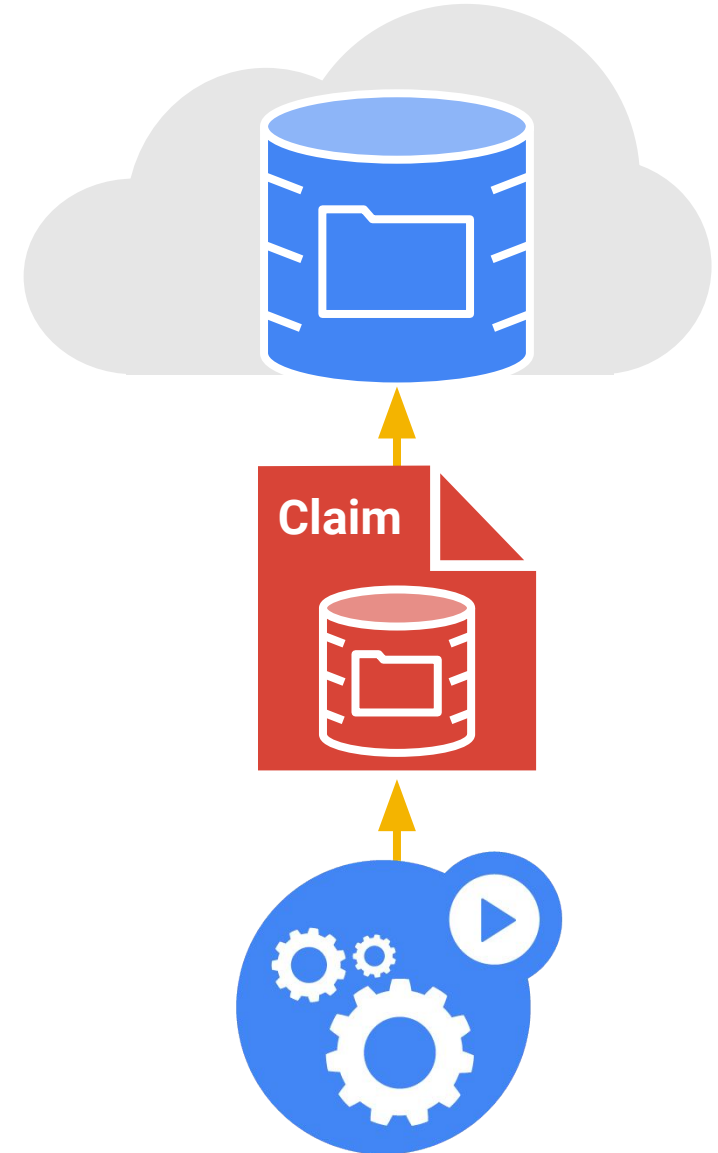
Admin provisions them, users claim them

- **NEW: auto-provisioning**

Independent lifetime from consumers

- lives until user is done with it
- can be handed-off between pods

Dynamically “scheduled” and managed, like nodes and pods



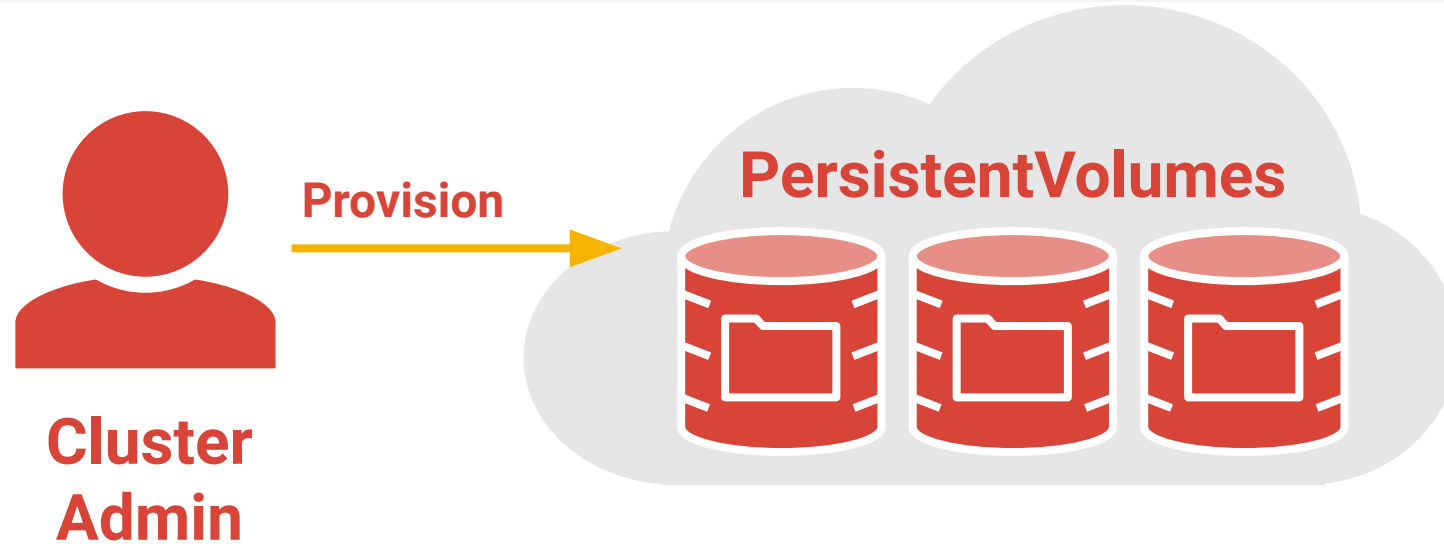
PersistentVolumes



**Cluster
Admin**



PersistentVolumes



PersistentVolumes



**Cluster
Admin**



User

PersistentVolumes



**Cluster
Admin**



User

Create



PVClaim

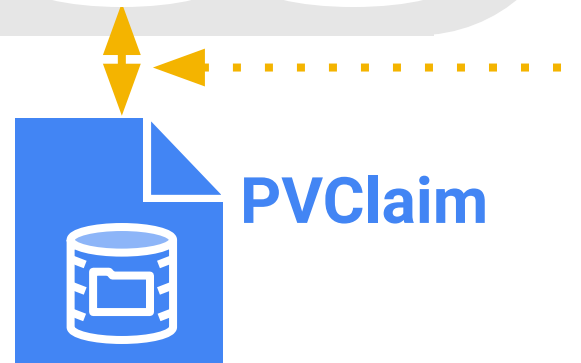
PersistentVolumes



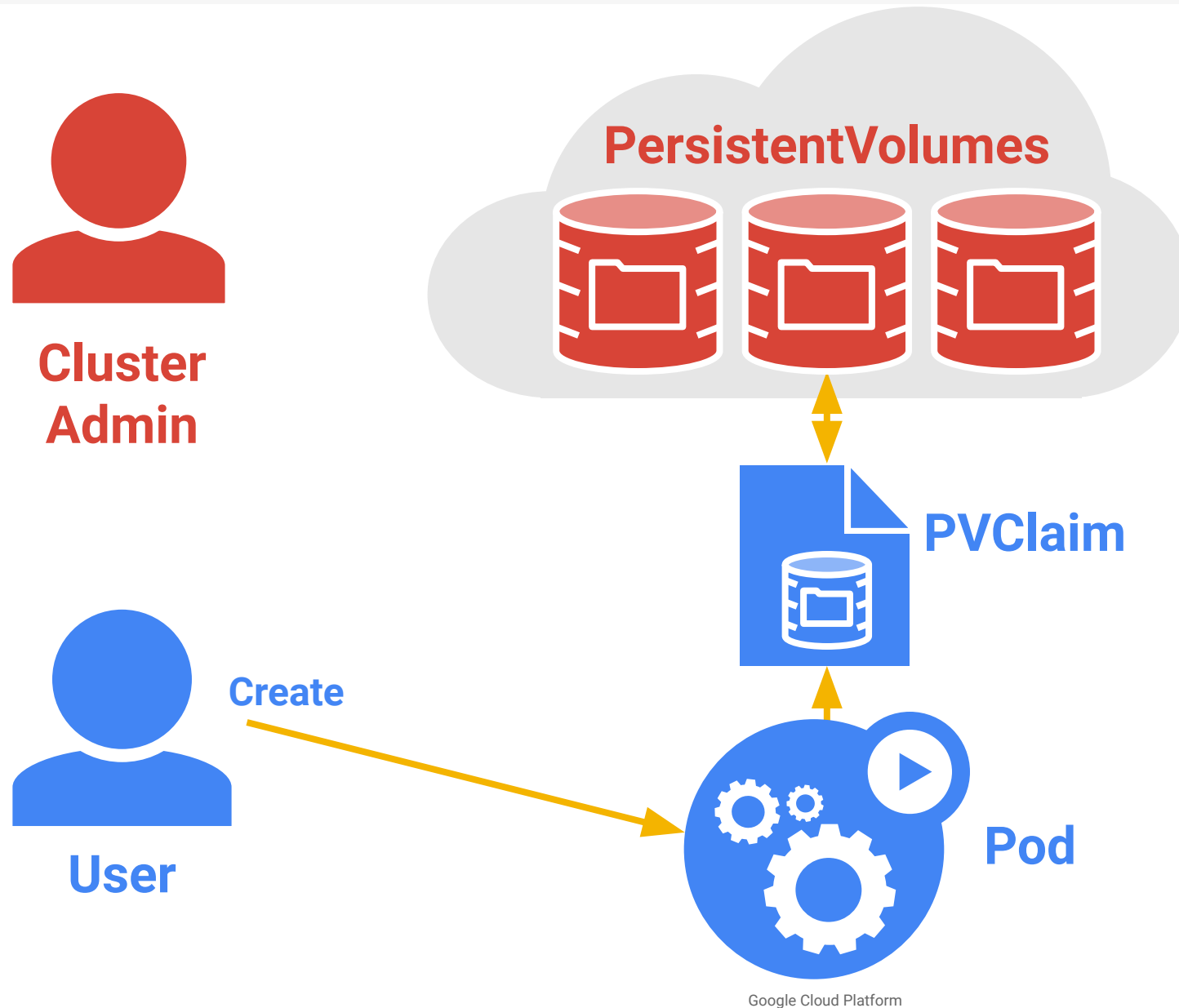
**Cluster
Admin**



User



PersistentVolumes



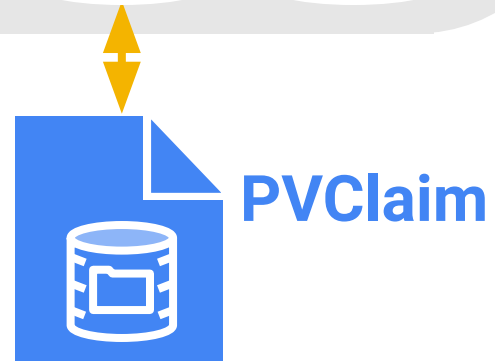
PersistentVolumes



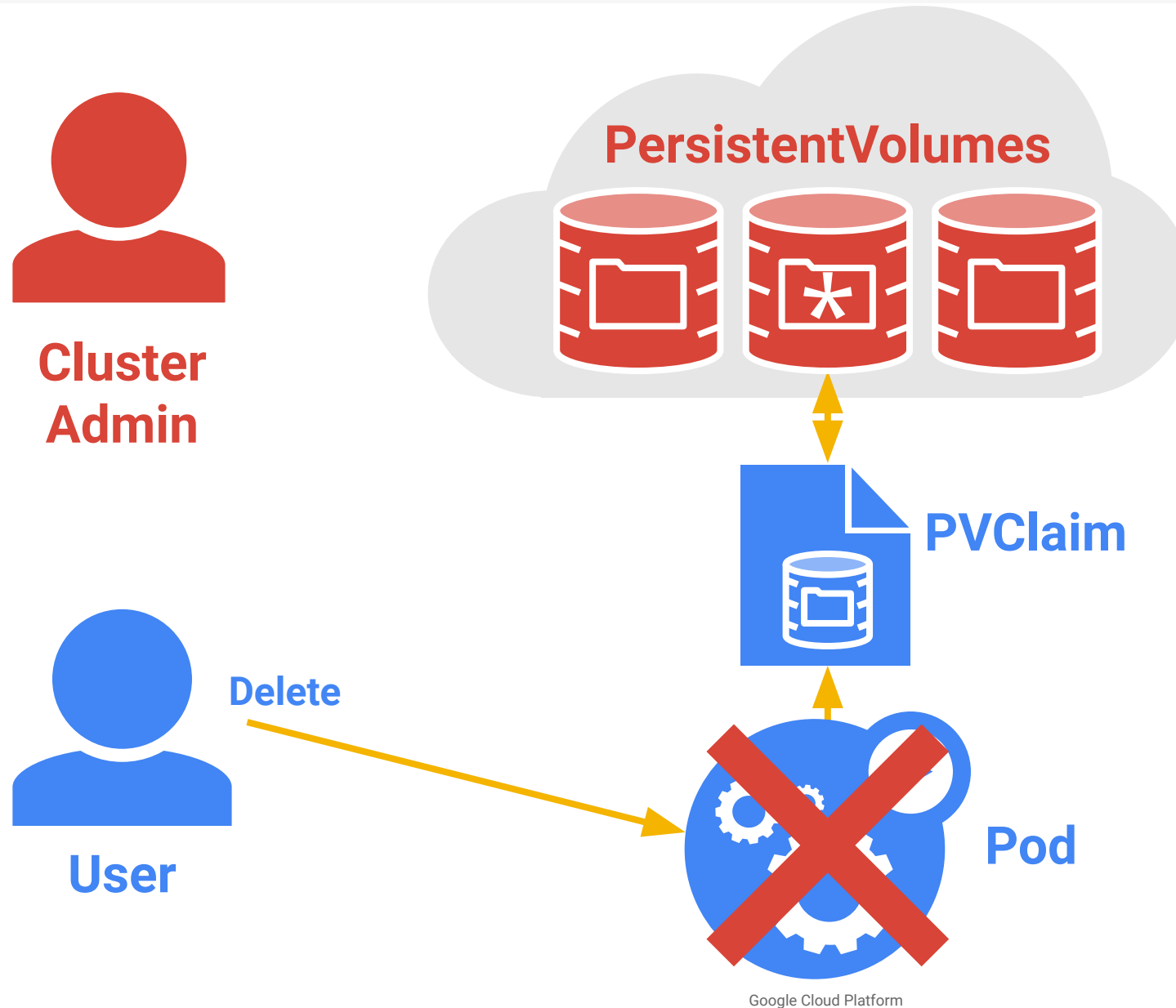
**Cluster
Admin**



User



PersistentVolumes



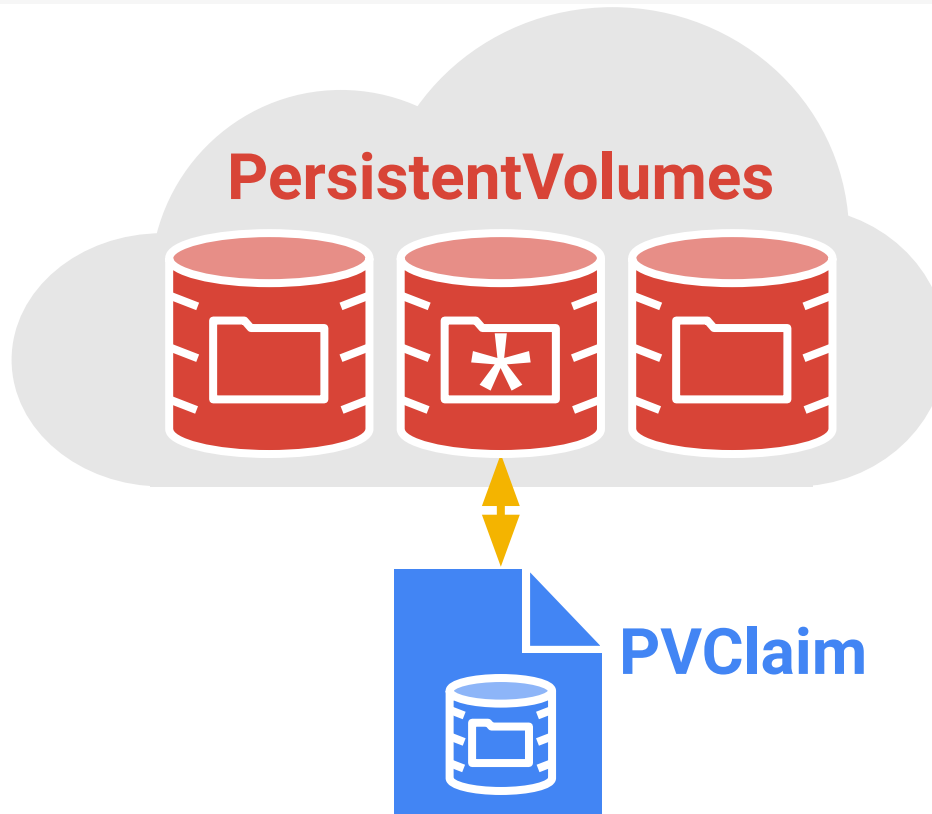
PersistentVolumes



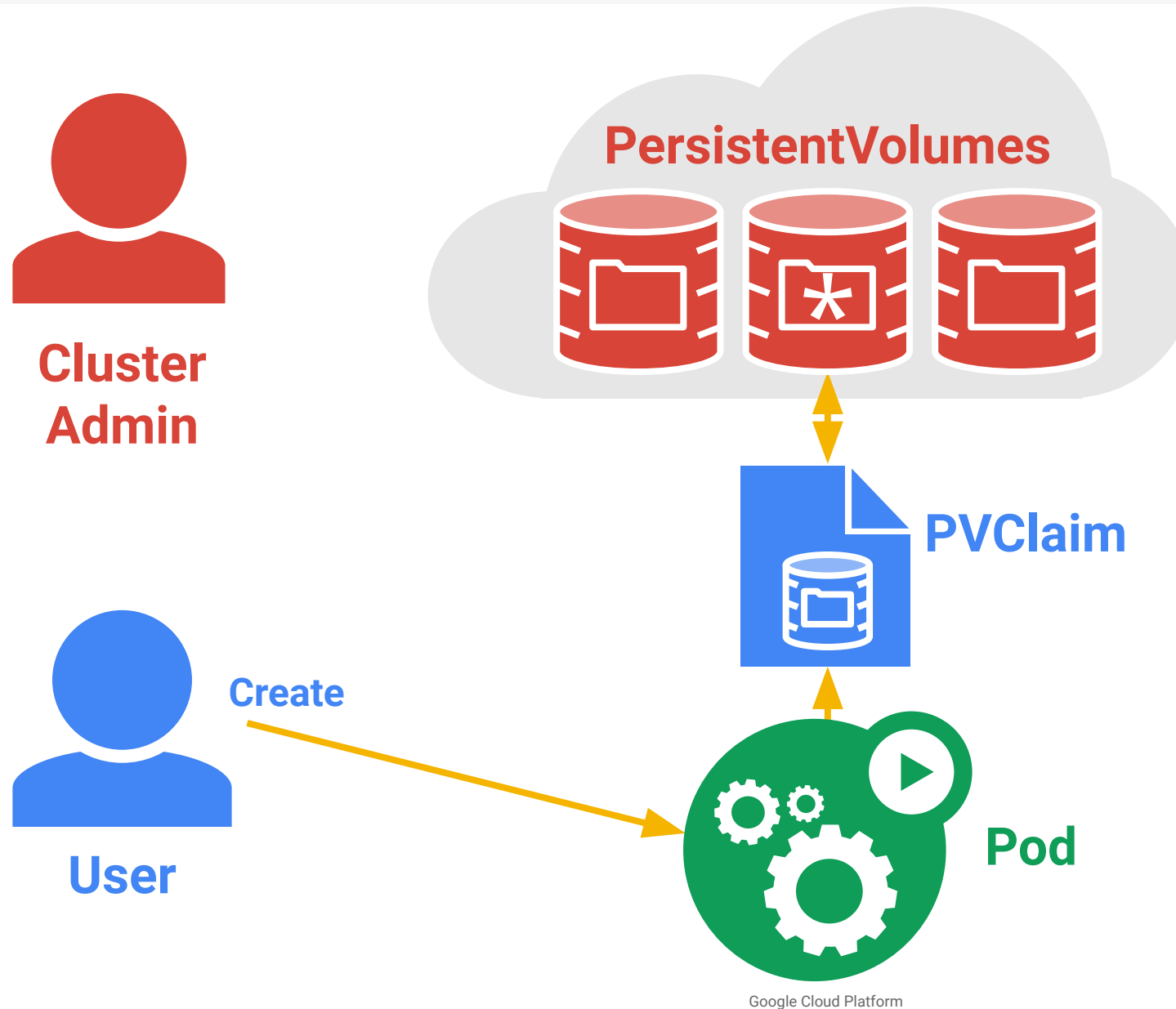
Cluster Admin



User



PersistentVolumes



PersistentVolumes



**Cluster
Admin**



User

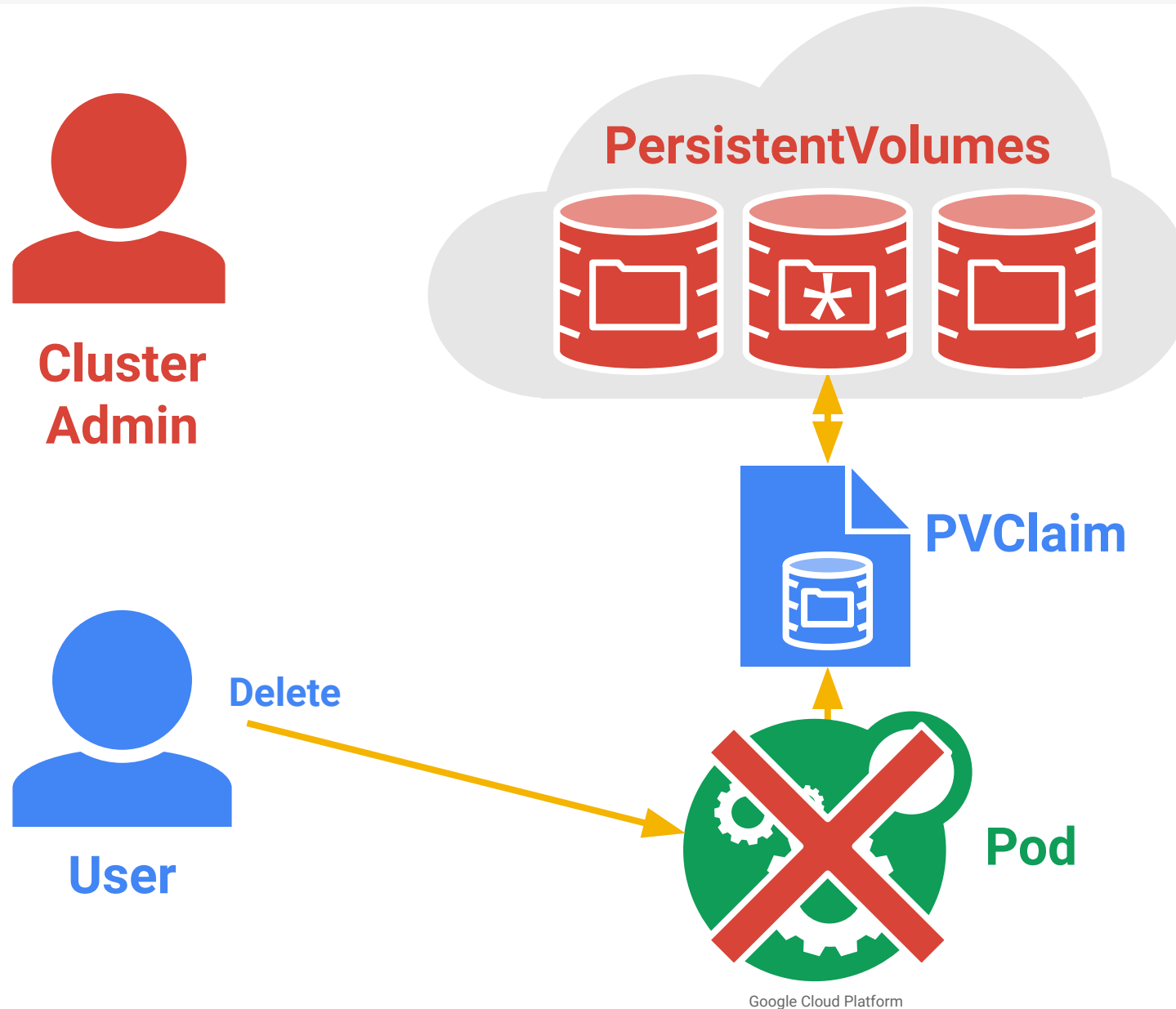


PVClaim

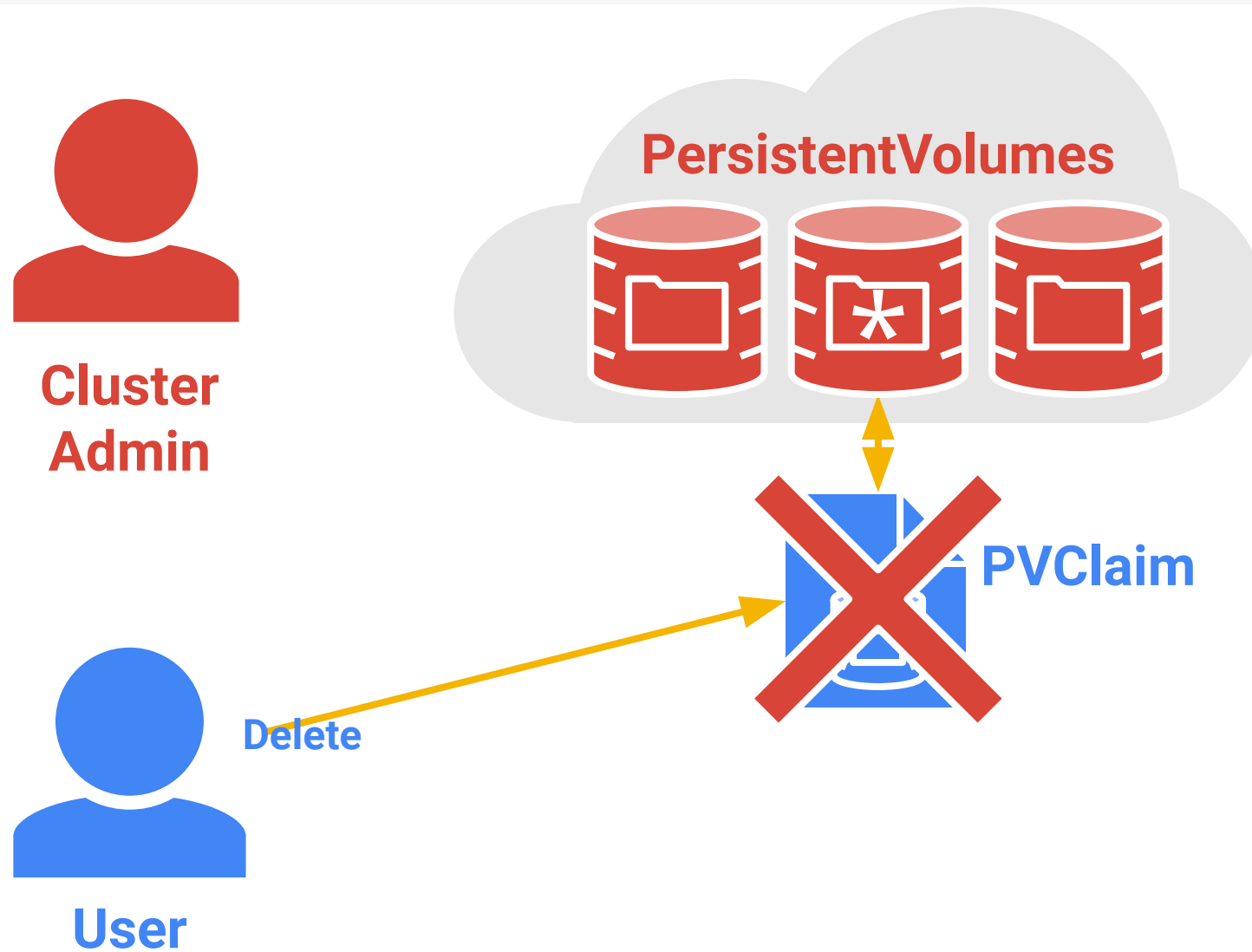


Pod

PersistentVolumes



PersistentVolumes



PersistentVolumes



**Cluster
Admin**



User



Recycler

Namespaces



Namespaces

Problem: I have too much stuff!

- name collisions in the API
- poor isolation between users
- don't want to expose things like Secrets

Solution: Slice up the cluster

- create new Namespaces as needed
 - per-user, per-app, per-department, etc.
- part of the API - NOT private machines
- most API objects are namespaced
 - part of the REST URL path
- Namespaces are just another API object
- One-step cleanup - delete the Namespace
- Obvious hook for policy enforcement (e.g. quota)

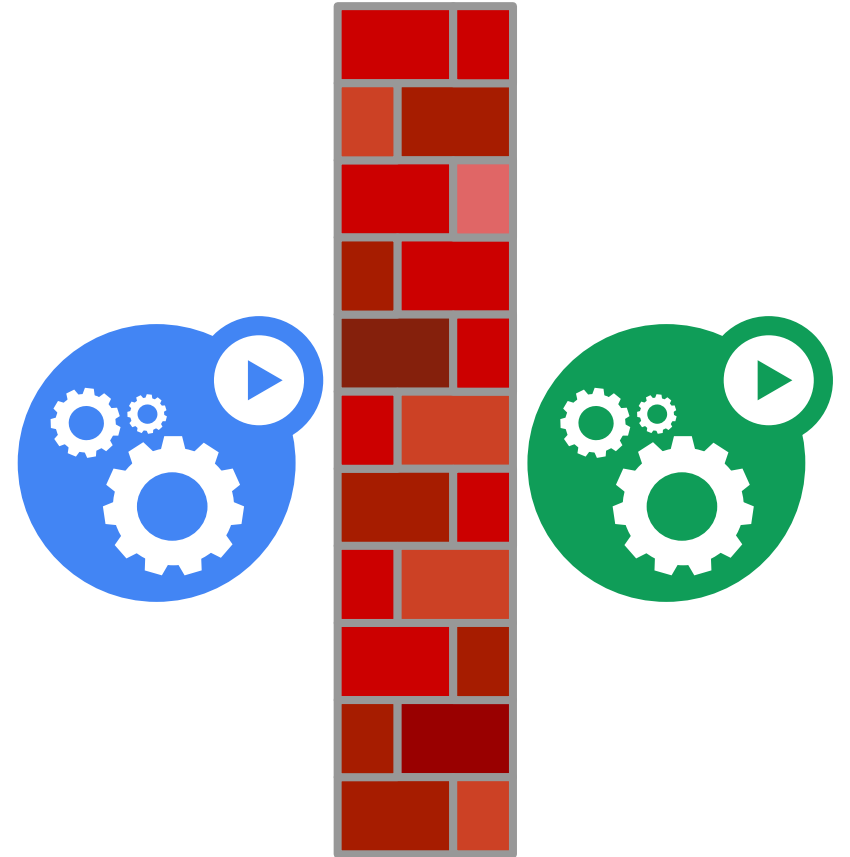


Resource Isolation

Resource Isolation

Principles:

- Apps must not be able to affect each other's performance
 - if so it is an **isolation failure**
- Repeated runs of the same app should see ~equal behavior
- QoS levels drives resource decisions in (soft) real-time
- Correct in all cases, optimal in some
 - reduce unreliable components
- SLOs are the lingua franca



Strong isolation

Pros:

- Sharing - users don't worry about interference (aka the noisy neighbor problem)
- Predictable - allows us to offer strong SLAs to apps

Cons:

- Stranding - arbitrary slices mean some resources get lost
- Confusing - how do I know how much I need?
 - analog: what size VM should I use?
 - smart auto-scaling is needed!
- Expensive - you pay for certainty

In reality this is a multi-dimensional bin-packing problem: CPU, memory, disk space, IO bandwidth, network bandwidth, ...

Requests and Limits

Request:

- how much of a resource you are asking to use, with a strong guarantee of availability
 - CPU (seconds/second)
 - RAM (bytes)
- scheduler will not over-commit requests

Limit:

- max amount of a resource you can access

Repercussions:

- Usage > Request: resources **might** be available
- Usage > Limit: throttled or killed



Quality of Service

Defined in terms of Request and Limit

Guaranteed: highest protection

- `request > 0 && limit == request`

Burstable: medium protection

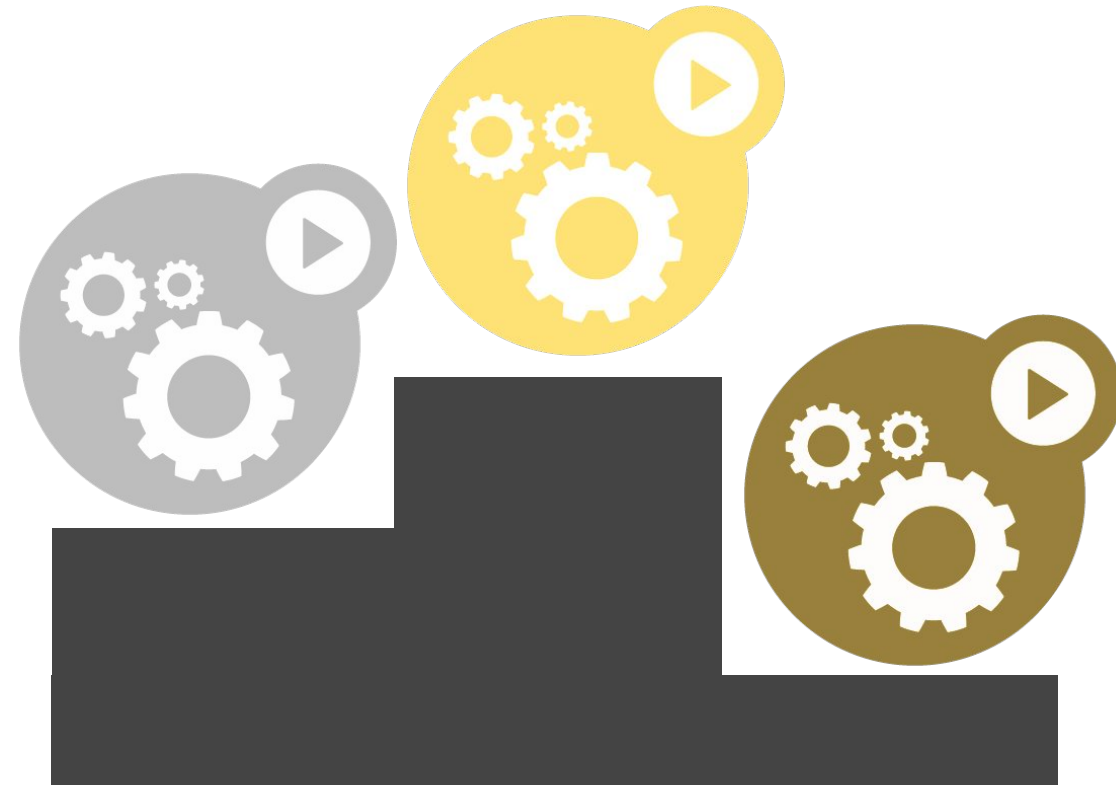
- `request > 0 && limit > request`

Best Effort: lowest protection

- `request == 0`

What does “protection” mean?

- OOM score
- CPU scheduling



Quota and Limits

ResourceQuota

Admission control: apply limits in **aggregate**

Per-namespace: ensure no user/app/department abuses the cluster

Reminiscent of disk quota by design

Applies to each type of resource

- CPU and memory for now

Disallows pods without resources



LimitRange

Admission control: limit the limits

- min and max
- ratio of limit/request

Default values for unspecified limits

Per-namespace

Together with ResourceQuota gives cluster admins powerful tools



Network Plugins

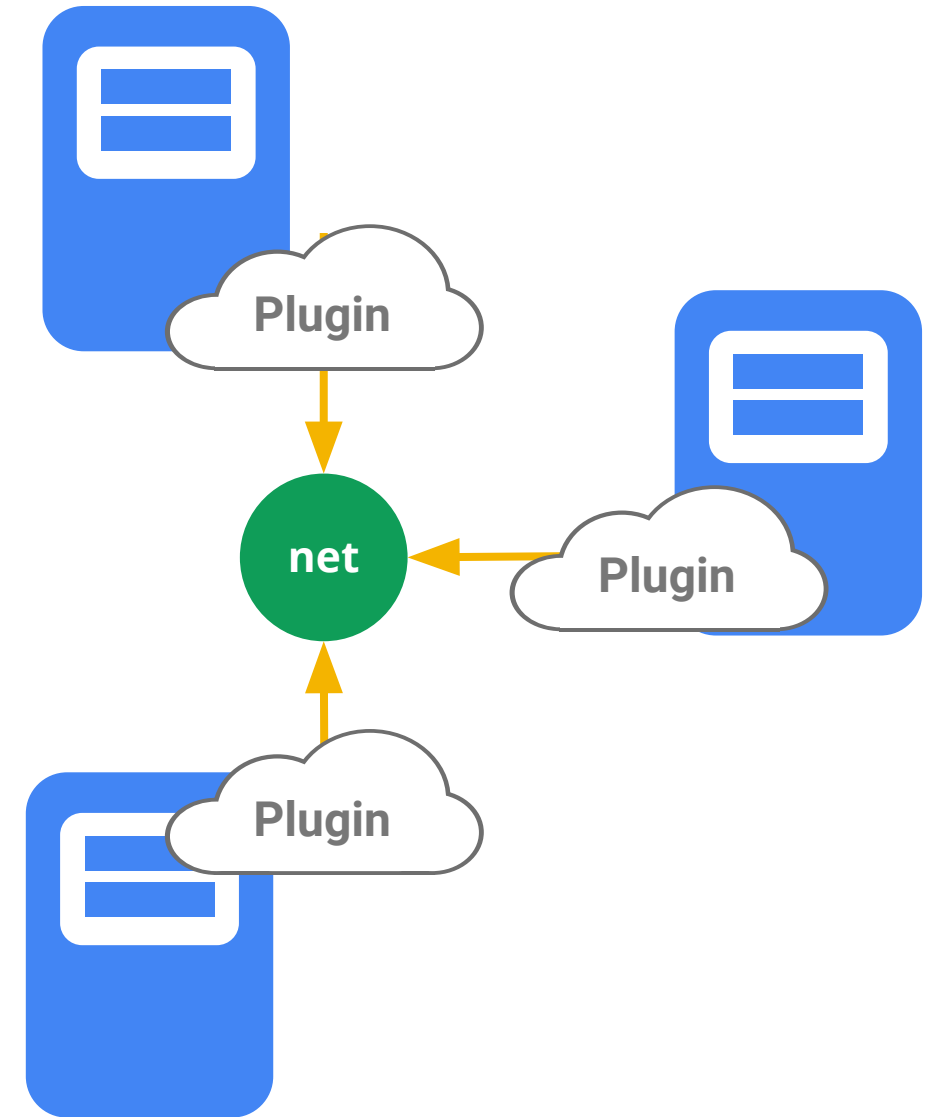
Network Plugins

Default Network plugin included in Kubernetes v1.9

Uses **CNI**:

- Simple exec interface
- Not using Docker libnetwork

Cluster admins can customize their installs



Network Isolation

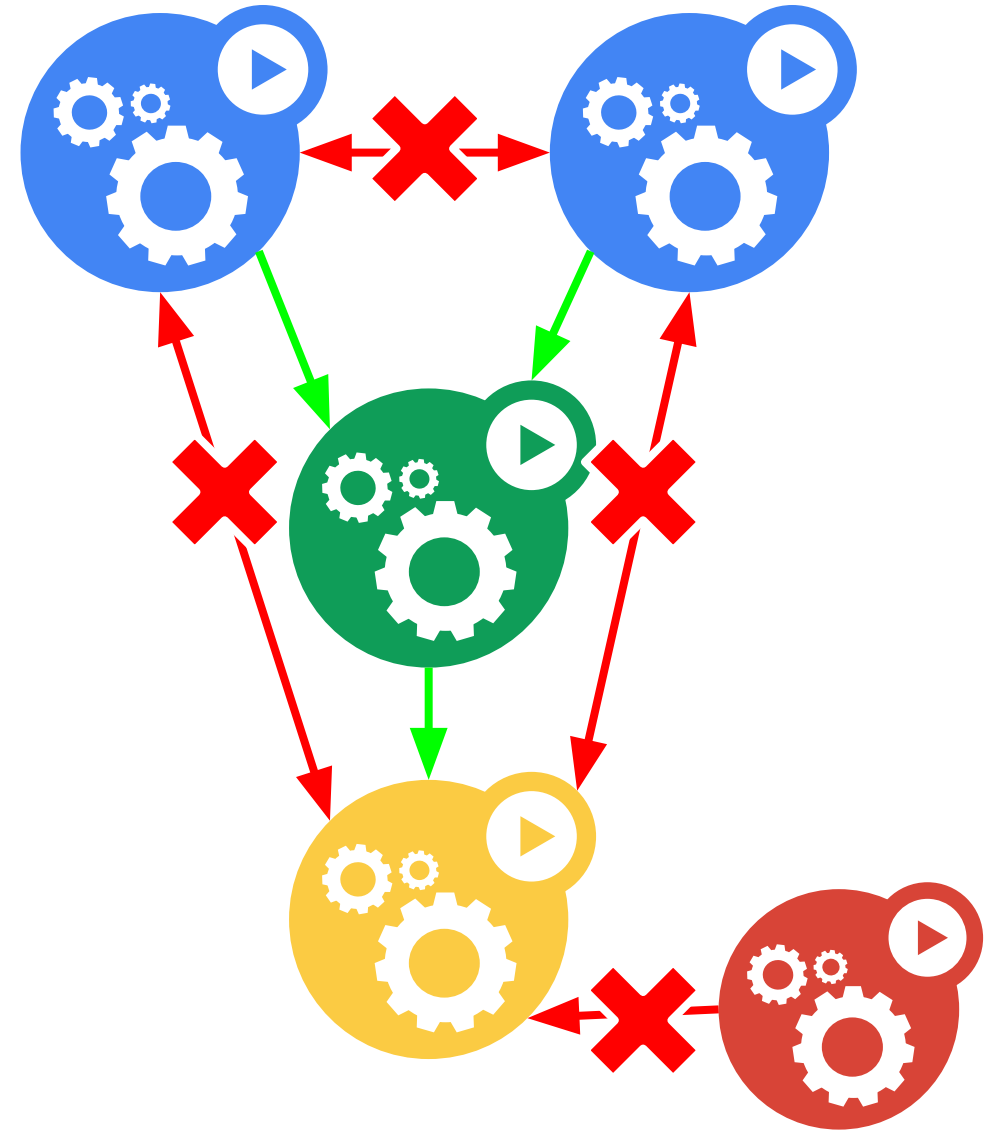
Network Isolation

Describe the DAG of your app, enforce it in the network

Restrict Pod-to-Pod traffic or across Namespaces

Designed by the network SIG

- implementations for Calico, OpenShift, Romana, OpenContrail (so far)



Node Drain



Node Drain

Goal: Evacuate a node for maintenance

- e.g. kernel upgrades

CLI: `kubectl drain`

- disallow scheduling
- allow grace period for pods to terminate
- kill pods

When done: `kubectl uncordon`

- the node rejoins the cluster



Cluster Auto-Scaling

Cluster Autoscaler

Add nodes when needed

- there are pending pods
- some pending pods would fit if we add a node

Remove nodes when not needed

- after removal, all pods must fit remaining nodes

Status: Works on GCE, GKE and AWS



Primary concepts

Container: A sealed application package (Docker)

Pod: A small group of tightly coupled Containers

example: content syncer & web server

Controller: A loop that drives current state towards desired state

example: replication controller

Service: A set of running pods that work together

example: load-balanced backends

Labels: Identifying metadata attached to other objects

example: phase=canary vs. phase=prod

Selector: A query against labels, producing a set result

example: all pods where label phase == prod

Design principles

Declarative > imperative: State your desired results, let the system actuate

Control loops: Observe, rectify, repeat

Simple > Complex: Try to do as little as possible

Modularity: Components, interfaces, & plugins

Legacy compatible: Requiring apps to change is a non-starter

Network-centric: IP addresses are cheap

No grouping: Labels are the only groups

Cattle > Pets: Manage your workload in bulk

Open > Closed: Open Source, standards, REST, JSON, etc.



Control loops

Drive **current state** -> **desired state**

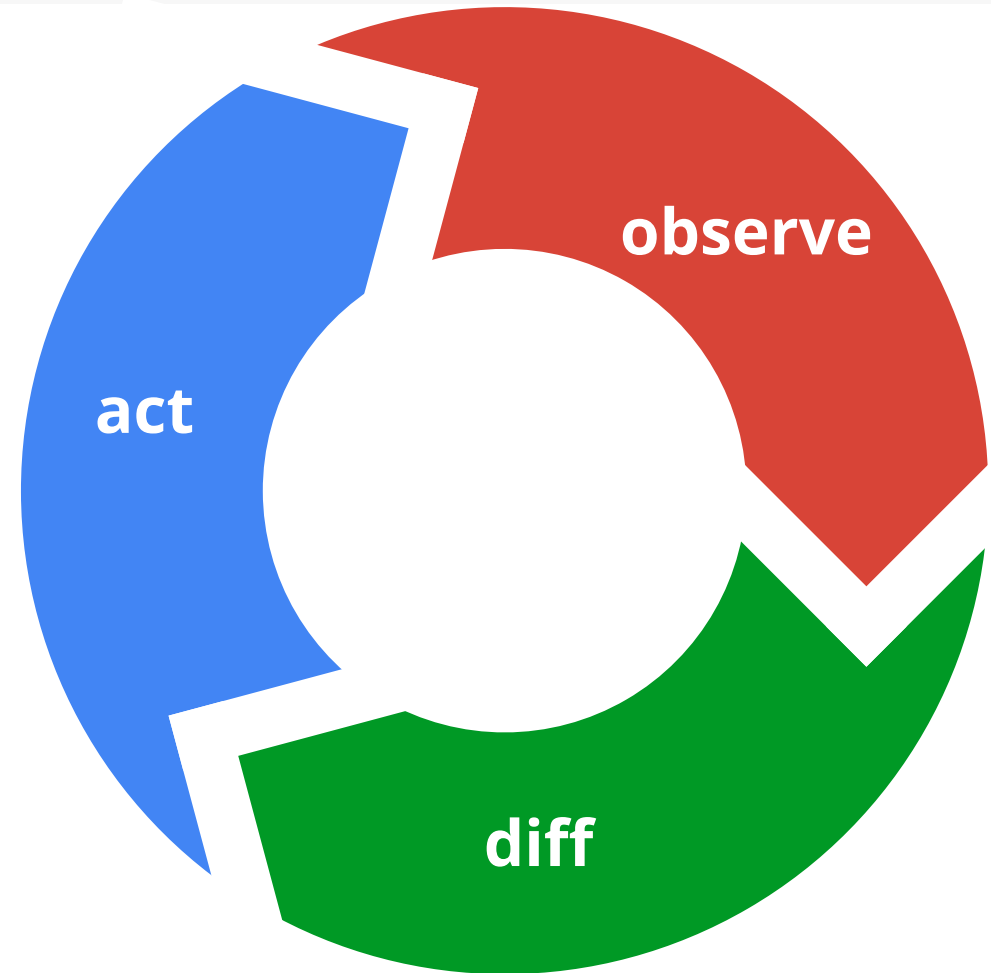
Act independently

APIs - **no shortcuts** or back doors

Observed state is truth

Recurring pattern in the system

Example: ReplicaSetController



Modularity

Loose coupling is a goal **everywhere**

- simpler
- composable
- extensible

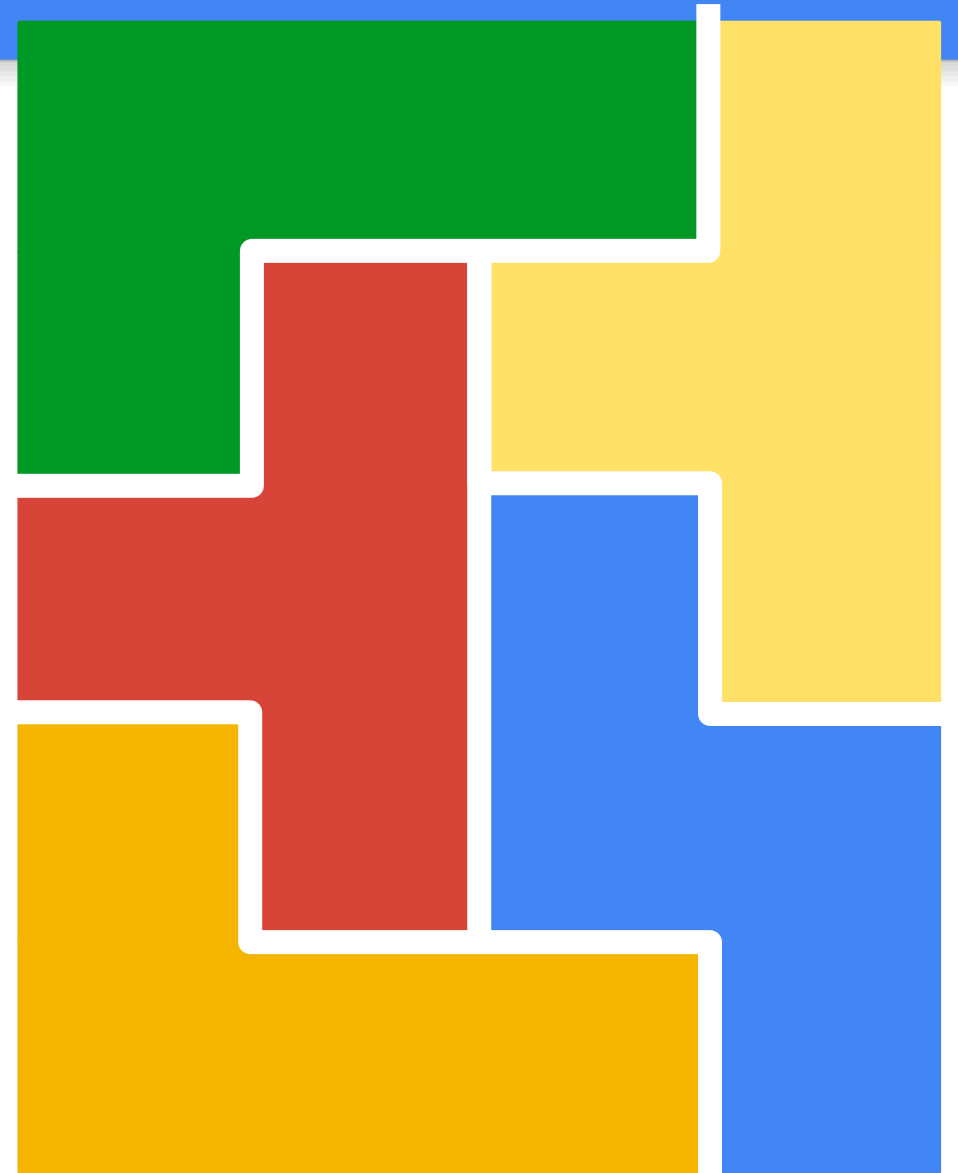
Code-level plugins where possible

Multi-process where possible

Isolate risk by interchangeable parts

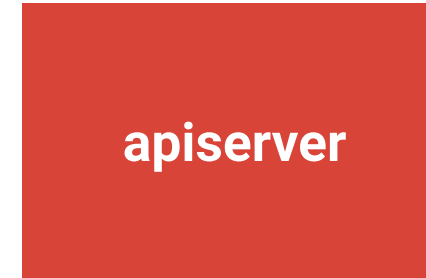
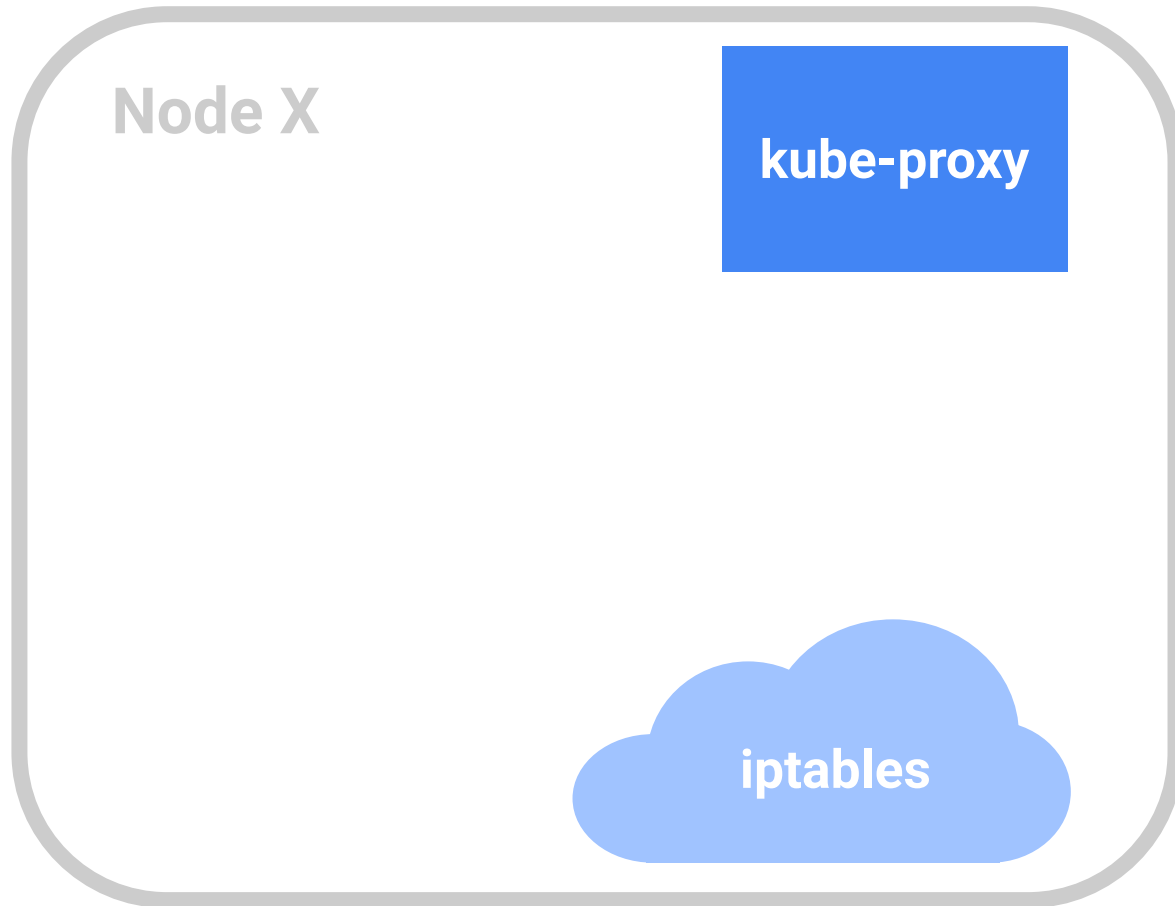
Example: ReplicationController

Example: Scheduler

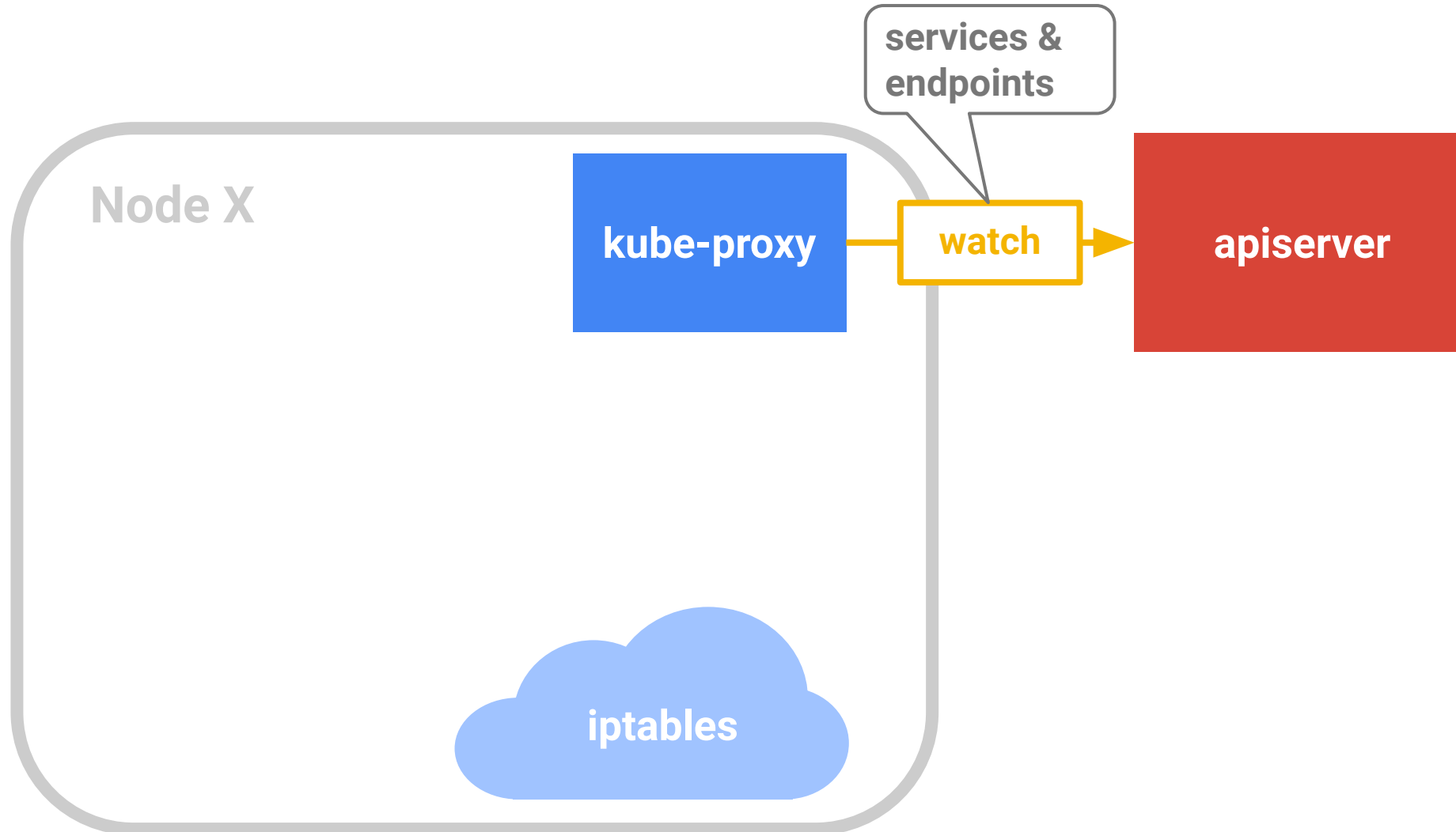


Kube Proxy iptables

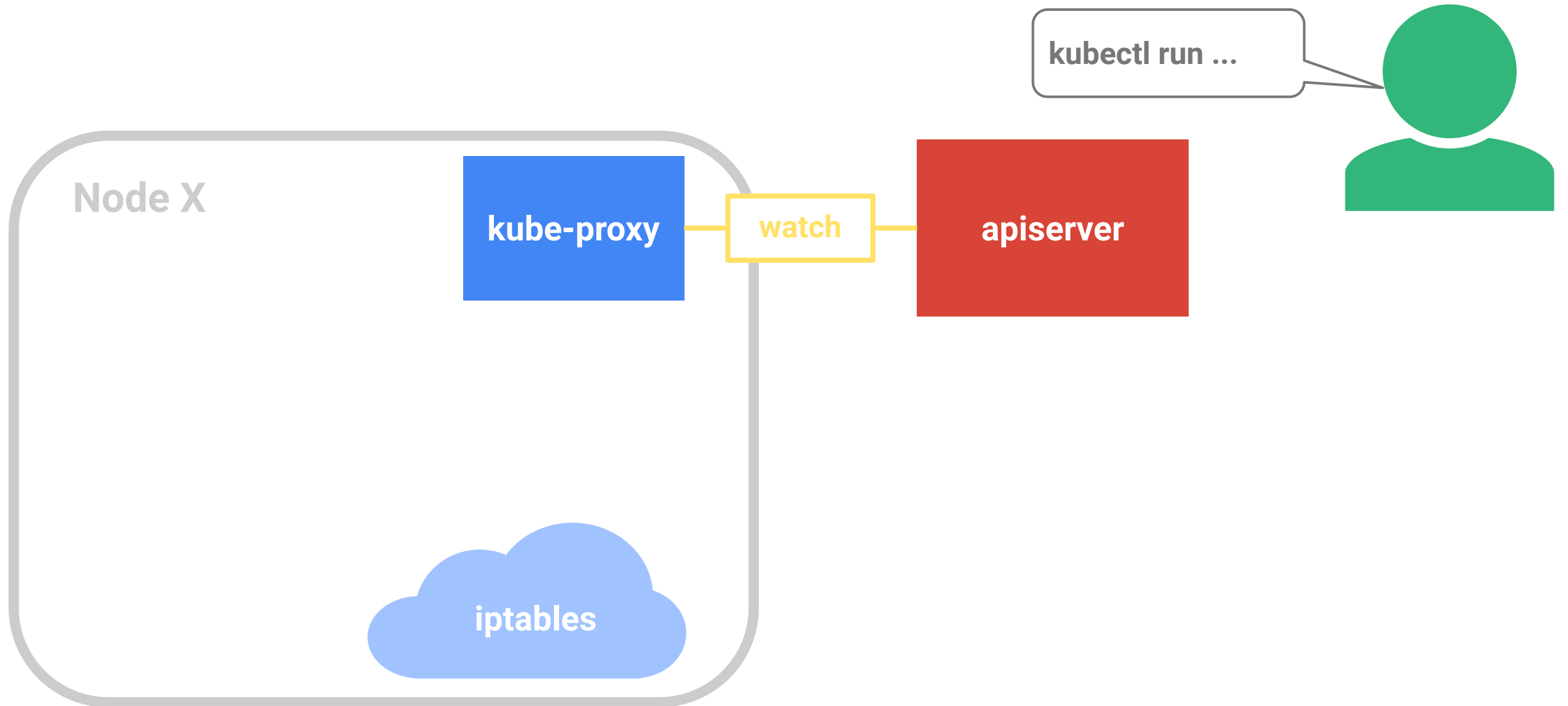
iptables kube-proxy



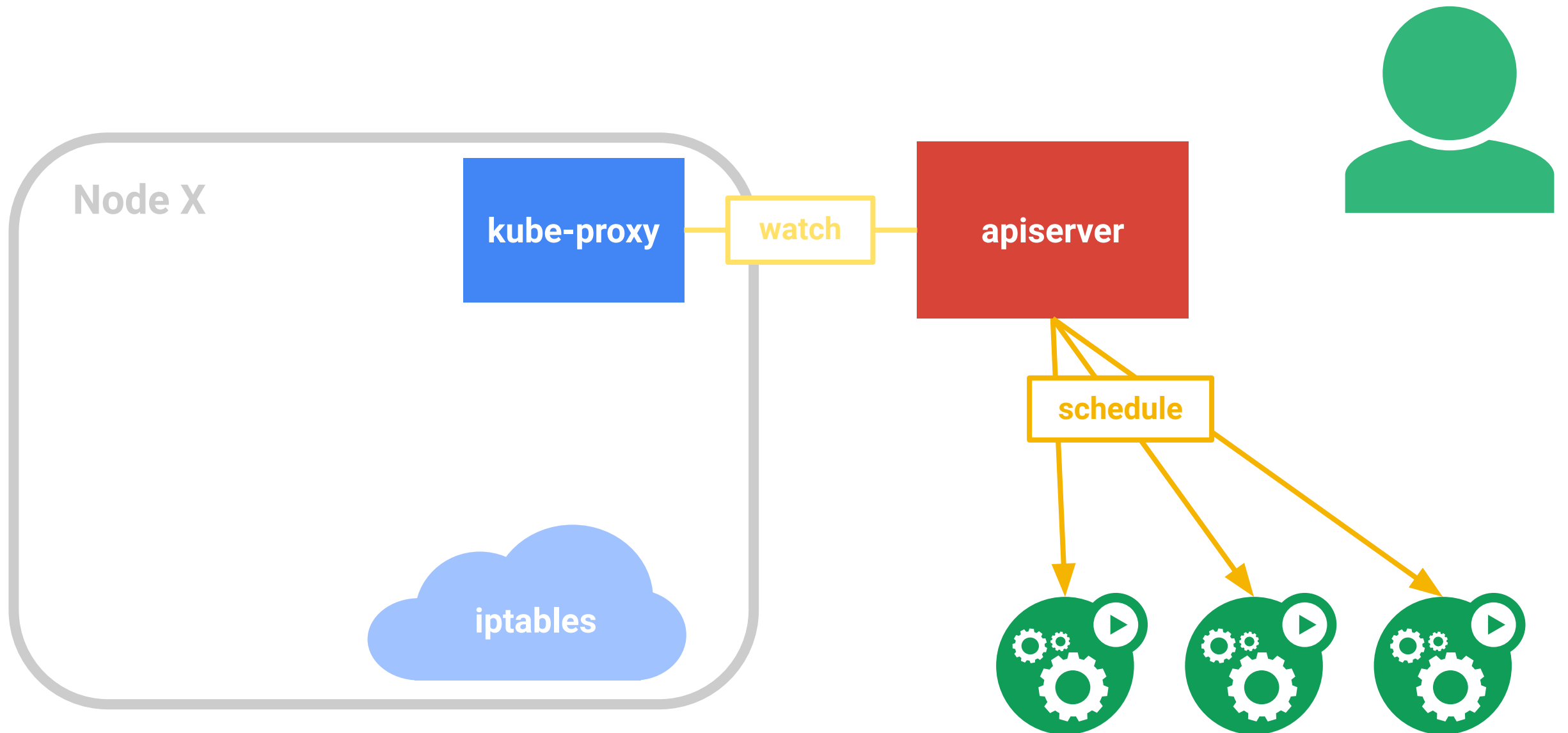
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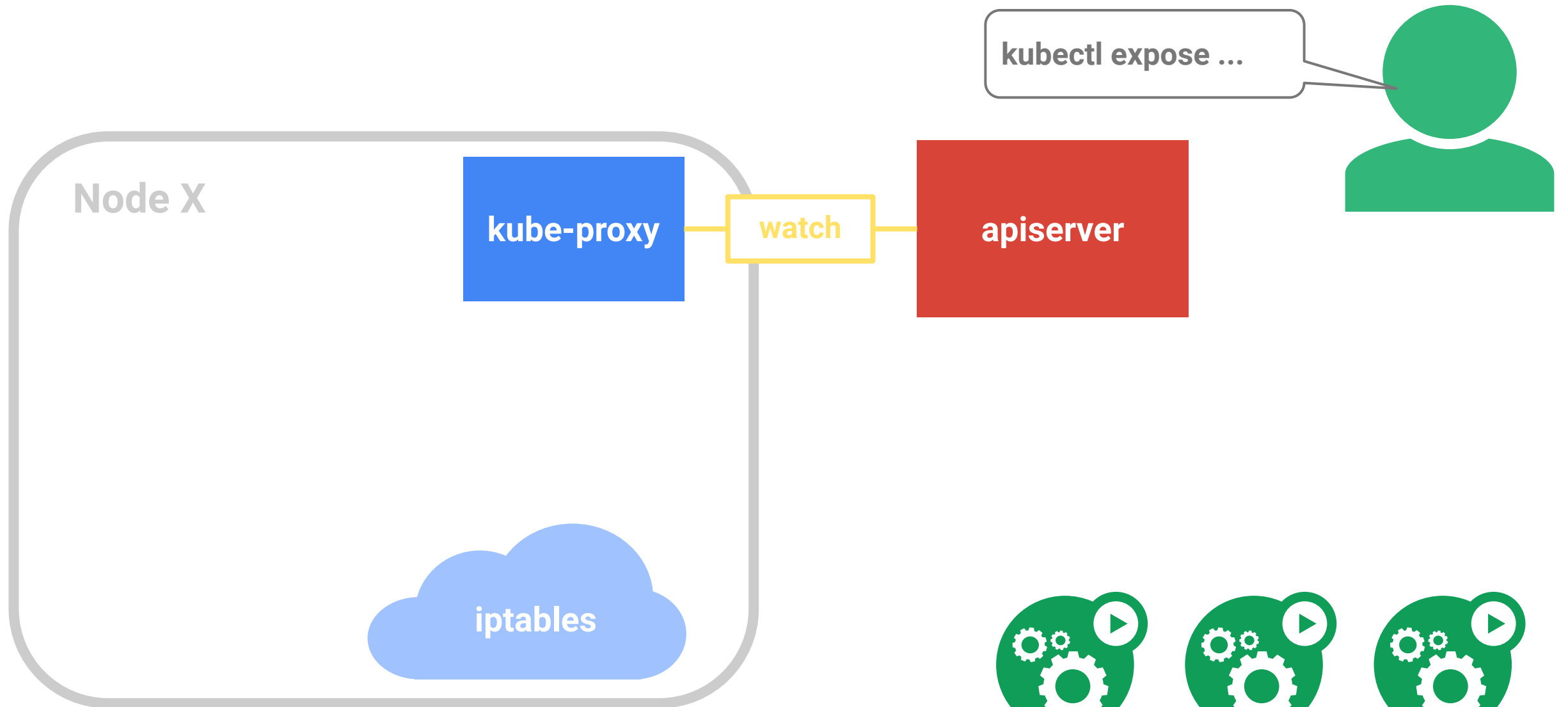
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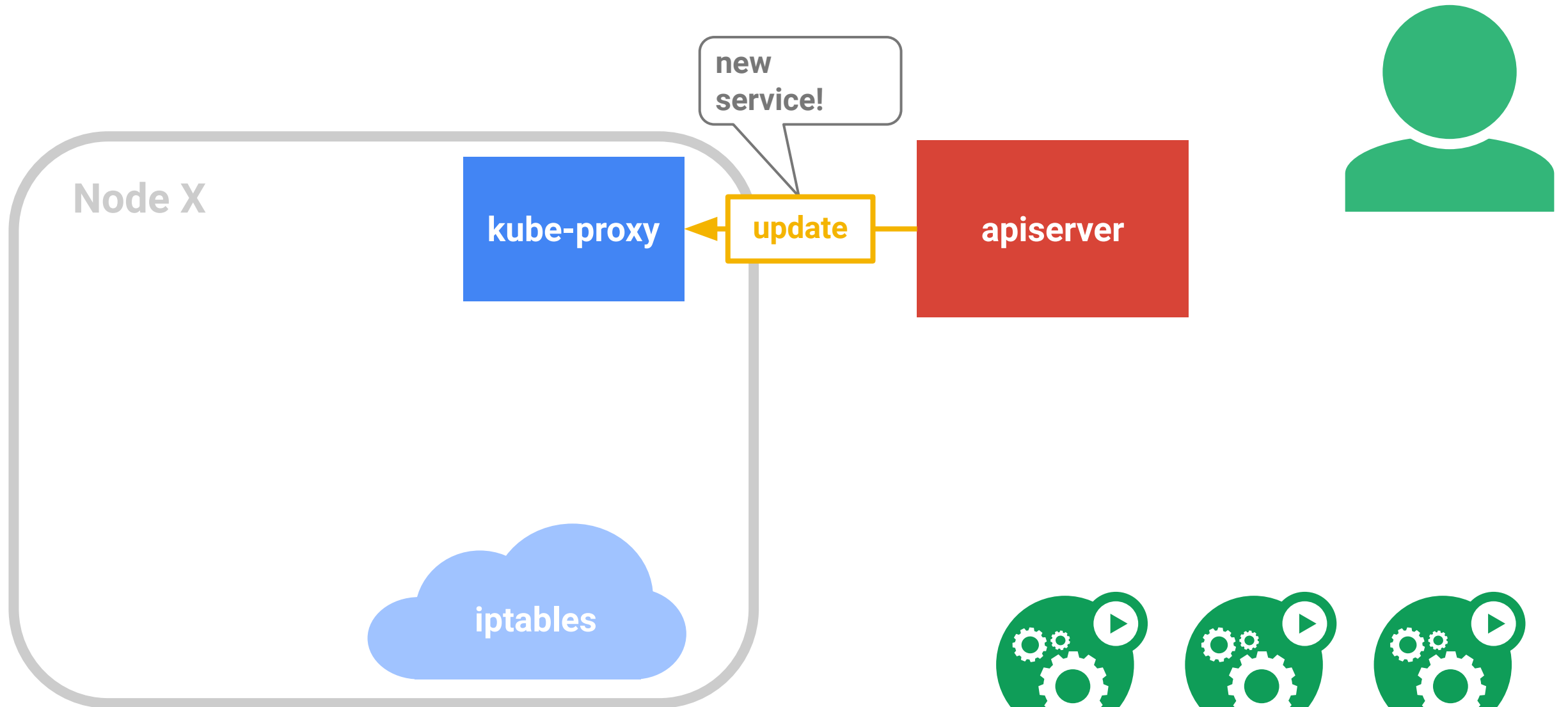
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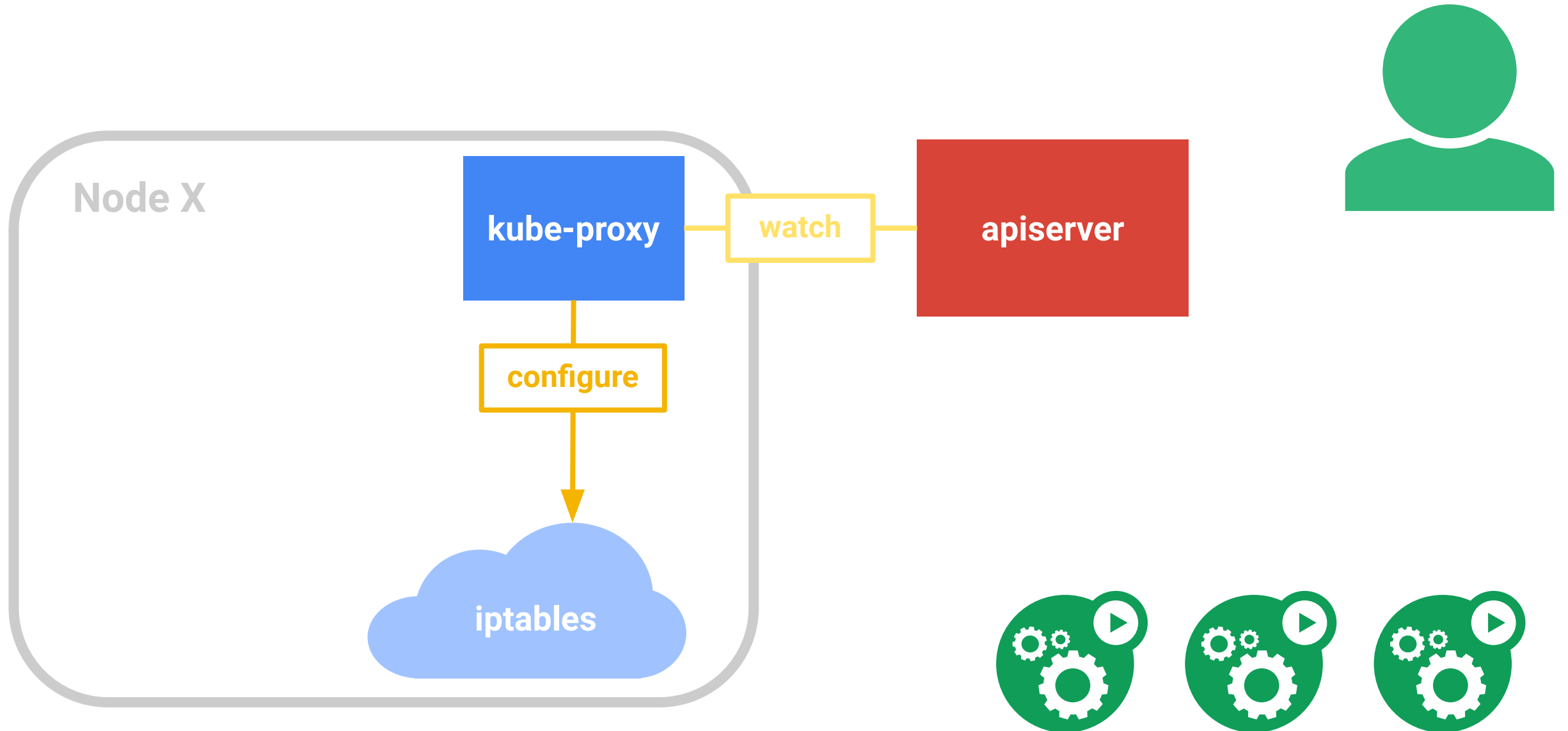
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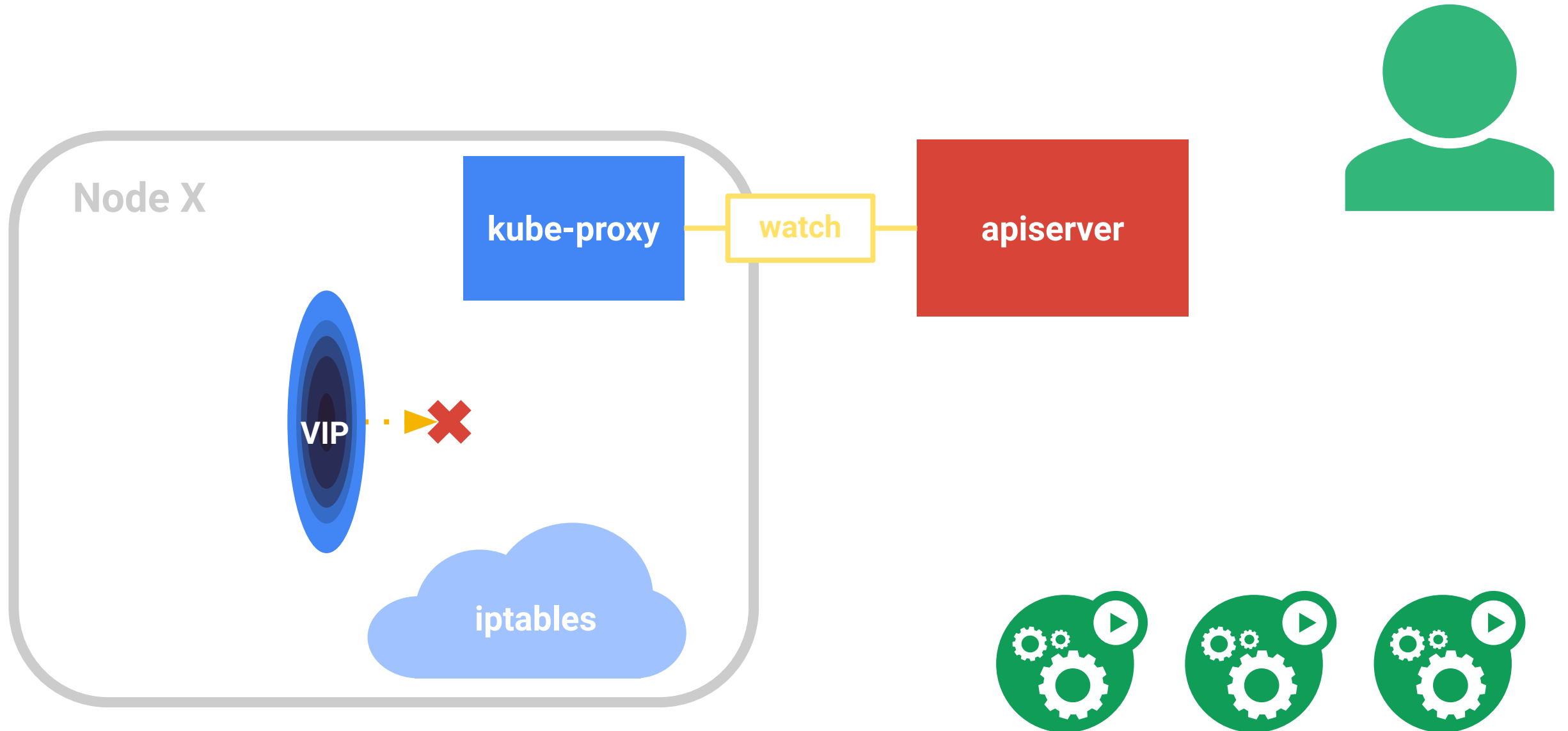
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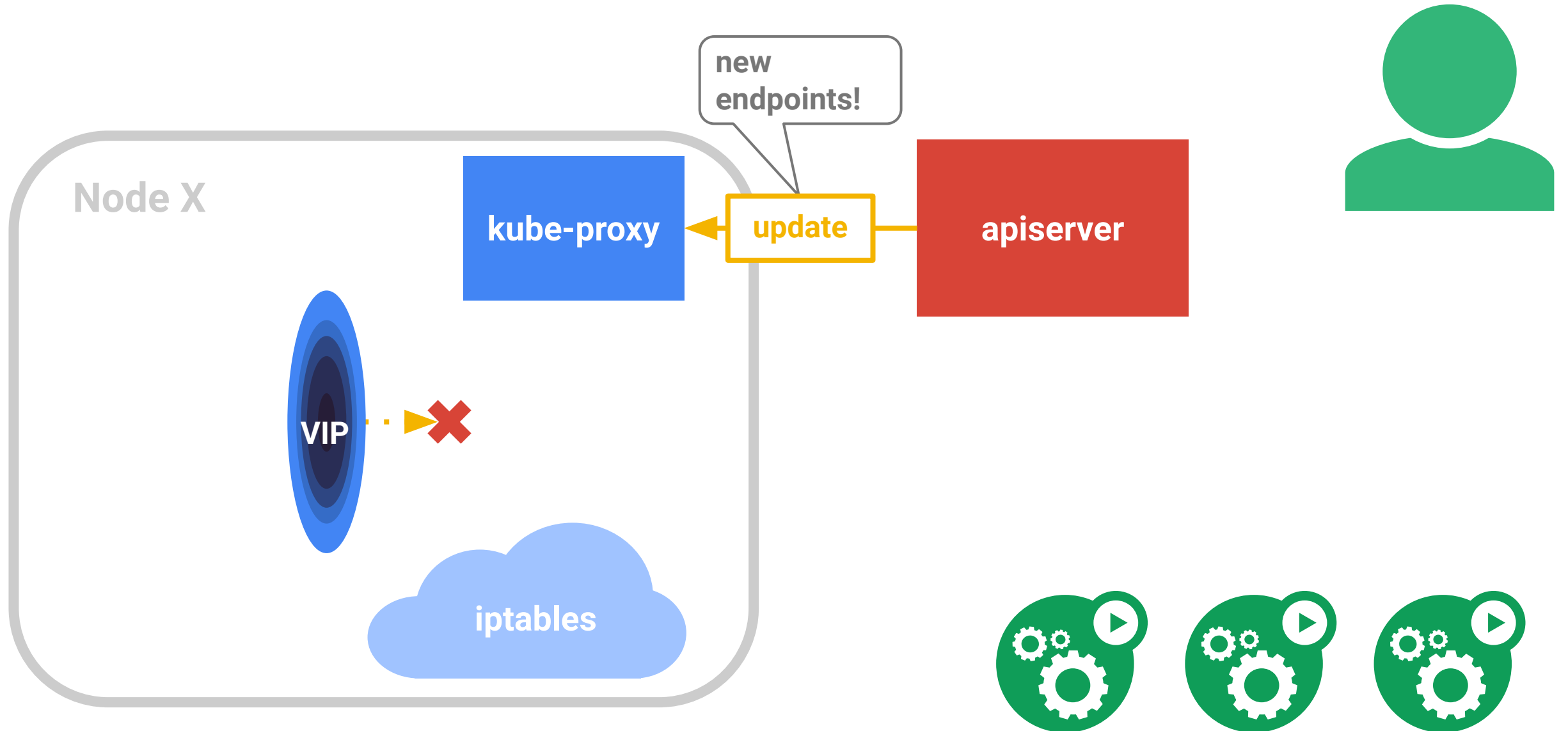
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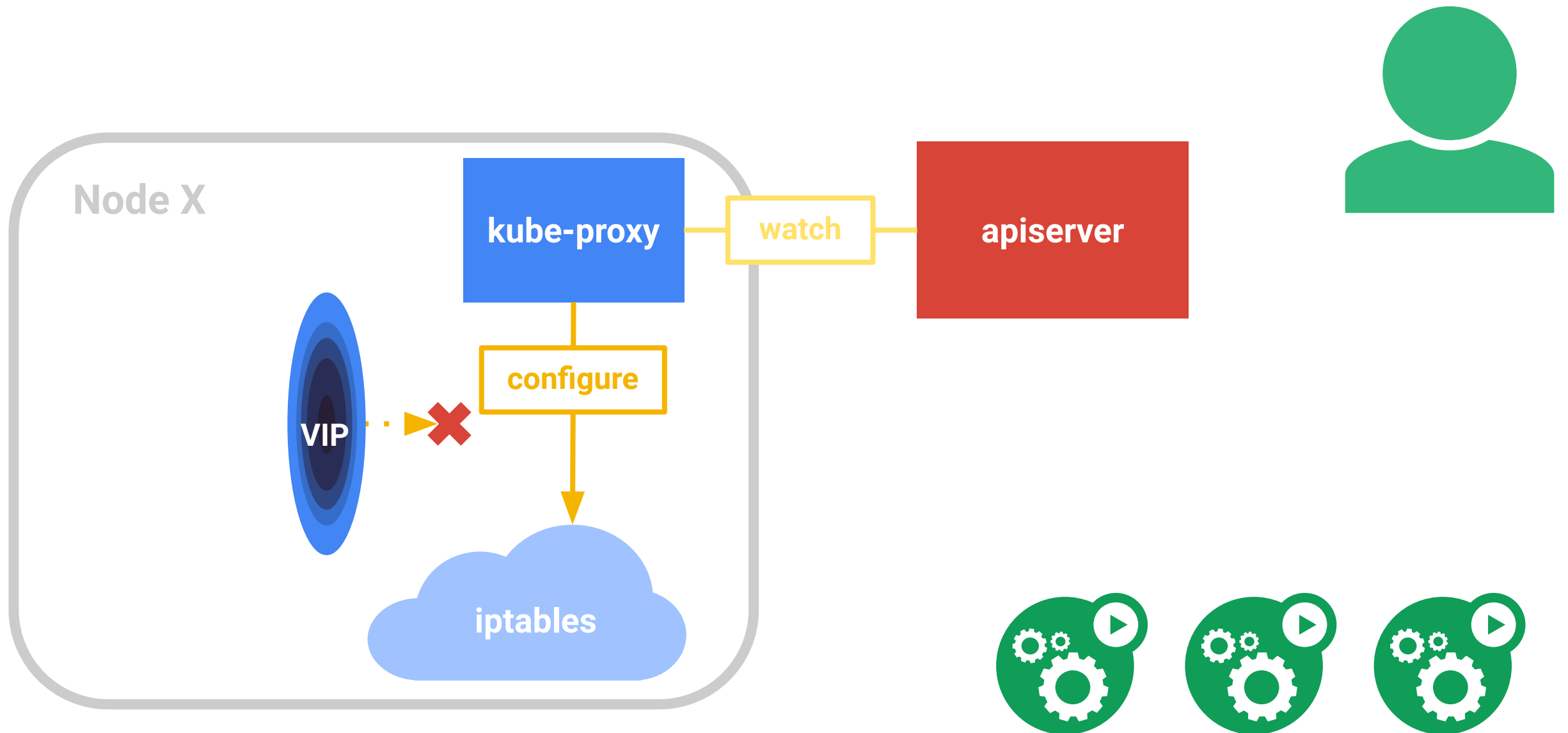
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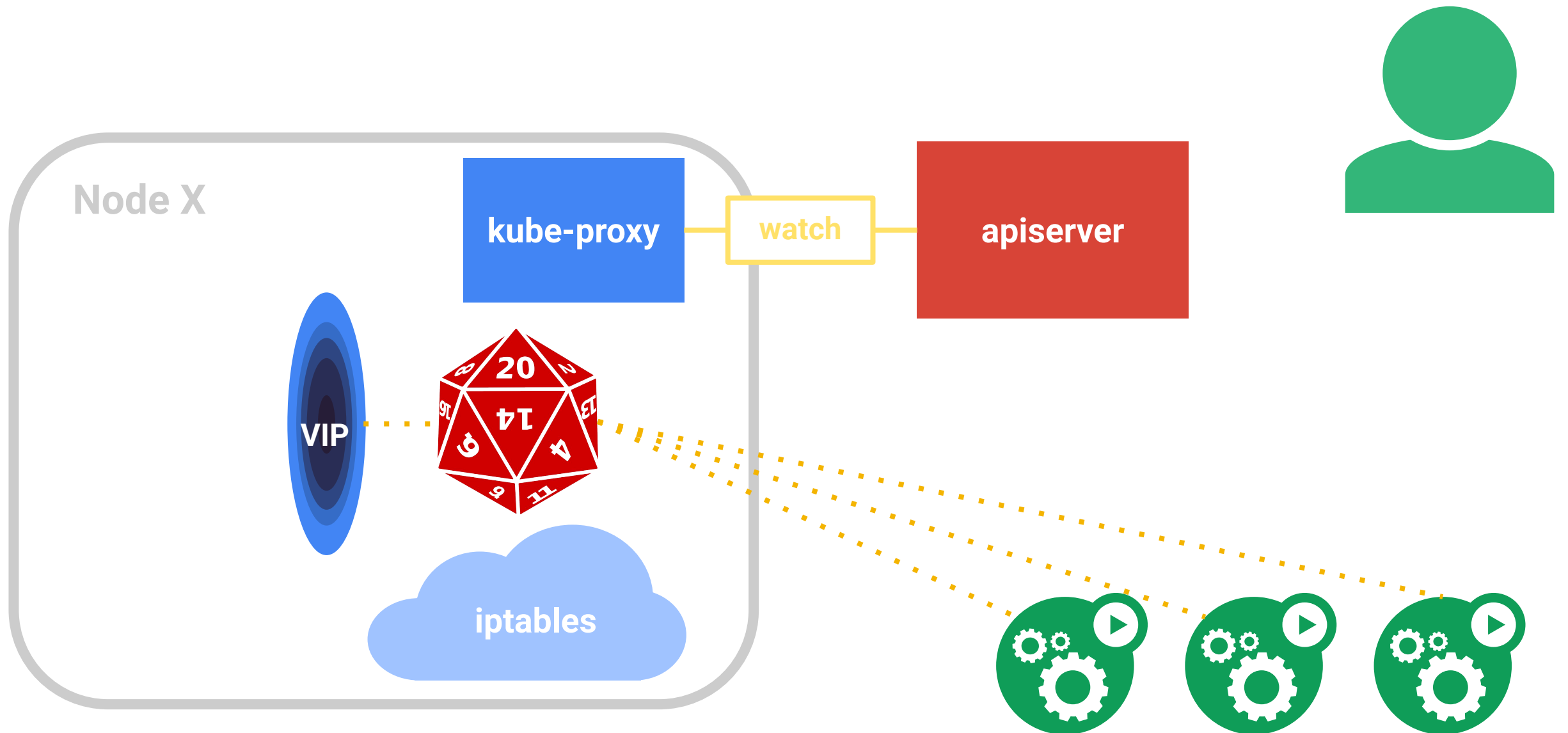
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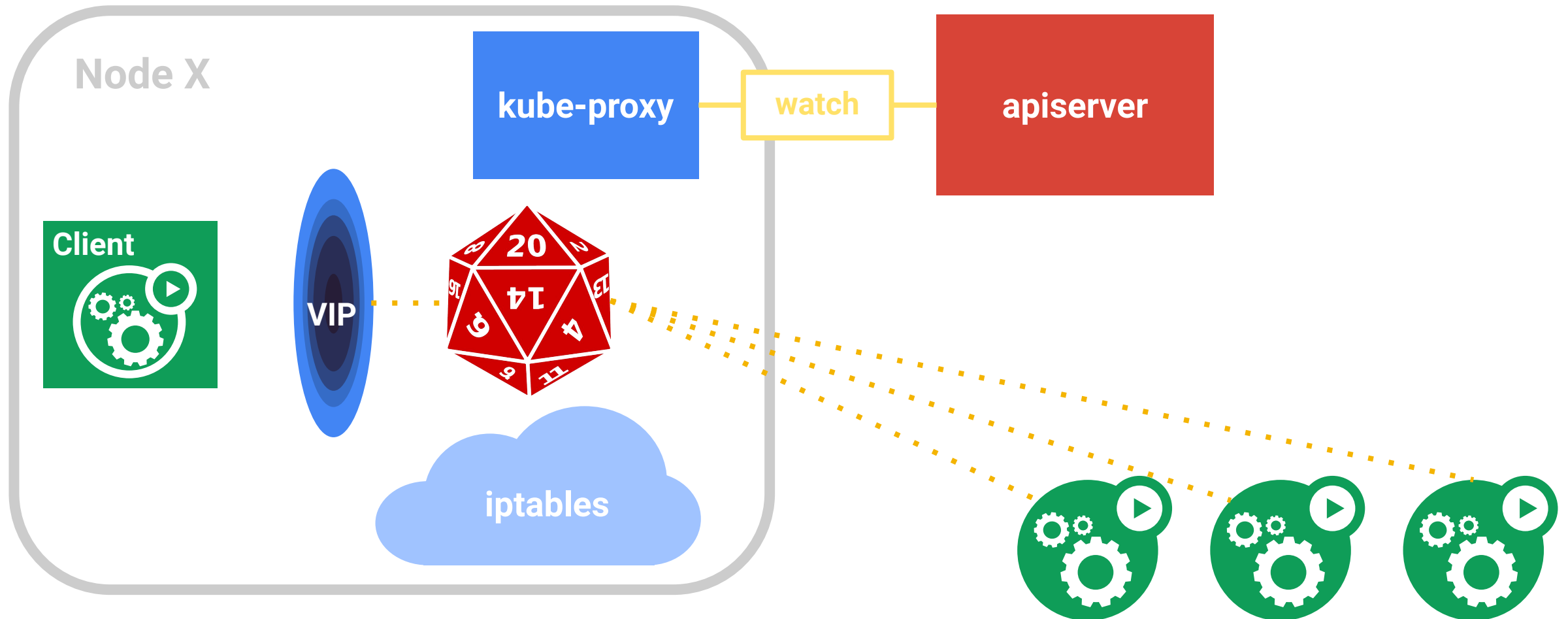
iptables kube-proxy



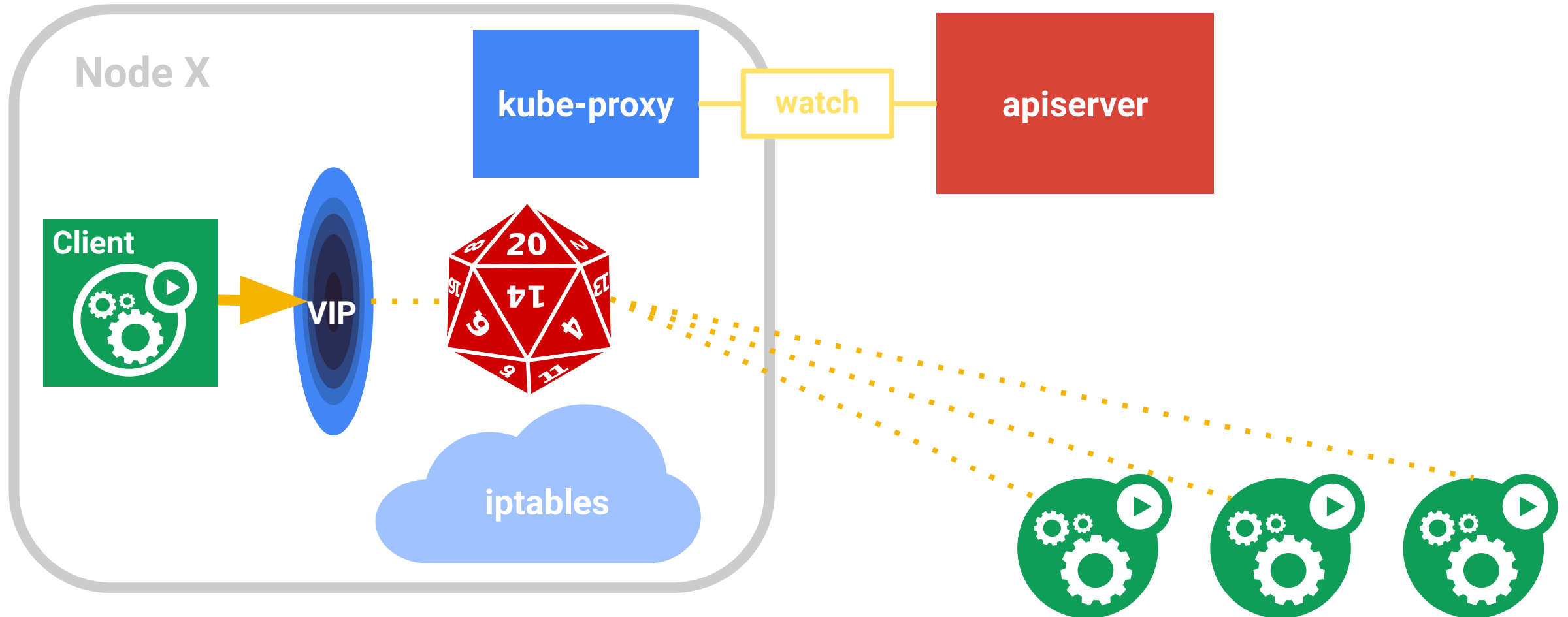
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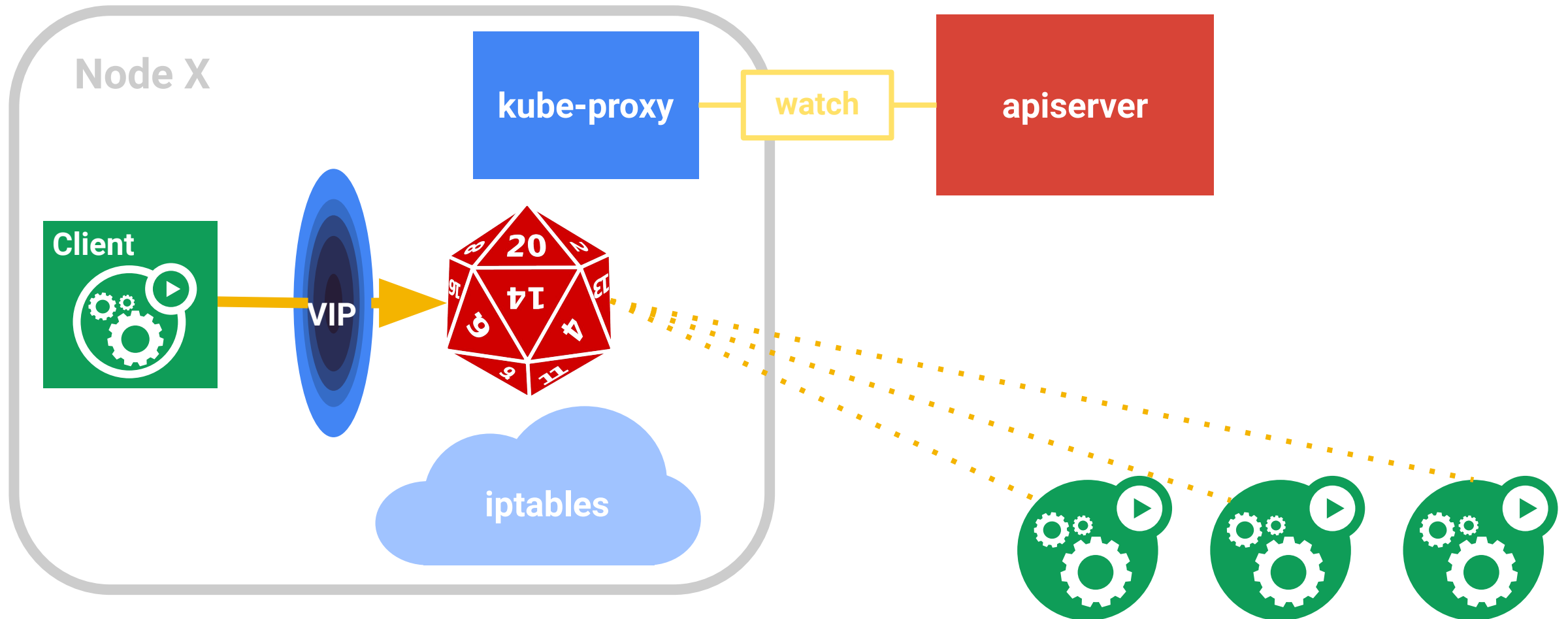
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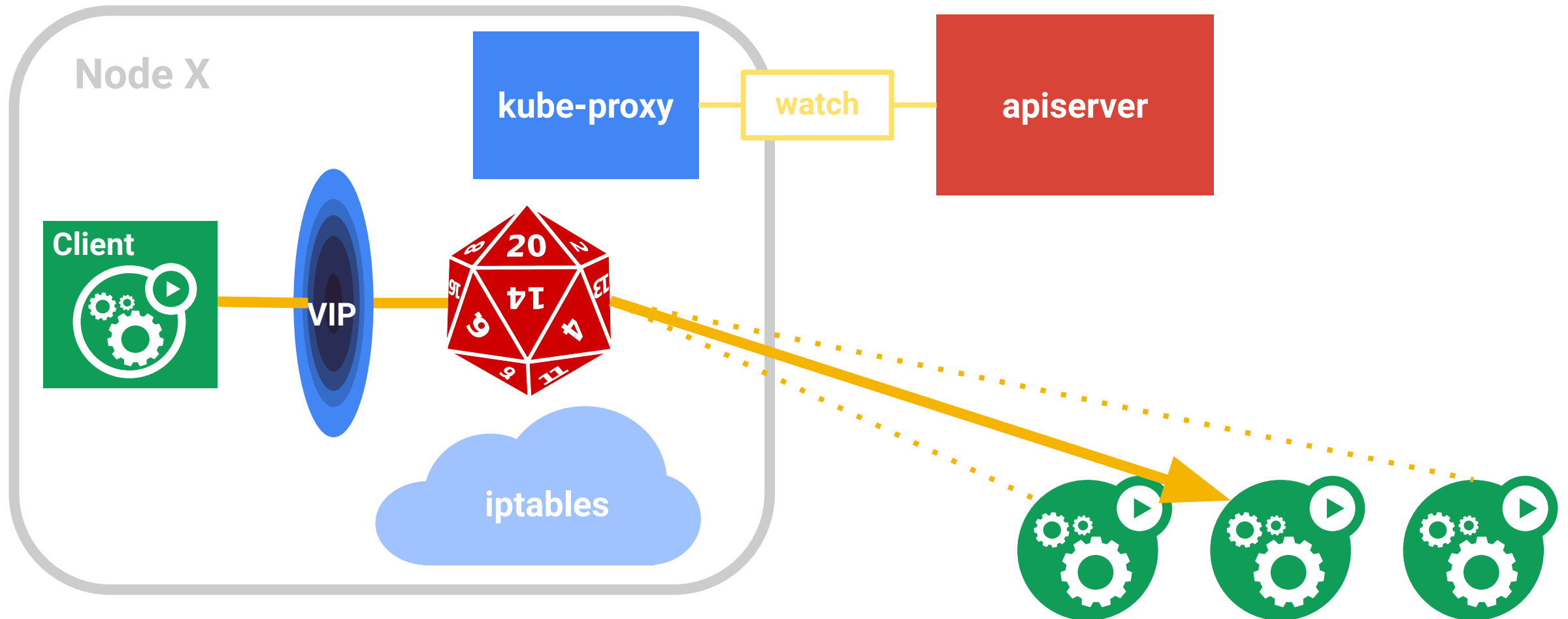
iptables kube-proxy



iptables kube-proxy



iptables kube-proxy



Icons





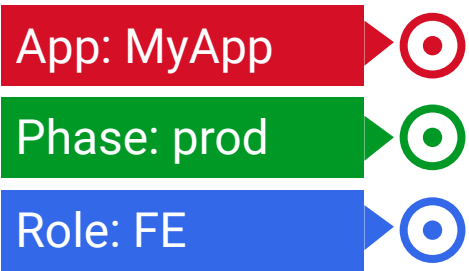
volume



container



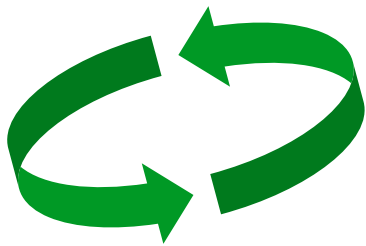
pod



labels



node



controller

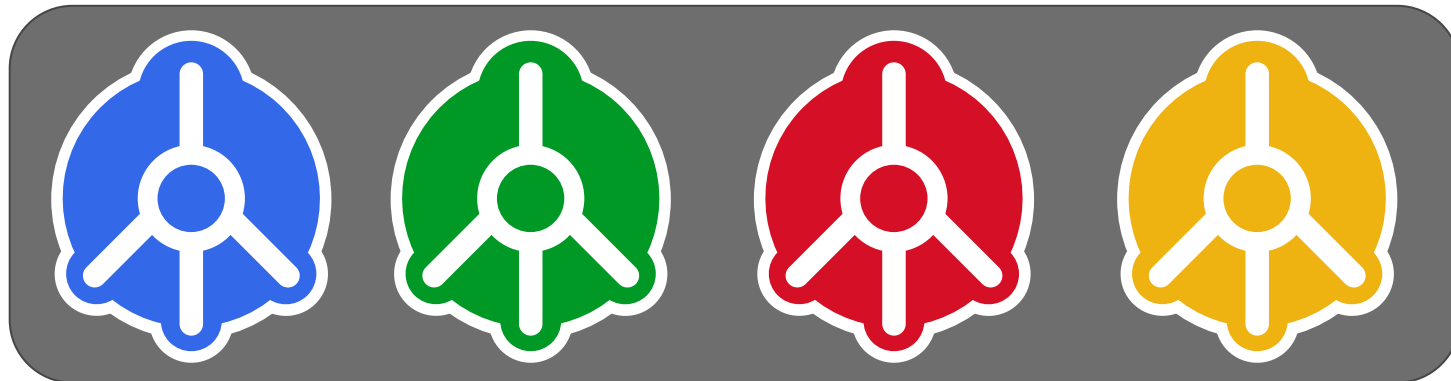


YAMLs



replication controller or replica set

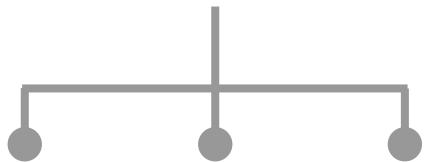
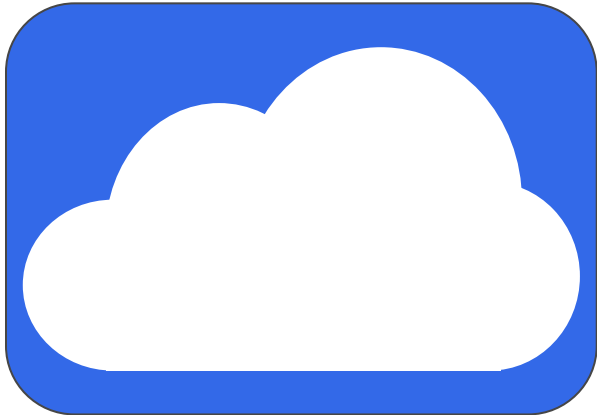




service (new)



deployment



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